tau+ (-15)
Particle 0
tauVisibleHadronic
signal
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (118.844, -90.904, -324.512, 357.350)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2519 simE=12.323084
subgraph recHits: 0 missing=2519 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 244 dE=840.526611
raw GEN: GenParticle #19 key=2
raw SIM: n/a

Primary
Vertex 9
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.282, -0.215, -0.214, 0.366)
nIn: 1 nOut: 5
raw GEN: GenVertex #11 key=-11
raw SIM: SimVertex #88 key=9

tau- (15)
Particle 1
tauVisibleHadronic
signal
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-118.844, 90.904, 324.512, 357.350)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 584 simE=0.357787
subgraph recHits: 0 missing=584 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 20 dE=33.602230
raw GEN: GenParticle #20 key=3
raw SIM: n/a

Primary
Vertex 6
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.165, 0.127, 1.006, 0.354)
nIn: 1 nOut: 3
raw GEN: GenVertex #8 key=-8
raw SIM: SimVertex #86 key=7

tau+ (-15)
Particle 2
tauVisibleLeptonic
signal
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-15.115, 14.241, -137.910, 139.476)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 611 simE=0.134844
subgraph recHits: 0 missing=611 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 15 dE=33.116207
raw GEN: GenParticle #21 key=4
raw SIM: n/a

Primary
Vertex 0
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.119, 0.113, -0.550, 0.375)
nIn: 1 nOut: 3
raw GEN: GenVertex #2 key=-2
raw SIM: SimVertex #80 key=1

tau- (15)
Particle 3
tauVisibleHadronic
signal
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (15.115, -14.241, 137.910, 139.476)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 952 simE=0.257999
subgraph recHits: 0 missing=952 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 25 dE=49.678024
raw GEN: GenParticle #22 key=5
raw SIM: n/a

Primary
Vertex 1
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.138, -0.129, 1.800, 0.380)
nIn: 1 nOut: 3
raw GEN: GenVertex #3 key=-3
raw SIM: SimVertex #81 key=2

tau+ (-15)
Particle 4
tauVisibleLeptonic
signal
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-6.800, -5.445, 72.632, 73.174)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 154 simE=0.023114
subgraph recHits: 0 missing=154 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 25 dE=73.133179
raw GEN: GenParticle #23 key=6
raw SIM: n/a

Primary
Vertex 2
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.020, -0.018, 0.785, 0.346)
nIn: 1 nOut: 3
raw GEN: GenVertex #4 key=-4
raw SIM: SimVertex #82 key=3

tau- (15)
Particle 5
tauVisibleHadronic
signal
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (6.800, 5.445, -72.632, 73.174)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 636 simE=1.141405
subgraph recHits: 0 missing=636 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 89 dE=186.351166
raw GEN: GenParticle #24 key=7
raw SIM: n/a

decay
Vertex 3
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (0.008, 0.005, 0.484, 0.340)
nIn: 1 nOut: 4
raw GEN: GenVertex #5 key=-5
raw SIM: n/a

tau+ (-15)
Particle 6
tauVisibleLeptonic
signal
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-39.092, -71.318, 281.549, 293.065)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 101 simE=0.037210
subgraph recHits: 0 missing=101 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 22 dE=37.520306
raw GEN: GenParticle #25 key=8
raw SIM: n/a

Primary
Vertex 4
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.060, -0.112, 0.994, 0.353)
nIn: 1 nOut: 3
raw GEN: GenVertex #6 key=-6
raw SIM: SimVertex #84 key=5

tau- (15)
Particle 7
tauVisibleLeptonic
signal
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (39.092, 71.318, -281.549, 293.065)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 47 simE=0.011304
subgraph recHits: 0 missing=47 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 23 dE=53.008003
raw GEN: GenParticle #26 key=9
raw SIM: n/a

Primary
Vertex 5
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.140, 0.252, -0.446, 0.372)
nIn: 1 nOut: 3
raw GEN: GenVertex #7 key=-7
raw SIM: SimVertex #85 key=6

tau+ (-15)
Particle 8
tauVisibleHadronic
signal
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-159.902, 129.762, -282.200, 349.352)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 474 simE=75.926926
subgraph recHits: 0 missing=474 recoE=0.000000
direct tracker simHits: 4 dE=4.000000
subgraph tracker simHits: 104 dE=405.194977
raw GEN: GenParticle #27 key=10
raw SIM: SimTrack #848 key=1

decay
Vertex 7
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (-3.102, 2.518, -4.926, 0.564)
nIn: 1 nOut: 4
raw GEN: GenVertex #9 key=-9
raw SIM: n/a

tau- (15)
Particle 9
tauVisibleHadronic
signal
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (159.902, -129.762, 282.200, 349.352)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 913 simE=94.252060
subgraph recHits: 0 missing=913 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 34 dE=105.694298
raw GEN: GenParticle #28 key=11
raw SIM: n/a

Primary
Vertex 8
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (1.304, -1.057, 2.850, 0.433)
nIn: 1 nOut: 4

raw GEN: GenVertex #10 key=-10
raw SIM: SimVertex #87 key=8

anti-nu_tau (-16)
Particle 10
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-2.817, 3.148, -24.998, 25.352)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #30 key=12
raw SIM: n/a

e+ (-11)
Particle 11
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: e+ (-11)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-3.670, 3.026, -34.857, 35.180)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 164 simE=0.030953
direct recHits: 0 missing=164 recoE=0.000000
subgraph simHits: 611 simE=0.134844
subgraph recHits: 0 missing=611 recoE=0.000000
direct tracker simHits: 33 dE=33.116207
subgraph tracker simHits: 15 dE=33.116207
checkpointId: 0
x4@checkpoint: (-28.505, 28.443, -293.500, 0.000)
p4@checkpoint: (-0.651, 0.885, -7.952, 8.027)
raw GEN: GenParticle #31 key=13
raw SIM: SimTrack #849 key=3

Bremsstrahlung
Vertex 262
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-5.653, 4.774, -53.573, 2.160)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #503 key=424

Bremsstrahlung
Vertex 263
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-8.832, 7.563, -84.533, 3.202)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #504 key=425

Bremsstrahlung
Vertex 264
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-17.929, 16.148, -175.927, 6.279)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #506 key=427

Bremsstrahlung
Vertex 265
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-17.939, 16.159, -176.033, 6.283)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #507 key=428

nu_e (12)
Particle 12
stableLegsFromInitialState
stableDecayProducts
pid: nu_e (12)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-8.628, 8.067, -78.056, 78.945)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #32 key=14
raw SIM: n/a

nu_tau (16)
Particle 13
stableLegsFromInitialState
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (4.497, -4.104, 35.764, 36.278)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #33 key=15
raw SIM: n/a

pi0 (111)
Particle 14
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (5.415, -5.482, 52.154, 52.720)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 662 simE=0.204957
subgraph recHits: 0 missing=662 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 6 dE=12.462387
raw GEN: GenParticle #34 key=16
raw SIM: n/a

Primary
Vertex 10
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.138, -0.129, 1.801, 0.380)
nIn: 1 nOut: 2
raw GEN: GenVertex #12 key=-12
raw SIM: SimVertex #89 key=10

pi- (-211)
Particle 15
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (5.203, -4.654, 49.993, 50.478)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 289 simE=0.051907
direct recHits: 0 missing=289 recoE=0.000000
subgraph simHits: 291 simE=0.053043
subgraph recHits: 0 missing=291 recoE=0.000000
direct tracker simHits: 37 dE=36.553421
subgraph tracker simHits: 21 dE=37.215637
checkpointId: 0
x4@checkpoint: (31.276, -26.255, 293.500, 0.000)
p4@checkpoint: (5.453, -4.294, 49.869, 50.350)
raw GEN: GenParticle #35 key=17
raw SIM: SimTrack #850 key=6

Ionisation

Vertex 259
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.654, -1.481, 16.344, 0.870)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #497 key=418

anti-nu_tau (-16)
Particle 16
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-5.633, -5.208, 63.449, 63.911)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #36 key=18
raw SIM: n/a

e+ (-11)
Particle 17
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: e+ (-11)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.837, -0.139, 5.661, 5.725)
nProdVtx: 1 nDecayVtx: 10
nParents: 1 nChildren: 10
nCheckpoints: 1
direct simHits: 95 simE=0.015243
direct recHits: 0 missing=95 recoE=0.000000
subgraph simHits: 154 simE=0.023114
subgraph recHits: 0 missing=154 recoE=0.000000
direct tracker simHits: 52 dE=51.133175
subgraph tracker simHits: 25 dE=73.133179
checkpointId: 0
x4@checkpoint: (-42.441, 8.360, 293.500, 0.000)
p4@checkpoint: (-0.466, 0.313, 3.717, 3.759)
raw GEN: GenParticle #37 key=19
raw SIM: SimTrack #851 key=8

Bremsstrahlung
Vertex 245
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-6.227, -0.780, 42.444, 1.751)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #462 key=383

Bremsstrahlung
Vertex 246
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-6.242, -0.781, 42.545, 1.754)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #463 key=384

Bremsstrahlung
Vertex 247
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-16.573, -0.695, 111.265, 4.072)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #464 key=385

Bremsstrahlung
Vertex 248
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-16.775, -0.675, 112.615, 4.118)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #465 key=386

Bremsstrahlung
Vertex 249
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-16.803, -0.672, 112.801, 4.124)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #466 key=387

Bremsstrahlung
Vertex 250
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-17.163, -0.633, 115.185, 4.205)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #469 key=390

Bremsstrahlung
Vertex 251
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-27.967, 1.578, 187.627, 6.649)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #473 key=394

Bremsstrahlung
Vertex 252
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-30.268, 2.320, 203.609, 7.188)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #475 key=396

Bremsstrahlung
Vertex 257
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (13.060, -31.203, 128.880, 4.993)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #495 key=416

Annihilation

Vertex 258
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-5.637, -0.841, 40.840, 1.718)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #496 key=417

nu_e (12)
Particle 18
stableLegsFromInitialState
stableDecayProducts
pid: nu_e (12)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.330, -0.098, 3.522, 3.539)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #38 key=20
raw SIM: n/a

nu_tau (16)
Particle 19
stableLegsFromInitialState
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (1.273, 1.354, -12.398, 12.537)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #39 key=21
raw SIM: n/a

pi0 (111)
Particle 20
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (0.200, 0.335, -2.554, 2.587)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 49 simE=1.009305
subgraph recHits: 0 missing=49 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 32 dE=59.063396
raw GEN: GenParticle #40 key=22
raw SIM: n/a

decay
Vertex 11
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (0.008, 0.005, 0.484, 0.340)
nIn: 1 nOut: 2
raw GEN: GenVertex #13 key=-13
raw SIM: n/a

pi0 (111)
Particle 21
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (0.469, 0.317, -6.220, 6.247)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 18 simE=0.002201
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 55 dE=114.287750
raw GEN: GenParticle #41 key=23
raw SIM: n/a

Primary
Vertex 12
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.008, 0.005, 0.484, 0.340)
nIn: 1 nOut: 2
raw GEN: GenVertex #14 key=-14
raw SIM: SimVertex #90 key=11

pi- (-211)
Particle 22
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (4.859, 3.439, -51.459, 51.803)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 569 simE=0.129901
direct recHits: 0 missing=569 recoE=0.000000
subgraph simHits: 569 simE=0.129901
subgraph recHits: 0 missing=569 recoE=0.000000
direct tracker simHits: 13 dE=13.000000
subgraph tracker simHits: 13 dE=13.000000
checkpointId: 0
x4@checkpoint: (27.095, 20.559, -293.500, 0.000)
p4@checkpoint: (4.618, 3.737, -51.413, 51.755)
raw GEN: GenParticle #42 key=24
raw SIM: SimTrack #852 key=11

anti-nu_tau (-16)
Particle 23
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-9.714, -16.747, 68.156, 70.853)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #43 key=25
raw SIM: n/a

mu+ (-13)
Particle 24
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: mu+ (-13)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-10.555, -20.491, 80.278, 83.522)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 101 simE=0.037210
direct recHits: 0 missing=101 recoE=0.000000
subgraph simHits: 101 simE=0.037210
subgraph recHits: 0 missing=101 recoE=0.000000
direct tracker simHits: 37 dE=37.520306
subgraph tracker simHits: 22 dE=37.520306
checkpointId: 0
x4@checkpoint: (-40.041, -73.941, 293.500, 0.000)
p4@checkpoint: (-11.365, -20.032, 80.245, 83.485)
raw GEN: GenParticle #44 key=26
raw SIM: SimTrack #853 key=13

Ionisation

Vertex 239
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-85.899, -150.564, 605.961, 21.344)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #455 key=376

Ionisation

Vertex 240
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-113.141, -193.867, 785.144, 27.560)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #456 key=377

Ionisation

Vertex 241
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-133.192, -226.217, 918.462, 32.185)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #457 key=378

Ionisation

Vertex 242
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-139.916, -237.350, 964.230, 33.772)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #458 key=379

Compton
Vertex 243
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-131.762, -244.421, 1011.075, 35.584)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #459 key=380

nu_mu (14)
Particle 25
stableLegsFromInitialState
stableDecayProducts
pid: nu_mu (14)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-18.823, -34.079, 133.115, 138.691)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #45 key=27
raw SIM: n/a

nu_tau (16)
Particle 26
stableLegsFromInitialState
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (16.471, 29.072, -113.417, 118.236)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #46 key=28
raw SIM: n/a

mu- (13)
Particle 27
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: mu- (13)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (11.913, 22.858, -90.161, 93.773)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 47 simE=0.011304
direct recHits: 0 missing=47 recoE=0.000000
subgraph simHits: 47 simE=0.011304
subgraph recHits: 0 missing=47 recoE=0.000000
direct tracker simHits: 52 dE=53.008007
subgraph tracker simHits: 23 dE=53.008003
checkpointId: 0
x4@checkpoint: (37.515, 75.235, -293.500, 0.000)
p4@checkpoint: (11.082, 23.258, -90.122, 93.732)
raw GEN: GenParticle #47 key=29
raw SIM: SimTrack #854 key=16

Ionisation

Vertex 236
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (96.642, 206.203, -793.788, 27.902)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #452 key=373

Ionisation

Vertex 237
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.765, 248.414, -953.464, 33.448)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #453 key=374

Compton
Vertex 238
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (97.920, 209.938, -797.852, 28.077)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #454 key=375

anti-nu_mu (-14)
Particle 28
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_mu (-14)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (10.708, 19.388, -77.971, 81.056)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #48 key=30
raw SIM: n/a

nu_tau (16)
Particle 29
stableLegsFromInitialState
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-93.537, 71.328, 255.947, 281.684)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #49 key=31
raw SIM: n/a

pi0 (111)
Particle 30
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-13.816, 10.722, 36.733, 40.684)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 206 simE=0.159267
subgraph recHits: 0 missing=206 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #50 key=32
raw SIM: n/a

Primary
Vertex 13
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.165, 0.127, 1.006, 0.355)
nIn: 1 nOut: 2
raw GEN: GenVertex #15 key=-15
raw SIM: SimVertex #91 key=12

pi- (-211)
Particle 31
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-11.490, 8.855, 31.832, 34.982)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 384 simE=0.198520
direct recHits: 0 missing=384 recoE=0.000000
subgraph simHits: 384 simE=0.198520
subgraph recHits: 0 missing=384 recoE=0.000000
direct tracker simHits: 33 dE=33.602230
subgraph tracker simHits: 33 dE=33.602230
checkpointId: 0
x4@checkpoint: (-101.241, 70.379, 271.554, 0.000)
p4@checkpoint: (-12.261, 7.657, 31.803, 34.934)
raw GEN: GenParticle #51 key=33
raw SIM: SimTrack #855 key=19

anti-nu_tau (-16)
Particle 32
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-104.486, 84.660, -183.662, 227.631)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #52 key=34
raw SIM: n/a

pi0 (111)
Particle 33
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-3.640, 2.974, -6.569, 8.078)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 100 simE=7.527984
subgraph recHits: 0 missing=100 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 2 dE=5.277207
raw GEN: GenParticle #53 key=35
raw SIM: SimTrack #1414 key=713228

decay
Vertex 14
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (-3.102, 2.518, -4.926, 0.564)
nIn: 1 nOut: 2
raw GEN: GenVertex #16 key=-16
raw SIM: n/a

pi0 (111)
Particle 34
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-32.776, 26.973, -58.117, 71.968)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 412 simE=67.785294
subgraph recHits: 0 missing=412 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 101 dE=351.189484
raw GEN: GenParticle #54 key=36

raw SIM: SimTrack #1413 key=713227 decay
Vertex 15
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (-3.102, 2.519, -4.926, 0.564)
nIn: 1 nOut: 2
raw GEN: GenVertex #17 key=-17
raw SIM: n/a

pi+ (211)
Particle 35
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-19.001, 15.156, -33.852, 41.674)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 33 simE=0.612131
direct recHits: 0 missing=33 recoE=0.000000
subgraph simHits: 34 simE=0.613643
subgraph recHits: 0 missing=34 recoE=0.000000
direct tracker simHits: 43 dE=44.728310
subgraph tracker simHits: 19 dE=44.728310
checkpointId: 0
x4@checkpoint: (-94.259, 79.487, -171.161, 0.000)
p4@checkpoint: (-18.100, 16.141, -33.817, 41.614)
raw GEN: GenParticle #55 key=37
raw SIM: SimTrack #1412 key=713226

Bremsstrahlung
Vertex 533
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.495, 74.363, -160.411, 6.948)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #847 key=768

nu_tau (16)
Particle 36
stableLegsFromInitialState
stableDecayProducts
pid: nu_tau (16)
status: 1

statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (85.296, -68.850, 149.929, 185.727)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #56 key=38
raw SIM: n/a

pi- (-211)
Particle 37
stableLegsFromInitialState
stableDecayProducts
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1

statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (3.130, -2.443, 5.360, 6.672)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 105 simE=2.028012
subgraph recHits: 0 missing=105 recoE=0.000000
direct tracker simHits: 11 dE=11.000000
subgraph tracker simHits: 19 dE=29.167646
raw GEN: GenParticle #57 key=39
raw SIM: SimTrack #856 key=21

HadronInelastic

Vertex 233
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (19.082, -14.038, 32.574, 1.667)
nIn: 1 nOut: 4
raw GEN: n/a
raw SIM: SimVertex #449 key=370

pi- (-211)
Particle 38
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1

statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (32.663, -26.895, 57.648, 71.509)
nProdVtx: 1 nDecayVtx: 7
nParents: 1 nChildren: 15
nCheckpoints: 1
direct simHits: 580 simE=27.845217
direct recHits: 0 missing=580 recoE=0.000000
subgraph simHits: 598 simE=27.880608
subgraph recHits: 0 missing=598 recoE=0.000000
direct tracker simHits: 46 dE=46.875011
subgraph tracker simHits: 22 dE=46.875015
checkpointId: 0
x4@checkpoint: (96.405, -76.869, 168.553, 0.000)
p4@checkpoint: (33.481, -25.814, 57.603, 71.453)
raw GEN: GenParticle #58 key=40
raw SIM: SimTrack #857 key=22

Annihilation

Vertex 226
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.764, -80.022, 212.523, 9.104)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #440 key=361

HadronInelastic

Vertex 227
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (70.538, -90.799, 194.075, 14.380)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #441 key=362

HadronInelastic

Vertex 228
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (81.802, -81.800, 221.832, 14.581)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #442 key=363

HadronInelastic

Vertex 229
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-7.343, 19.519, 138.685, 19.385)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #443 key=364

HadronInelastic

Vertex 230
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-25.928, -24.310, 17.630, 35.848)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #444 key=365

HadronInelastic

Vertex 231
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (70.522, -9.413, 223.973, 17.490)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #445 key=366

NuclearCapture
Vertex 232
domain: SIM
reason: NuclearCapture
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (87.380, -75.821, 191.843, 133.926)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #447 key=368

pi+ (211)
Particle 39
stableLegsFromInitialState
stableDecayProducts
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1

statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (38.814, -31.573, 69.263, 85.444)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 19
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 415 simE=64.343491
subgraph recHits: 0 missing=415 recoE=0.000000
direct tracker simHits: 26 dE=26.361404
subgraph tracker simHits: 15 dE=29.651627
raw GEN: GenParticle #59 key=41
raw SIM: SimTrack #858 key=23

HadronInelastic

Vertex 207
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.592, -74.823, 162.134, 6.987)
nIn: 1 nOut: 15
raw GEN: n/a
raw SIM: SimVertex #414 key=335

HadronAtRest

Vertex 208
domain: SIM
reason: HadronAtRest
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (92.421, -69.221, 161.631, 7.367)
nIn: 1 nOut: 4
raw GEN: n/a
raw SIM: SimVertex #415 key=336

anti-nu_tau (-16)
Particle 40
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (36.073, -27.227, -98.556, 108.425)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #60 key=42
raw SIM: n/a

pi0 (111)
Particle 41
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (21.850, -17.151, -60.442, 66.519)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 813 simE=0.306941
subgraph recHits: 0 missing=813 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 43 dE=103.155655
raw GEN: GenParticle #61 key=43
raw SIM: n/a

Primary
Vertex 16
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.282, -0.215, -0.215, 0.366)
nIn: 1 nOut: 2
raw GEN: GenVertex #18 key=-18
raw SIM: SimVertex #92 key=13

pi+ (211)
Particle 42
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (11.659, -9.097, -32.057, 35.304)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 325 simE=0.180746
direct recHits: 0 missing=325 recoE=0.000000
subgraph simHits: 325 simE=0.180746
subgraph recHits: 0 missing=325 recoE=0.000000
direct tracker simHits: 39 dE=39.804661
subgraph tracker simHits: 22 dE=39.804657
checkpointId: 0
x4@checkpoint: (93.540, -80.332, -267.011, 0.000)
p4@checkpoint: (10.693, -10.157, -32.018, 35.252)
raw GEN: GenParticle #62 key=44
raw SIM: SimTrack #859 key=25

HadronInelastic

Vertex 202
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (173.080, -152.729, -499.502, 18.713)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #409 key=330

Compton
Vertex 206
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (186.810, -266.434, -708.553, 26.700)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #413 key=334

pi+ (211)
Particle 43
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (17.441, -13.636, -47.682, 52.571)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 341 simE=0.400252
direct recHits: 0 missing=341 recoE=0.000000
subgraph simHits: 341 simE=0.400252
subgraph recHits: 0 missing=341 recoE=0.000000
direct tracker simHits: 43 dE=42.241753
subgraph tracker simHits: 43 dE=42.241753
checkpointId: 0
x4@checkpoint: (94.768, -78.879, -265.106, 0.000)
p4@checkpoint: (16.550, -14.652, -47.637, 52.515)
raw GEN: GenParticle #63 key=45
raw SIM: SimTrack #860 key=26

pi- (-211)
Particle 44
stableLegsFromInitialState
stableDecayProducts
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (31.822, -23.794, -85.776, 94.532)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 18
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1207 simE=11.435136
subgraph recHits: 0 missing=1207 recoE=0.000000
direct tracker simHits: 57 dE=54.507980
subgraph tracker simHits: 224 dE=655.324463
raw GEN: GenParticle #64 key=46
raw SIM: SimTrack #861 key=27

HadronInelastic

Vertex 123
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.221, -1.664, -5.440, 0.558)
nIn: 1 nOut: 17
raw GEN: n/a
raw SIM: SimVertex #287 key=208

Bremsstrahlung
Vertex 124
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-98.173, -60.833, 53.293, 4.912)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #288 key=209

gamma (22)
Particle 45
stableLegsFromInitialState
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (2.439, -2.494, 24.055, 24.307)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 328 simE=0.092106
subgraph recHits: 0 missing=328 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (29.718, -30.372, 293.500, 0.000)
p4@checkpoint: (2.439, -2.494, 24.055, 24.307)
raw GEN: GenParticle #65 key=47
raw SIM: SimTrack #862 key=28

PairConversion

Vertex 122
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (32.031, -32.737, 316.313, 10.981)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #286 key=207

gamma (22)
Particle 46
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (2.976, -2.988, 28.099, 28.413)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 424 simE=0.112851
subgraph recHits: 0 missing=424 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 6 dE=12.462387
raw GEN: GenParticle #66 key=48
raw SIM: SimTrack #863 key=29

PairConversion

Vertex 120
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (21.484, -21.565, 203.344, 7.178)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #276 key=197

gamma (22)
Particle 47
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.071, 0.186, -1.643, 1.655)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 41 simE=0.008685
subgraph recHits: 0 missing=41 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 7 dE=9.000000
raw GEN: GenParticle #67 key=49
raw SIM: SimTrack #864 key=30

PairConversion

Vertex 119
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (7.512, 19.780, -174.388, 6.216)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #273 key=194

gamma (22)
Particle 48
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.129, 0.150, -0.911, 0.932)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 8 simE=1.000619
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 25 dE=50.063396
raw GEN: GenParticle #68 key=50
raw SIM: SimTrack #865 key=31

PairConversion

Vertex 116
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (4.717, 5.447, -32.676, 1.472)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #261 key=182

gamma (22)
Particle 49
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.402, 0.246, -5.571, 5.591)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 6 simE=0.001229
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 29 dE=66.050575
raw GEN: GenParticle #69 key=51
raw SIM: SimTrack #866 key=32

PairConversion

Vertex 111
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (3.931, 2.408, -53.848, 2.159)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #243 key=164

gamma (22)
Particle 50
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.066, 0.071, -0.649, 0.656)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 12 simE=0.000971
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 28 dE=48.237175
raw GEN: GenParticle #70 key=52
raw SIM: SimTrack #867 key=33

PairConversion

Vertex 106
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (7.006, 7.480, -68.066, 2.652)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #228 key=149

gamma (22)
Particle 51
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-13.154, 10.173, 34.908, 38.666)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 160 simE=0.142198
subgraph recHits: 0 missing=160 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #71 key=53
raw SIM: SimTrack #868 key=34

PairConversion

Vertex 76
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.305, 70.607, 242.864, 9.291)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #179 key=100

gamma (22)
Particle 52
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.662, 0.549, 1.825, 2.018)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 49 simE=0.017069
subgraph recHits: 0 missing=49 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #72 key=54
raw SIM: SimTrack #869 key=35

PairConversion

Vertex 58
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-87.236, 72.367, 241.107, 9.208)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #142 key=63

gamma (22)
Particle 53
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-3.039, 2.511, -5.584, 6.835)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 76 simE=6.371028
subgraph recHits: 0 missing=76 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 2 dE=5.277207
raw GEN: GenParticle #73 key=55
raw SIM: SimTrack #1416 key=713230

PairConversion

Vertex 266
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.084, 73.602, -162.956, 7.016)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #514 key=435

gamma (22)
Particle 54
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.601, 0.463, -0.985, 1.243)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 31 simE=1.156956
subgraph recHits: 0 missing=31 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #74 key=56
raw SIM: SimTrack #1415 key=713229

PairConversion

Vertex 303
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.865, 70.920, -150.418, 6.690)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #563 key=484

Annihilation

Vertex 304
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.084, 70.885, -154.007, 6.854)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #565 key=486

gamma (22)
Particle 55
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-17.092, 14.146, -30.424, 37.654)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 332 simE=35.333977
subgraph recHits: 0 missing=332 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 101 dE=327.058441
raw GEN: GenParticle #75 key=57
raw SIM: SimTrack #1478 key=715870

PairConversion

Vertex 308
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-8.291, 6.816, -14.164, 0.945)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #570 key=491

gamma (22)
Particle 56
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-15.683, 12.827, -27.694, 34.314)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 180 simE=32.451336
subgraph recHits: 0 missing=180 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 12 dE=24.130983
raw GEN: GenParticle #76 key=58
raw SIM: SimTrack #1477 key=715869

PairConversion

Vertex 474
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-28.666, 23.436, -50.074, 2.430)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #769 key=690

gamma (22)
Particle 57
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (17.356, -13.554, -47.920, 52.737)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 656 simE=0.238252
subgraph recHits: 0 missing=656 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 37 dE=80.217590
raw GEN: GenParticle #77 key=59
raw SIM: SimTrack #870 key=36

PairConversion

Vertex 27
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.214, -1.723, -5.548, 0.562)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #107 key=28

gamma (22)
Particle 58
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (4.494, -3.597, -12.522, 13.781)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 200 simE=0.068689
subgraph recHits: 0 missing=200 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 14 dE=22.938061
raw GEN: GenParticle #78 key=60
raw SIM: SimTrack #871 key=37

PairConversion

Vertex 17
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.119, -33.702, -116.800, 4.646)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #93 key=14

e- (11)
Particle 59
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (2.378, -1.903, -6.626, 7.293)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 68 simE=0.015221
direct recHits: 0 missing=68 recoE=0.000000
subgraph simHits: 81 simE=0.018806
subgraph recHits: 0 missing=81 recoE=0.000000
direct tracker simHits: 14 dE=13.395922
subgraph tracker simHits: 10 dE=13.395923
checkpointId: 0
x4@checkpoint: (102.142, -69.065, -269.118, 0.000)
p4@checkpoint: (2.461, -1.009, -5.786, 6.368)
raw GEN: n/a
raw SIM: SimTrack #872 key=38

Bremsstrahlung
Vertex 24
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.783, -35.023, -121.425, 4.816)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #102 key=23

Bremsstrahlung
Vertex 25
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.786, -35.024, -121.432, 4.816)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #103 key=24

Bremsstrahlung
Vertex 26
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (48.402, -38.570, -134.112, 5.281)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #105 key=26

e+ (-11)
Particle 60
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (2.116, -1.694, -5.895, 6.489)
nProdVtx: 1 nDecayVtx: 6
nParents: 1 nChildren: 6
nCheckpoints: 1
direct simHits: 44 simE=0.012212
direct recHits: 0 missing=44 recoE=0.000000
subgraph simHits: 121 simE=0.049883
subgraph recHits: 0 missing=121 recoE=0.000000
direct tracker simHits: 11 dE=9.542139
subgraph tracker simHits: 10 dE=9.542139
checkpointId: 0
x4@checkpoint: (80.587, -93.320, -274.435, 0.000)
p4@checkpoint: (0.144, -0.705, -1.575, 1.732)
raw GEN: n/a
raw SIM: SimTrack #873 key=39

Bremsstrahlung
Vertex 18
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.851, -35.101, -121.639, 4.824)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #94 key=15

Bremsstrahlung
Vertex 19
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.905, -35.145, -121.791, 4.829)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #95 key=16

Bremsstrahlung
Vertex 20
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (55.783, -46.361, -157.287, 6.133)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #96 key=17

Bremsstrahlung
Vertex 21
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (77.561, -82.659, -250.127, 9.540)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #97 key=18

Bremsstrahlung
Vertex 22
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.529, -88.902, -264.503, 10.067)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #100 key=21

Bremsstrahlung
Vertex 23
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.544, -88.957, -264.627, 10.072)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #101 key=22

gamma (22)
Particle 61
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.052, -0.042, -0.145, 0.160)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.002595
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.002595
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.900, -77.499, -267.341, 0.000)
p4@checkpoint: (0.052, -0.042, -0.145, 0.160)
raw GEN: n/a
raw SIM: SimTrack #874 key=40

gamma (22)
Particle 62
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.111, -0.908, -3.115, 3.430)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 51 simE=0.027872
direct recHits: 0 missing=51 recoE=0.000000
subgraph simHits: 51 simE=0.027872
subgraph recHits: 0 missing=51 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.845, -77.567, -267.397, 0.000)
p4@checkpoint: (1.111, -0.908, -3.115, 3.430)
raw GEN: n/a
raw SIM: SimTrack #875 key=41

gamma (22)
Particle 63
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.028, -0.031, -0.090, 0.100)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000874
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000874
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (90.213, -84.050, -268.196, 0.000)
p4@checkpoint: (0.028, -0.031, -0.090, 0.100)
raw GEN: n/a
raw SIM: SimTrack #876 key=42

gamma (22)
Particle 64
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.073, -0.210, -0.489, 0.537)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.003950
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=0.003950
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (81.124, -92.854, -273.891, 0.000)
p4@checkpoint: (0.073, -0.210, -0.489, 0.537)
raw GEN: n/a
raw SIM: SimTrack #877 key=44

gamma (22)
Particle 65
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.022, -0.080, -0.183, 0.201)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.002091
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.002091
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (80.712, -93.212, -274.304, 0.000)
p4@checkpoint: (0.022, -0.080, -0.183, 0.201)
raw GEN: n/a
raw SIM: SimTrack #880 key=47

gamma (22)
Particle 66
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.033, -0.121, -0.274, 0.301)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000289
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000289
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (80.709, -93.214, -274.264, 0.000)
p4@checkpoint: (0.033, -0.121, -0.274, 0.301)
raw GEN: n/a
raw SIM: SimTrack #881 key=48

gamma (22)
Particle 67
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.028, -0.022, -0.077, 0.085)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000349
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000349
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.620, -76.600, -267.806, 0.000)
p4@checkpoint: (0.028, -0.022, -0.077, 0.085)
raw GEN: n/a
raw SIM: SimTrack #882 key=8761

gamma (22)
Particle 68
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.065, -0.051, -0.179, 0.197)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.000712
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.000712
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.632, -76.584, -267.774, 0.000)
p4@checkpoint: (0.065, -0.051, -0.179, 0.197)
raw GEN: n/a
raw SIM: SimTrack #883 key=8762

gamma (22)
Particle 69
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.199, -0.149, -0.542, 0.596)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.002523
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.002523
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.525, -75.443, -268.133, 0.000)
p4@checkpoint: (0.199, -0.149, -0.542, 0.596)
raw GEN: n/a
raw SIM: SimTrack #885 key=8765

e- (11)
Particle 70
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (5.858, -4.575, -16.174, 17.800)
nProdVtx: 1 nDecayVtx: 6
nParents: 1 nChildren: 6
nCheckpoints: 1
direct simHits: 30 simE=0.003528
direct recHits: 0 missing=30 recoE=0.000000
subgraph simHits: 225 simE=0.063278
subgraph recHits: 0 missing=225 recoE=0.000000
direct tracker simHits: 43 dE=42.039921
subgraph tracker simHits: 26 dE=42.039925
checkpointId: 0
x4@checkpoint: (70.408, 28.393, -293.500, 0.000)
p4@checkpoint: (-0.256, 0.205, -0.758, 0.826)
raw GEN: n/a
raw SIM: SimTrack #887 key=18004

Bremsstrahlung
Vertex 52
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (4.661, -3.624, -12.296, 0.809)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #135 key=56

Bremsstrahlung
Vertex 53
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (9.535, -7.313, -25.629, 1.299)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #136 key=57

Bremsstrahlung
Vertex 54
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (9.914, -7.593, -26.656, 1.336)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #137 key=58

Bremsstrahlung
Vertex 55
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (20.252, -14.850, -54.181, 2.347)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #138 key=59

Bremsstrahlung
Vertex 56
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.713, -24.825, -110.039, 4.397)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #139 key=60

Bremsstrahlung
Vertex 57
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (52.874, -25.533, -130.327, 5.140)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #140 key=61

e+ (-11)
Particle 71
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (11.498, -8.978, -31.746, 34.937)
nProdVtx: 1 nDecayVtx: 17
nParents: 1 nChildren: 17
nCheckpoints: 1
direct simHits: 101 simE=0.035147
direct recHits: 0 missing=101 recoE=0.000000
subgraph simHits: 468 simE=0.174973
subgraph recHits: 0 missing=468 recoE=0.000000
direct tracker simHits: 40 dE=38.177647
subgraph tracker simHits: 20 dE=38.177650
checkpointId: 0
x4@checkpoint: (89.649, -84.652, -268.733, 0.000)
p4@checkpoint: (1.752, -2.368, -6.415, 7.059)
raw GEN: n/a
raw SIM: SimTrack #888 key=18005

Bremsstrahlung
Vertex 28
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (4.640, -3.623, -12.258, 0.808)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #109 key=30

Bremsstrahlung
Vertex 29
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (4.646, -3.628, -12.276, 0.809)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #110 key=31

Bremsstrahlung
Vertex 30
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (4.660, -3.638, -12.313, 0.810)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #111 key=32

Bremsstrahlung
Vertex 31
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (9.462, -7.428, -25.635, 1.299)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #112 key=33

Bremsstrahlung
Vertex 32
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (22.247, -17.728, -61.380, 2.611)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #113 key=34

Bremsstrahlung
Vertex 33
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (22.276, -17.751, -61.461, 2.614)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #114 key=35

Bremsstrahlung
Vertex 34
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.366, -35.476, -121.419, 4.815)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #115 key=36

Bremsstrahlung
Vertex 35
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.374, -35.482, -121.442, 4.816)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #116 key=37

Bremsstrahlung
Vertex 36
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.388, -35.494, -121.482, 4.817)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #117 key=38

Bremsstrahlung
Vertex 37
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.017, -38.672, -131.995, 5.203)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #118 key=39

Bremsstrahlung
Vertex 38
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (55.586, -46.508, -157.279, 6.131)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #119 key=40

Bremsstrahlung
Vertex 39
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (55.610, -46.532, -157.353, 6.134)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #120 key=41

Bremsstrahlung
Vertex 40
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (63.999, -54.740, -182.923, 7.072)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #121 key=42

Bremsstrahlung
Vertex 41
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (64.518, -55.267, -184.534, 7.132)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #122 key=43

Bremsstrahlung
Vertex 42
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (75.381, -66.974, -219.251, 8.406)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #123 key=44

Bremsstrahlung
Vertex 43
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.455, -76.586, -246.565, 9.409)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #124 key=45

Bremsstrahlung
Vertex 44
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.456, -76.587, -246.567, 9.409)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #125 key=46

gamma (22)
Particle 72
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.131, -0.887, -3.131, 3.446)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 46 simE=0.012431
direct recHits: 0 missing=46 recoE=0.000000
subgraph simHits: 46 simE=0.012431
subgraph recHits: 0 missing=46 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.014, -76.099, -267.974, 0.000)
p4@checkpoint: (1.131, -0.887, -3.131, 3.446)
raw GEN: n/a
raw SIM: SimTrack #890 key=18007

gamma (22)
Particle 73
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.094, -0.074, -0.261, 0.288)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.001640
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.001640
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.012, -76.103, -267.983, 0.000)
p4@checkpoint: (0.094, -0.074, -0.261, 0.288)
raw GEN: n/a
raw SIM: SimTrack #891 key=18008

gamma (22)
Particle 74
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, -0.002, -0.007, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.006113
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.006113
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.743, -77.692, -219.005, 0.000)
p4@checkpoint: (0.003, -0.002, -0.005, 0.006)
raw GEN: n/a
raw SIM: SimTrack #892 key=18010

gamma (22)
Particle 75
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.024, -0.019, -0.068, 0.075)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000388
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000388
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.628, -76.590, -267.947, 0.000)
p4@checkpoint: (0.024, -0.019, -0.068, 0.075)
raw GEN: n/a
raw SIM: SimTrack #893 key=18011

gamma (22)
Particle 76
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.271, -0.222, -0.762, 0.839)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 12 simE=0.002581
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 12 simE=0.002581
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.689, -77.759, -267.815, 0.000)
p4@checkpoint: (0.271, -0.222, -0.762, 0.839)
raw GEN: n/a
raw SIM: SimTrack #894 key=18013

gamma (22)
Particle 77
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.817, -0.668, -2.298, 2.529)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 46 simE=0.016740
direct recHits: 0 missing=46 recoE=0.000000
subgraph simHits: 46 simE=0.016740
subgraph recHits: 0 missing=46 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.676, -77.776, -267.812, 0.000)
p4@checkpoint: (0.817, -0.668, -2.298, 2.529)
raw GEN: n/a
raw SIM: SimTrack #895 key=18014

gamma (22)
Particle 78
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.369, -1.182, -3.938, 4.333)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 65 simE=0.018137
direct recHits: 0 missing=65 recoE=0.000000
subgraph simHits: 65 simE=0.018137
subgraph recHits: 0 missing=65 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.287, -79.453, -267.896, 0.000)
p4@checkpoint: (1.369, -1.182, -3.938, 4.333)
raw GEN: n/a
raw SIM: SimTrack #896 key=18016

gamma (22)
Particle 79
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.018, -0.016, -0.052, 0.057)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000249
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000249
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.293, -79.446, -267.846, 0.000)
p4@checkpoint: (0.018, -0.016, -0.052, 0.057)
raw GEN: n/a
raw SIM: SimTrack #897 key=18017

gamma (22)
Particle 80
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (4.044, -3.494, -11.630, 12.799)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 201 simE=0.062297
direct recHits: 0 missing=201 recoE=0.000000
subgraph simHits: 201 simE=0.062297
subgraph recHits: 0 missing=201 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.281, -79.460, -267.830, 0.000)
p4@checkpoint: (4.044, -3.494, -11.630, 12.799)
raw GEN: n/a
raw SIM: SimTrack #898 key=18018

gamma (22)
Particle 81
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.023, -0.021, -0.068, 0.075)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000155
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000155
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.717, -80.125, -268.002, 0.000)
p4@checkpoint: (0.023, -0.021, -0.068, 0.075)
raw GEN: n/a
raw SIM: SimTrack #899 key=18019

gamma (22)
Particle 82
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000456
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000456
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (92.605, -81.408, -268.019, 0.000)
p4@checkpoint: (0.000, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #900 key=18022

gamma (22)
Particle 83
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.522, -0.492, -1.563, 1.720)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 35 simE=0.008639
direct recHits: 0 missing=35 recoE=0.000000
subgraph simHits: 35 simE=0.008639
subgraph recHits: 0 missing=35 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (92.605, -81.408, -268.052, 0.000)
p4@checkpoint: (0.522, -0.492, -1.563, 1.720)
raw GEN: n/a
raw SIM: SimTrack #901 key=18023

gamma (22)
Particle 84
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.265, -0.268, -0.821, 0.903)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.003423
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 18 simE=0.003423
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.503, -82.644, -268.260, 0.000)
p4@checkpoint: (0.265, -0.268, -0.821, 0.903)
raw GEN: n/a
raw SIM: SimTrack #902 key=18024

gamma (22)
Particle 85
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.013, -0.013, -0.041, 0.045)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000583
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000583
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.417, -82.739, -268.185, 0.000)
p4@checkpoint: (0.013, -0.013, -0.041, 0.045)
raw GEN: n/a
raw SIM: SimTrack #903 key=18025

gamma (22)
Particle 86
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.066, -0.075, -0.218, 0.240)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.002872
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.002872
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (90.304, -83.952, -268.351, 0.000)
p4@checkpoint: (0.066, -0.075, -0.218, 0.240)
raw GEN: n/a
raw SIM: SimTrack #904 key=18027

gamma (22)
Particle 87
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.004, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000109
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000109
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.786, -84.506, -268.631, 0.000)
p4@checkpoint: (0.001, -0.001, -0.004, 0.004)
raw GEN: n/a
raw SIM: SimTrack #905 key=18030

gamma (22)
Particle 88
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.121, -0.151, -0.421, 0.463)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 14 simE=0.003011
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #906 key=18031

PairConversion

Vertex 45
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.513, -76.659, -246.766, 9.416)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #126 key=47

e- (11)
Particle 89
pid: e- (11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.061, -0.076, -0.212, 0.234)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 6 simE=0.001208
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 7 simE=0.001545
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.710, -80.133, -270.822, 0.000)
p4@checkpoint: (0.090, 0.027, -0.196, 0.217)
raw GEN: n/a
raw SIM: SimTrack #907 key=28025

Bremsstrahlung
Vertex 51
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (86.710, -79.276, -255.895, 9.751)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #134 key=55

e+ (-11)
Particle 90
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.060, -0.075, -0.209, 0.230)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 1 simE=0.000120
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 7 simE=0.001466
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (81.149, -77.703, -293.500, 0.000)
p4@checkpoint: (0.030, -0.010, -0.071, 0.078)
raw GEN: n/a
raw SIM: SimTrack #908 key=28026

Ionisation

Vertex 46
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.518, -76.665, -246.785, 9.417)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #127 key=48

Ionisation

Vertex 47
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (78.070, -82.050, -276.340, 10.485)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #128 key=49

Bremsstrahlung
Vertex 48
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (78.065, -82.044, -276.359, 10.486)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #129 key=50

e- (11)
Particle 91
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.026, -0.042, -0.097, 0.109)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 3 simE=0.000595
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 4 simE=0.001144
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (83.901, -76.845, -293.500, 0.000)
p4@checkpoint: (0.014, -0.041, -0.083, 0.093)
raw GEN: n/a
raw SIM: SimTrack #909 key=28027

Bremsstrahlung
Vertex 50
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.363, -77.712, -249.379, 9.515)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #132 key=53

e- (11)
Particle 92
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.003, 0.003, -0.026, 0.026)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000113
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (78.012, -82.130, -293.500, 0.000)
p4@checkpoint: (-0.001, -0.001, -0.006, 0.006)
raw GEN: n/a
raw SIM: SimTrack #910 key=28031

Bremsstrahlung
Vertex 49
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (78.028, -82.036, -276.563, 10.493)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #130 key=51

gamma (22)
Particle 93
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000088
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000088
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (73.501, -76.708, -293.500, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #911 key=28032

gamma (22)
Particle 94
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.002, -0.014, 0.014)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000113
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000113
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (75.480, -79.931, -293.500, 0.000)
p4@checkpoint: (-0.002, 0.002, -0.014, 0.014)
raw GEN: n/a
raw SIM: SimTrack #912 key=28132

gamma (22)
Particle 95
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, -0.003, -0.009, 0.010)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000549
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000549
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.819, -82.293, -255.307, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #914 key=28176

gamma (22)
Particle 96
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000337
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000337
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (92.127, -81.948, -269.362, 0.000)
p4@checkpoint: (0.001, -0.000, -0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #916 key=28294

gamma (22)
Particle 97
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (3.388, -2.619, -9.327, 10.263)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 132 simE=0.040413
direct recHits: 0 missing=132 recoE=0.000000
subgraph simHits: 132 simE=0.040413
subgraph recHits: 0 missing=132 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.532, -75.435, -267.995, 0.000)
p4@checkpoint: (3.388, -2.619, -9.327, 10.263)
raw GEN: n/a
raw SIM: SimTrack #917 key=63827

gamma (22)
Particle 98
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.017, -0.012, -0.045, 0.049)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000140
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000140
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (98.955, -73.558, -268.597, 0.000)
p4@checkpoint: (0.017, -0.012, -0.045, 0.049)
raw GEN: n/a
raw SIM: SimTrack #918 key=63828

gamma (22)
Particle 99
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000487
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000487
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (99.066, -73.409, -268.342, 0.000)
p4@checkpoint: (0.001, -0.001, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #919 key=63829

gamma (22)
Particle 100
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.944, -1.297, -5.089, 5.600)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 52 simE=0.014403
direct recHits: 0 missing=52 recoE=0.000000
subgraph simHits: 52 simE=0.014403
subgraph recHits: 0 missing=52 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (101.948, -69.350, -268.085, 0.000)
p4@checkpoint: (1.944, -1.297, -5.089, 5.600)
raw GEN: n/a
raw SIM: SimTrack #920 key=63832

gamma (22)
Particle 101
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.391, -0.084, -0.872, 0.959)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.003817
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.003817
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (116.446, -40.537, -272.346, 0.000)
p4@checkpoint: (0.391, -0.084, -0.872, 0.959)
raw GEN: n/a
raw SIM: SimTrack #921 key=63834

gamma (22)
Particle 102
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.028, 0.002, -0.062, 0.068)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000490
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.000490
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.459, -21.229, -281.704, 0.000)
p4@checkpoint: (0.028, 0.002, -0.062, 0.068)
raw GEN: n/a
raw SIM: SimTrack #922 key=63836

e- (11)
Particle 103
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.374, 0.310, 1.031, 1.140)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 18 simE=0.004535
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 25 simE=0.008651
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.634, 77.827, 262.033, 0.000)
p4@checkpoint: (-0.291, 0.127, 0.663, 0.735)
raw GEN: n/a
raw SIM: SimTrack #924 key=87785

Bremsstrahlung
Vertex 67
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.321, 73.238, 244.065, 9.317)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #158 key=79

Bremsstrahlung
Vertex 68
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.799, 73.611, 245.345, 9.364)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #159 key=80

Bremsstrahlung
Vertex 69
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.932, 73.714, 245.699, 9.377)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #160 key=81

Bremsstrahlung
Vertex 70
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.995, 73.764, 245.864, 9.383)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #162 key=83

PairConversion

Vertex 71
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.614, 74.247, 247.475, 9.443)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #170 key=91

e+ (-11)
Particle 104
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.289, 0.240, 0.794, 0.878)
nProdVtx: 1 nDecayVtx: 8
nParents: 1 nChildren: 8
nCheckpoints: 1
direct simHits: 6 simE=0.001742
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 25 simE=0.008418
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-79.780, 86.180, 293.500, 0.000)
p4@checkpoint: (0.068, -0.045, 0.181, 0.198)
raw GEN: n/a
raw SIM: SimTrack #925 key=87786

Bremsstrahlung
Vertex 59
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.924, 73.870, 245.824, 9.382)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #144 key=65

Bremsstrahlung
Vertex 60
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.933, 73.879, 245.850, 9.383)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #145 key=66

Bremsstrahlung
Vertex 61
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.334, 74.284, 247.056, 9.428)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #149 key=70

Bremsstrahlung
Vertex 62
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.466, 74.424, 247.467, 9.443)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #151 key=72

Bremsstrahlung
Vertex 63
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.470, 74.429, 247.482, 9.444)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #152 key=73

Bremsstrahlung
Vertex 64
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.637, 74.612, 248.013, 9.463)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #153 key=74

Bremsstrahlung
Vertex 65
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.458, 74.414, 247.445, 9.442)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #154 key=75

Bremsstrahlung
Vertex 66
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.304, 73.239, 243.937, 9.313)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #157 key=78

gamma (22)
Particle 105
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.003, 0.009, 0.010)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001202
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001202
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.920, 77.474, 249.248, 0.000)
p4@checkpoint: (-0.001, 0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #927 key=87793

gamma (22)
Particle 106
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001745
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001745
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.564, 79.123, 261.848, 0.000)
p4@checkpoint: (-0.001, 0.001, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #928 key=87794

gamma (22)
Particle 107
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.062, 0.066, 0.194, 0.214)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.002273
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.002273
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.264, 79.481, 262.395, 0.000)
p4@checkpoint: (-0.062, 0.066, 0.194, 0.214)
raw GEN: n/a
raw SIM: SimTrack #932 key=87803

gamma (22)
Particle 108
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.115, 0.124, 0.362, 0.399)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000241
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.000241
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.207, 79.548, 262.438, 0.000)
p4@checkpoint: (-0.115, 0.124, 0.362, 0.399)
raw GEN: n/a
raw SIM: SimTrack #934 key=87806

gamma (22)
Particle 109
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000212
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000212
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.301, 79.437, 262.654, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #935 key=87807

gamma (22)
Particle 110
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000125
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000125
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.061, 79.721, 262.528, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #936 key=87808

gamma (22)
Particle 111
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000416
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000416
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-101.784, 69.591, 247.663, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #937 key=88505

gamma (22)
Particle 112
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000461
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000461
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.160, 75.914, 254.996, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #940 key=88763

gamma (22)
Particle 113
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.003, 0.012, 0.013)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000516
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000516
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.054, 78.534, 262.152, 0.000)
p4@checkpoint: (-0.004, 0.003, 0.012, 0.013)
raw GEN: n/a
raw SIM: SimTrack #941 key=88768

gamma (22)
Particle 114
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.020, 0.015, 0.052, 0.058)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 1 simE=0.000003
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 3 simE=0.000503
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-81.945, 92.130, 282.744, 0.000)
p4@checkpoint: (0.001, 0.001, 0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #942 key=88773

Compton
Vertex 73
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.902, 73.691, 245.620, 9.374)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #174 key=95

gamma (22)
Particle 115
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000588
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000588
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-82.890, 91.281, 257.851, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #943 key=88776

gamma (22)
Particle 116
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.003, 0.009, 0.010)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000499
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000499
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.001, 78.599, 261.816, 0.000)
p4@checkpoint: (-0.003, 0.003, 0.009, 0.010)
raw GEN: n/a
raw SIM: SimTrack #945 key=88781

e+ (-11)
Particle 117
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.007, 0.006, 0.019, 0.021)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.002010
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.823, 74.924, 293.500, 0.000)
p4@checkpoint: (-0.004, -0.003, 0.008, 0.009)
raw GEN: n/a
raw SIM: SimTrack #953 key=89765

Bremsstrahlung
Vertex 72
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.671, 74.295, 247.628, 9.449)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #171 key=92

gamma (22)
Particle 118
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002010
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002010
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.362, 73.008, 260.803, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #954 key=89766

e- (11)
Particle 119
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.020, 0.014, 0.049, 0.055)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000500
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #958 key=90098

Bremsstrahlung
Vertex 74
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.095, 73.736, 245.996, 9.389)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #177 key=98

Bremsstrahlung
Vertex 75
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.204, 73.589, 246.258, 9.399)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #178 key=99

gamma (22)
Particle 120
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, -0.001, 0.012, 0.012)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000348
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000348
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.907, 70.857, 281.994, 0.000)
p4@checkpoint: (-0.004, -0.001, 0.012, 0.012)
raw GEN: n/a
raw SIM: SimTrack #961 key=90109

gamma (22)
Particle 121
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000152
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000152
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.240, 73.173, 265.480, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #962 key=90112

e- (11)
Particle 122
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.384, 0.297, 1.020, 1.129)
nProdVtx: 1 nDecayVtx: 8
nParents: 1 nChildren: 8
nCheckpoints: 1
direct simHits: 9 simE=0.001862
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 43 simE=0.028018
subgraph recHits: 0 missing=43 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.116, 56.592, 293.500, 0.000)
p4@checkpoint: (0.080, -0.034, 0.173, 0.194)
raw GEN: n/a
raw SIM: SimTrack #963 key=90183

Bremsstrahlung
Vertex 81
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.330, 70.627, 242.932, 9.293)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #187 key=108

Bremsstrahlung
Vertex 82
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.381, 70.665, 243.069, 9.298)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #188 key=109

Bremsstrahlung
Vertex 83
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.542, 70.787, 243.503, 9.314)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #191 key=112

Bremsstrahlung
Vertex 84
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.581, 70.814, 243.602, 9.318)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #192 key=113

Bremsstrahlung
Vertex 85
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.597, 70.825, 243.639, 9.319)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #193 key=114

Bremsstrahlung
Vertex 86
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.047, 71.093, 244.656, 9.357)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #194 key=115

Bremsstrahlung
Vertex 87
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.207, 71.174, 244.998, 9.370)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #195 key=116

Bremsstrahlung
Vertex 88
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.610, 71.352, 245.837, 9.402)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #196 key=117

e+ (-11)
Particle 123
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-12.770, 9.875, 33.888, 37.537)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 92 simE=0.095352
direct recHits: 0 missing=92 recoE=0.000000
subgraph simHits: 120 simE=0.114180
subgraph recHits: 0 missing=120 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.519, 75.451, 259.404, 0.000)
p4@checkpoint: (-9.324, 7.325, 24.893, 27.572)
raw GEN: n/a
raw SIM: SimTrack #964 key=90184

Bremsstrahlung
Vertex 77
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.344, 70.638, 242.968, 9.294)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #180 key=101

Bremsstrahlung
Vertex 78
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.505, 70.762, 243.394, 9.310)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #183 key=104

Bremsstrahlung
Vertex 79
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.016, 71.157, 244.749, 9.360)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #184 key=105

Bremsstrahlung
Vertex 80
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.021, 71.161, 244.762, 9.361)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #185 key=106

gamma (22)
Particle 124
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000377
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000377
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.537, 75.429, 259.397, 0.000)
p4@checkpoint: (-0.001, 0.001, 0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #965 key=90185

gamma (22)
Particle 125
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-3.160, 2.443, 8.377, 9.280)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 35 simE=0.016717
direct recHits: 0 missing=35 recoE=0.000000
subgraph simHits: 35 simE=0.016717
subgraph recHits: 0 missing=35 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.538, 75.427, 259.388, 0.000)
p4@checkpoint: (-3.160, 2.443, 8.377, 9.280)
raw GEN: n/a
raw SIM: SimTrack #968 key=90189

gamma (22)
Particle 126
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.098, 0.076, 0.261, 0.289)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000808
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000808
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.535, 75.431, 259.396, 0.000)
p4@checkpoint: (-0.098, 0.076, 0.261, 0.289)
raw GEN: n/a
raw SIM: SimTrack #969 key=90190

gamma (22)
Particle 127
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.066, 0.051, 0.174, 0.193)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000926
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.000926
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.534, 75.432, 259.399, 0.000)
p4@checkpoint: (-0.066, 0.051, 0.174, 0.193)
raw GEN: n/a
raw SIM: SimTrack #970 key=90191

gamma (22)
Particle 128
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.221, 0.173, 0.589, 0.652)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 26 simE=0.008330
subgraph recHits: 0 missing=26 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #972 key=136143

PairConversion

Vertex 89
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.387, 70.671, 243.083, 9.299)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #198 key=119

gamma (22)
Particle 129
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.066, 0.051, 0.181, 0.199)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000003
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000003
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.547, 75.415, 260.055, 0.000)
p4@checkpoint: (-0.066, 0.051, 0.181, 0.199)
raw GEN: n/a
raw SIM: SimTrack #973 key=136144

gamma (22)
Particle 130
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000282
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000282
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-93.980, 79.816, 246.007, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #976 key=136149

gamma (22)
Particle 131
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.002, 0.008, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001141
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001141
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.858, 75.012, 258.779, 0.000)
p4@checkpoint: (-0.003, 0.002, 0.008, 0.009)
raw GEN: n/a
raw SIM: SimTrack #977 key=136151

gamma (22)
Particle 132
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000095
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000095
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.784, 75.108, 258.658, 0.000)
p4@checkpoint: (-0.001, 0.001, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #978 key=136153

gamma (22)
Particle 133
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000277
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000277
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-98.252, 74.495, 257.959, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #979 key=136155

gamma (22)
Particle 134
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, 0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001935
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001935
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-98.481, 74.192, 258.324, 0.000)
p4@checkpoint: (-0.001, 0.000, 0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #980 key=136156

gamma (22)
Particle 135
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, 0.003, 0.014, 0.016)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.014093
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.014093
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-98.765, 73.813, 258.467, 0.000)
p4@checkpoint: (-0.007, 0.003, 0.014, 0.016)
raw GEN: n/a
raw SIM: SimTrack #981 key=136157

e- (11)
Particle 136
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.144, 0.112, 0.381, 0.422)
nProdVtx: 1 nDecayVtx: 6
nParents: 1 nChildren: 6
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 17 simE=0.004995
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.375, 71.260, 293.500, 0.000)
p4@checkpoint: (-0.009, 0.007, 0.007, 0.013)
raw GEN: n/a
raw SIM: SimTrack #983 key=136729

Bremsstrahlung
Vertex 94
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.430, 70.704, 243.197, 9.303)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #204 key=125

Bremsstrahlung
Vertex 95
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.462, 70.729, 243.282, 9.306)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #205 key=126

Bremsstrahlung
Vertex 96
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.481, 70.744, 243.336, 9.308)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #206 key=127

Bremsstrahlung
Vertex 98
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.498, 70.815, 243.445, 9.312)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #216 key=137

Annihilation

Vertex 99
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.260, 71.073, 245.086, 9.388)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #218 key=139

Bremsstrahlung
Vertex 100
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.811, 70.559, 246.181, 9.408)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #219 key=140

e+ (-11)
Particle 137
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.075, 0.061, 0.209, 0.230)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 9 simE=0.003335
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-91.127, 71.078, 293.500, 0.000)
p4@checkpoint: (-0.016, -0.004, 0.048, 0.051)
raw GEN: n/a
raw SIM: SimTrack #984 key=136730

Bremsstrahlung
Vertex 90
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.470, 70.743, 243.327, 9.308)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #199 key=120

Bremsstrahlung
Vertex 91
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.495, 70.767, 243.403, 9.310)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #200 key=121

Bremsstrahlung
Vertex 92
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.516, 70.787, 243.470, 9.313)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #201 key=122

Bremsstrahlung
Vertex 93
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.830, 71.410, 245.005, 9.369)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #202 key=123

gamma (22)
Particle 138
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000174
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000174
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.992, 76.128, 259.660, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #985 key=136733

gamma (22)
Particle 139
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000336
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000336
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.891, 70.880, 249.370, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #986 key=136734

gamma (22)
Particle 140
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.049, 0.045, 0.149, 0.163)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.002725
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.002725
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.139, 75.941, 260.605, 0.000)
p4@checkpoint: (-0.049, 0.045, 0.149, 0.163)
raw GEN: n/a
raw SIM: SimTrack #987 key=136735

gamma (22)
Particle 141
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000100
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000100
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.959, 77.426, 260.438, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #988 key=136739

gamma (22)
Particle 142
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.090, 0.069, 0.237, 0.263)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 12 simE=0.002803
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #990 key=136981

PairConversion

Vertex 101
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.530, 70.780, 243.459, 9.313)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #221 key=142

gamma (22)
Particle 143
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000061
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000061
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.762, 75.136, 257.223, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #991 key=136984

gamma (22)
Particle 144
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.014, 0.010, 0.049, 0.052)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #992 key=136985

Annihilation

Vertex 97
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.866, 70.818, 245.837, 9.402)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #210 key=131

gamma (22)
Particle 145
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.345, 71.651, 257.093, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #996 key=137010

gamma (22)
Particle 146
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000547
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000547
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.083, 78.499, 247.505, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1002 key=137061

gamma (22)
Particle 147
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000468
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000468
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-102.714, 68.210, 235.232, 0.000)
p4@checkpoint: (-0.001, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1004 key=137086

gamma (22)
Particle 148
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000606
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000606
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.566, 79.122, 245.961, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1006 key=137101

e- (11)
Particle 149
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.032, 0.024, 0.085, 0.094)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.000796
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1008 key=137130

Bremsstrahlung
Vertex 104
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.594, 70.832, 243.632, 9.319)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #225 key=146

Bremsstrahlung
Vertex 105
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.073, 69.849, 245.928, 9.414)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #227 key=148

e+ (-11)
Particle 150
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.059, 0.045, 0.151, 0.168)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 5 simE=0.000935
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 7 simE=0.002007
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.433, 78.073, 262.604, 0.000)
p4@checkpoint: (0.014, 0.062, 0.137, 0.151)
raw GEN: n/a
raw SIM: SimTrack #1009 key=137131

Bremsstrahlung
Vertex 102
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.583, 70.819, 243.591, 9.318)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #222 key=143

Bremsstrahlung
Vertex 103
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.275, 71.317, 245.405, 9.384)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #223 key=144

gamma (22)
Particle 151
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000332
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000332
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.658, 75.272, 258.242, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1010 key=137133

gamma (22)
Particle 152
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000740
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000740
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.965, 74.871, 259.753, 0.000)
p4@checkpoint: (-0.001, 0.001, 0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1011 key=137136

gamma (22)
Particle 153
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.019, 0.019, 0.058, 0.064)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000667
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000667
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.943, 76.191, 259.724, 0.000)
p4@checkpoint: (-0.019, 0.019, 0.058, 0.064)
raw GEN: n/a
raw SIM: SimTrack #1013 key=137414

gamma (22)
Particle 154
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000129
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000129
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.451, 21.276, 245.884, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1015 key=137422

e- (11)
Particle 155
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.047, 0.052, -0.460, 0.465)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 12 simE=0.000971
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 17 dE=15.723337
subgraph tracker simHits: 13 dE=23.675375
checkpointId: 0
x4@checkpoint: (4.798, 16.951, -293.500, 0.000)
p4@checkpoint: (-0.004, -0.009, -0.059, 0.060)
raw GEN: n/a
raw SIM: SimTrack #1016 key=137567

Bremsstrahlung
Vertex 107
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.380, 9.735, -85.539, 3.242)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #233 key=154

Bremsstrahlung
Vertex 108
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (7.421, 15.581, -125.820, 4.601)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #234 key=155

Bremsstrahlung
Vertex 109
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (6.460, 16.550, -135.062, 4.913)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #236 key=157

Bremsstrahlung
Vertex 110
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (6.105, 16.881, -174.402, 6.248)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #239 key=160

e+ (-11)
Particle 156
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.019, 0.020, -0.189, 0.191)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 25 dE=24.561802
subgraph tracker simHits: 25 dE=24.561802
checkpointId: 0
x4@checkpoint: (7.325, 6.589, -293.500, 0.000)
p4@checkpoint: (0.007, 0.006, -0.058, 0.058)
raw GEN: n/a
raw SIM: SimTrack #1017 key=137568

gamma (22)
Particle 157
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000161
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000161
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 8 dE=7.952036
subgraph tracker simHits: 8 dE=7.952036
checkpointId: 0
x4@checkpoint: (-42.987, 115.564, -252.849, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1022 key=137642

gamma (22)
Particle 158
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.002, -0.019, 0.019)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000103
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000103
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-8.091, 34.725, -293.500, 0.000)
p4@checkpoint: (-0.002, 0.002, -0.019, 0.019)
raw GEN: n/a
raw SIM: SimTrack #1023 key=137644

gamma (22)
Particle 159
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.035, 0.026, -0.284, 0.287)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.000658
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.000658
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-12.893, 31.317, -293.500, 0.000)
p4@checkpoint: (-0.035, 0.026, -0.284, 0.287)
raw GEN: n/a
raw SIM: SimTrack #1025 key=137647

gamma (22)
Particle 160
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000050
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000050
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-45.613, 114.553, -269.346, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1028 key=137653

e- (11)
Particle 161
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.134, 0.082, -1.857, 1.864)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 34 dE=34.000000
subgraph tracker simHits: 34 dE=34.000000
checkpointId: 0
x4@checkpoint: (7.055, 19.863, -293.500, 0.000)
p4@checkpoint: (-0.067, 0.072, -1.123, 1.127)
raw GEN: n/a
raw SIM: SimTrack #1032 key=138267

e+ (-11)
Particle 162
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.270, 0.163, -3.714, 3.727)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 6 simE=0.001229
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 23 dE=23.000000
subgraph tracker simHits: 20 dE=32.050579
checkpointId: 0
x4@checkpoint: (12.497, 5.268, -293.500, 0.000)
p4@checkpoint: (0.017, 0.001, -0.242, 0.243)
raw GEN: n/a
raw SIM: SimTrack #1033 key=138268

Bremsstrahlung
Vertex 112
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (4.981, 3.021, -68.030, 2.634)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #244 key=165

Bremsstrahlung
Vertex 113
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.012, 3.038, -68.439, 2.647)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #245 key=166

Bremsstrahlung
Vertex 114
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (10.689, 5.370, -140.032, 5.044)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #246 key=167

Bremsstrahlung
Vertex 115
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (13.528, 4.692, -175.880, 6.244)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #247 key=168

gamma (22)
Particle 163
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.005, 0.003, -0.065, 0.065)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000039
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000039
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (21.925, 12.417, -293.500, 0.000)
p4@checkpoint: (0.005, 0.003, -0.065, 0.065)
raw GEN: n/a
raw SIM: SimTrack #1034 key=138269

gamma (22)
Particle 164
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=1.050578
subgraph tracker simHits: 1 dE=1.050578
checkpointId: 0
x4@checkpoint: (100.586, -0.866, -293.500, 0.000)
p4@checkpoint: (0.000, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1035 key=138271

gamma (22)
Particle 165
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.271, 0.081, -3.278, 3.290)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.001191
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.001191
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (23.368, 9.166, -293.500, 0.000)
p4@checkpoint: (0.271, 0.081, -3.278, 3.290)
raw GEN: n/a
raw SIM: SimTrack #1036 key=138273

gamma (22)
Particle 166
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.006, -0.005, -0.092, 0.092)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 8 dE=8.000000
subgraph tracker simHits: 8 dE=8.000000
raw GEN: n/a
raw SIM: SimTrack #1037 key=138274

e- (11)
Particle 167
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.119, 0.138, -0.838, 0.858)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 7 simE=0.000619
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 17 dE=17.010075
subgraph tracker simHits: 12 dE=18.063400
checkpointId: 0
x4@checkpoint: (-16.071, 12.496, -293.500, 0.000)
p4@checkpoint: (0.046, -0.112, -0.591, 0.604)
raw GEN: n/a
raw SIM: SimTrack #1052 key=142736

Bremsstrahlung
Vertex 117
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.166, 19.561, -103.413, 3.885)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #265 key=186

Bremsstrahlung
Vertex 118
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.162, 19.576, -103.484, 3.887)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #266 key=187

e+ (-11)
Particle 168
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.010, 0.012, -0.073, 0.075)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 32 dE=32.000000
subgraph tracker simHits: 32 dE=32.000000
checkpointId: 0
x4@checkpoint: (6.506, 6.930, -293.500, 0.000)
p4@checkpoint: (0.031, 0.001, -0.046, 0.055)
raw GEN: n/a
raw SIM: SimTrack #1053 key=142737

gamma (22)
Particle 169
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=1.053327
subgraph tracker simHits: 1 dE=1.053327
checkpointId: 0
x4@checkpoint: (-18.751, 74.527, -293.500, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1057 key=142748

gamma (22)
Particle 170
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, 0.026, -0.122, 0.125)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.000619
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.000619
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-2.716, 59.285, -293.500, 0.000)
p4@checkpoint: (-0.007, 0.026, -0.122, 0.125)
raw GEN: n/a
raw SIM: SimTrack #1058 key=142749

e- (11)
Particle 171
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.063, 0.165, -1.465, 1.475)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 41 simE=0.008685
direct recHits: 0 missing=41 recoE=0.000000
subgraph simHits: 41 simE=0.008685
subgraph recHits: 0 missing=41 recoE=0.000000
direct tracker simHits: 4 dE=3.306748
subgraph tracker simHits: 4 dE=3.306748
checkpointId: 0
x4@checkpoint: (6.480, 33.030, -293.500, 0.000)
p4@checkpoint: (-0.081, 0.141, -1.447, 1.456)
raw GEN: n/a
raw SIM: SimTrack #1065 key=143044

e+ (-11)
Particle 172
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.007, 0.021, -0.179, 0.180)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 7 dE=5.693253
subgraph tracker simHits: 7 dE=5.693253
checkpointId: 0
x4@checkpoint: (9.239, 22.443, -293.500, 0.000)
p4@checkpoint: (0.028, -0.000, -0.169, 0.171)
raw GEN: n/a
raw SIM: SimTrack #1066 key=143045

e- (11)
Particle 173
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (2.921, -2.933, 27.580, 27.889)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 285 simE=0.066917
direct recHits: 0 missing=285 recoE=0.000000
subgraph simHits: 409 simE=0.111469
subgraph recHits: 0 missing=409 recoE=0.000000
direct tracker simHits: 4 dE=4.124860
subgraph tracker simHits: 4 dE=4.124860
checkpointId: 0
x4@checkpoint: (31.195, -30.984, 293.500, 0.000)
p4@checkpoint: (1.833, -1.701, 16.656, 16.843)
raw GEN: n/a
raw SIM: SimTrack #1069 key=145375

Bremsstrahlung
Vertex 121
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (28.408, -28.357, 267.980, 9.358)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #280 key=201

e+ (-11)
Particle 174
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.055, -0.055, 0.519, 0.525)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 15 simE=0.001382
direct recHits: 0 missing=15 recoE=0.000000
subgraph simHits: 15 simE=0.001382
subgraph recHits: 0 missing=15 recoE=0.000000
direct tracker simHits: 8 dE=8.337528
subgraph tracker simHits: 8 dE=8.337528
checkpointId: 0
x4@checkpoint: (20.702, -33.709, 293.500, 0.000)
p4@checkpoint: (-0.068, -0.044, 0.507, 0.514)
raw GEN: n/a
raw SIM: SimTrack #1070 key=145376

gamma (22)
Particle 175
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.182, -1.132, 10.896, 11.019)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 205 simE=0.044551
direct recHits: 0 missing=205 recoE=0.000000
subgraph simHits: 205 simE=0.044551
subgraph recHits: 0 missing=205 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (31.176, -31.008, 293.500, 0.000)
p4@checkpoint: (1.182, -1.132, 10.896, 11.019)
raw GEN: n/a
raw SIM: SimTrack #1074 key=146177

e- (11)
Particle 176
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.294, -1.323, 12.762, 12.895)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 213 simE=0.045647
direct recHits: 0 missing=213 recoE=0.000000
subgraph simHits: 213 simE=0.045647
subgraph recHits: 0 missing=213 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (32.125, -32.833, 317.238, 0.000)
p4@checkpoint: (1.293, -1.323, 12.760, 12.893)
raw GEN: n/a
raw SIM: SimTrack #1080 key=192657

e+ (-11)
Particle 177
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.145, -1.171, 11.293, 11.412)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 198 simE=0.046459
direct recHits: 0 missing=198 recoE=0.000000
subgraph simHits: 198 simE=0.046459
subgraph recHits: 0 missing=198 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (32.064, -32.771, 316.638, 0.000)
p4@checkpoint: (1.145, -1.171, 11.293, 11.411)
raw GEN: n/a
raw SIM: SimTrack #1081 key=192658

pi+ (211)
Particle 178
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (6.493, -4.084, -17.190, 18.824)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 242 simE=0.078731
direct recHits: 0 missing=242 recoE=0.000000
subgraph simHits: 243 simE=0.083124
subgraph recHits: 0 missing=243 recoE=0.000000
direct tracker simHits: 34 dE=33.873947
subgraph tracker simHits: 23 dE=33.873947
checkpointId: 0
x4@checkpoint: (97.994, -74.834, -277.246, 0.000)
p4@checkpoint: (5.589, -5.112, -17.153, 18.751)
raw GEN: n/a
raw SIM: SimTrack #1082 key=233564

Bremsstrahlung
Vertex 201
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.596, -11.539, -46.469, 2.057)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #407 key=328

pi- (-211)
Particle 179
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.764, -0.723, -2.398, 2.623)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 9
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 68 simE=0.538345
subgraph recHits: 0 missing=68 recoE=0.000000
direct tracker simHits: 79 dE=77.457520
subgraph tracker simHits: 81 dE=279.726532
raw GEN: n/a
raw SIM: SimTrack #1083 key=233565

HadronInelastic

Vertex 184
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.516, -4.631, -15.573, 0.927)
nIn: 1 nOut: 9
raw GEN: n/a
raw SIM: SimVertex #387 key=308

p (2212)
Particle 180
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.328, -0.734, -0.419, 1.305)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 14 simE=0.030434
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 59 dE=60.781868
subgraph tracker simHits: 16 dE=60.781868
raw GEN: n/a
raw SIM: SimTrack #1084 key=233566

HadronInelastic

Vertex 183
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-114.219, -22.001, -134.285, 13.695)
nIn: 1 nOut: 4
raw GEN: n/a
raw SIM: SimVertex #386 key=307

pi0 (111)
Particle 181
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.019, 0.000, -0.174, 0.221)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 12 simE=0.204615
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1085 key=233567

Decay
Vertex 182
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.221, -1.664, -5.440, 0.558)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #385 key=306

n (2112)
Particle 182
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.039, -0.110, 0.016, 0.947)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.006520
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.006520
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (42.891, -115.599, 11.412, 0.000)
p4@checkpoint: (0.039, -0.110, 0.016, 0.947)
raw GEN: n/a
raw SIM: SimTrack #1086 key=233569

pi0 (111)
Particle 183
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (5.078, -3.565, -12.825, 14.247)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 164 simE=8.493608
subgraph recHits: 0 missing=164 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 28 dE=53.547081
raw GEN: n/a
raw SIM: SimTrack #1087 key=233570

Decay
Vertex 142
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.221, -1.664, -5.440, 0.558)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #324 key=245

pi- (-211)
Particle 184
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (16.335, -12.285, -44.059, 48.569)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 492 simE=0.235661
direct recHits: 0 missing=492 recoE=0.000000
subgraph simHits: 492 simE=0.235661
subgraph recHits: 0 missing=492 recoE=0.000000
direct tracker simHits: 43 dE=42.897320
subgraph tracker simHits: 27 dE=42.897320
checkpointId: 0
x4@checkpoint: (100.863, -70.920, -265.274, 0.000)
p4@checkpoint: (17.088, -11.187, -43.993, 48.503)
raw GEN: n/a
raw SIM: SimTrack #1088 key=233571

HadronInelastic

Vertex 136
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (189.090, -141.277, -512.305, 19.215)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #317 key=238

PairConversion

Vertex 137
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (291.958, -159.631, -649.600, 24.962)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #318 key=239

PairConversion

Vertex 138
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (287.785, -158.627, -681.169, 25.872)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #319 key=240

HadronInelastic

Vertex 139
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (191.120, -121.312, -489.011, 18.364)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #320 key=241

HadronInelastic

Vertex 140
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (203.527, -130.037, -519.276, 19.493)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #321 key=242

pi0 (111)
Particle 185
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.720, -0.438, -1.717, 1.918)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 46 simE=0.515526
subgraph recHits: 0 missing=46 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 12 dE=19.784187
raw GEN: n/a
raw SIM: SimTrack #1089 key=233572

Decay
Vertex 129
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.221, -1.664, -5.440, 0.558)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #298 key=219

pi- (-211)
Particle 186
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.057, -0.638, -1.242, 1.405)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 33 simE=0.006248
direct recHits: 0 missing=33 recoE=0.000000
subgraph simHits: 33 simE=0.006248
subgraph recHits: 0 missing=33 recoE=0.000000
direct tracker simHits: 48 dE=48.526867
subgraph tracker simHits: 48 dE=48.526867
checkpointId: 0
x4@checkpoint: (109.074, -17.189, -293.500, 0.000)
p4@checkpoint: (0.232, 0.556, -1.224, 1.371)
raw GEN: n/a
raw SIM: SimTrack #1090 key=233573

pi0 (111)
Particle 187
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.835, -0.151, -2.592, 2.730)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 44 simE=0.016518
subgraph recHits: 0 missing=44 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 14 dE=19.798100
raw GEN: n/a
raw SIM: SimTrack #1091 key=233574

Decay
Vertex 127
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.221, -1.664, -5.440, 0.558)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #291 key=212

pi+ (211)
Particle 188
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.360, -0.839, -2.656, 3.102)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 59 simE=1.202021
direct recHits: 0 missing=59 recoE=0.000000
subgraph simHits: 61 simE=1.202679
subgraph recHits: 0 missing=61 recoE=0.000000
direct tracker simHits: 42 dE=41.880848
subgraph tracker simHits: 20 dE=41.880848
checkpointId: 0
x4@checkpoint: (68.040, -102.828, -213.599, 0.000)
p4@checkpoint: (0.231, -1.537, -2.605, 3.036)
raw GEN: n/a
raw SIM: SimTrack #1092 key=233575

Bremsstrahlung
Vertex 125
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (66.518, -94.690, -199.496, 8.115)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #289 key=210

Bremsstrahlung
Vertex 126
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (66.518, -94.695, -199.496, 8.116)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #290 key=211

n (2112)
Particle 189
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.016, -0.425, 0.059, 1.033)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 20 simE=0.086099
direct recHits: 0 missing=20 recoE=0.000000
subgraph simHits: 20 simE=0.086099
subgraph recHits: 0 missing=20 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (6.758, -123.115, 11.363, 0.000)
p4@checkpoint: (0.016, -0.425, 0.059, 1.033)
raw GEN: n/a
raw SIM: SimTrack #1093 key=233578

n (2112)
Particle 190
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.168, 0.040, 0.955)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.011955
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.011955
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (0.339, -123.300, 23.647, 0.000)
p4@checkpoint: (-0.003, -0.168, 0.040, 0.955)
raw GEN: n/a
raw SIM: SimTrack #1094 key=233579

n (2112)
Particle 191
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.116, 0.111, -0.032, 0.954)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.002992
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.002992
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.845, 85.495, -30.795, 0.000)
p4@checkpoint: (-0.116, 0.111, -0.032, 0.954)
raw GEN: n/a
raw SIM: SimTrack #1096 key=233581

n (2112)
Particle 192
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.065, 0.013, 0.104, 0.948)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000113
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000113
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.183, 5.375, 101.412, 0.000)
p4@checkpoint: (0.113, -0.004, 0.044, 0.947)
raw GEN: n/a
raw SIM: SimTrack #1097 key=233586

n (2112)
Particle 193
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.063, 0.037, -0.018, 0.943)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000055
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000055
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.619, 29.904, 54.430, 0.000)
p4@checkpoint: (-0.029, -0.017, 0.024, 0.940)
raw GEN: n/a
raw SIM: SimTrack #1098 key=233588

gamma (22)
Particle 194
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.003, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000397
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000397
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-9.085, 122.965, 91.579, 0.000)
p4@checkpoint: (-0.000, 0.003, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1099 key=233590

gamma (22)
Particle 195
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000243
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000243
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.054, -73.425, 52.975, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1100 key=233595

gamma (22)
Particle 196
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000079
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000079
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (66.668, -103.722, -198.325, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1101 key=234410

gamma (22)
Particle 197
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000580
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000580
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (66.077, -104.100, -198.123, 0.000)
p4@checkpoint: (-0.000, -0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1102 key=234411

gamma (22)
Particle 198
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.642, -0.079, -1.813, 1.925)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 41 simE=0.014296
direct recHits: 0 missing=41 recoE=0.000000
subgraph simHits: 41 simE=0.014296
subgraph recHits: 0 missing=41 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (104.156, -14.177, -293.500, 0.000)
p4@checkpoint: (0.642, -0.079, -1.813, 1.925)
raw GEN: n/a
raw SIM: SimTrack #1103 key=234414

gamma (22)
Particle 199
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.193, -0.072, -0.779, 0.805)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 3 simE=0.002222
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 14 dE=19.798100
raw GEN: n/a
raw SIM: SimTrack #1104 key=234415

PairConversion

Vertex 128
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (19.381, -8.058, -74.704, 2.948)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #292 key=213

e- (11)
Particle 200
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.181, -0.067, -0.733, 0.758)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.002222
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.002222
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 19 dE=19.798100
subgraph tracker simHits: 19 dE=19.798100
checkpointId: 0
x4@checkpoint: (26.094, 23.294, -293.500, 0.000)
p4@checkpoint: (-0.174, 0.001, -0.690, 0.712)
raw GEN: n/a
raw SIM: SimTrack #1105 key=234416

gamma (22)
Particle 201
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.213, -0.139, -0.388, 0.464)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.500319
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.500319
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.031, -67.732, -189.546, 0.000)
p4@checkpoint: (0.213, -0.139, -0.388, 0.464)
raw GEN: n/a
raw SIM: SimTrack #1113 key=239710

gamma (22)
Particle 202
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.507, -0.298, -1.329, 1.454)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 33 simE=0.015207
subgraph recHits: 0 missing=33 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 12 dE=19.784187
raw GEN: n/a
raw SIM: SimTrack #1114 key=239711

PairConversion

Vertex 130
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (14.269, -8.749, -37.015, 1.710)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #299 key=220

e- (11)
Particle 203
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.295, -0.174, -0.775, 0.847)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 26 simE=0.012874
subgraph recHits: 0 missing=26 recoE=0.000000
direct tracker simHits: 1 dE=0.901053
subgraph tracker simHits: 1 dE=0.901053
raw GEN: n/a
raw SIM: SimTrack #1115 key=239712

Bremsstrahlung
Vertex 132
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (14.333, -8.786, -37.184, 1.716)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #313 key=234

Bremsstrahlung
Vertex 133
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (14.483, -8.873, -37.579, 1.730)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #314 key=235

Bremsstrahlung
Vertex 134
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (14.639, -8.263, -44.435, 1.970)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #315 key=236

e+ (-11)
Particle 204
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.215, -0.125, -0.553, 0.607)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 1 simE=0.000266
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 7 simE=0.002334
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 19 dE=18.883131
subgraph tracker simHits: 11 dE=18.883133
checkpointId: 0
x4@checkpoint: (16.831, -28.020, -293.500, 0.000)
p4@checkpoint: (0.053, -0.045, -0.151, 0.166)
raw GEN: n/a
raw SIM: SimTrack #1116 key=239713

Bremsstrahlung
Vertex 131
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (22.860, -35.167, -105.823, 4.218)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #302 key=223

gamma (22)
Particle 205
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.047, -0.109, -0.268, 0.293)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.002068
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.002068
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-10.198, -111.252, -293.500, 0.000)
p4@checkpoint: (-0.047, -0.109, -0.268, 0.293)
raw GEN: n/a
raw SIM: SimTrack #1119 key=239717

gamma (22)
Particle 206
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000142
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000142
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-123.145, 6.190, -53.134, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1130 key=240562

gamma (22)
Particle 207
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.285, -0.168, -0.757, 0.826)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 24 simE=0.007598
subgraph recHits: 0 missing=24 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1131 key=240563

PairConversion

Vertex 135
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (97.486, -57.886, -257.825, 9.750)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #316 key=237

gamma (22)
Particle 208
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.005133
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.005133
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.199, -81.867, -214.191, 0.000)
p4@checkpoint: (-0.001, -0.001, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1132 key=240565

e- (11)
Particle 209
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.034, -0.020, -0.090, 0.098)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000127
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000127
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.494, -53.378, -293.500, 0.000)
p4@checkpoint: (-0.021, -0.027, -0.086, 0.093)
raw GEN: n/a
raw SIM: SimTrack #1133 key=240571

e+ (-11)
Particle 210
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.251, -0.149, -0.667, 0.728)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 23 simE=0.007472
direct recHits: 0 missing=23 recoE=0.000000
subgraph simHits: 23 simE=0.007472
subgraph recHits: 0 missing=23 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (105.044, -64.564, -280.735, 0.000)
p4@checkpoint: (0.180, -0.231, -0.662, 0.723)
raw GEN: n/a
raw SIM: SimTrack #1134 key=240572

pi+ (211)
Particle 211
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.063, -0.458, -1.376, 1.458)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1135 key=246452

e- (11)
Particle 212
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.014, -0.002, -0.019, 0.024)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1136 key=269602

e- (11)
Particle 213
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.005, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1137 key=269726

pi- (-211)
Particle 214
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.144, 0.099, -0.727, 0.761)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1138 key=272238

pi+ (211)
Particle 215
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.356, -0.339, -0.265, 0.576)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1139 key=277742

PairConversion

Vertex 141
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (141.583, -141.290, -606.206, 26.278)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #322 key=243

e+ (-11)
Particle 216
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.092, -0.013, 0.093)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1140 key=278326

gamma (22)
Particle 217
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (4.068, -2.869, -10.167, 11.320)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 153 simE=8.486023
subgraph recHits: 0 missing=153 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 28 dE=53.547081
raw GEN: n/a
raw SIM: SimTrack #1142 key=287377

PairConversion

Vertex 143
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (18.902, -13.427, -47.131, 2.106)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #325 key=246

gamma (22)
Particle 218
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.010, -0.696, -2.658, 2.927)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 11 simE=0.007585
direct recHits: 0 missing=11 recoE=0.000000
subgraph simHits: 11 simE=0.007585
subgraph recHits: 0 missing=11 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (101.446, -70.083, -266.675, 0.000)
p4@checkpoint: (1.010, -0.696, -2.658, 2.927)
raw GEN: n/a
raw SIM: SimTrack #1143 key=287378

e- (11)
Particle 219
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.464, -0.327, -1.158, 1.290)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 32 simE=0.004677
direct recHits: 0 missing=32 recoE=0.000000
subgraph simHits: 39 simE=0.006928
subgraph recHits: 0 missing=39 recoE=0.000000
direct tracker simHits: 37 dE=37.047714
subgraph tracker simHits: 18 dE=37.047710
checkpointId: 0
x4@checkpoint: (74.004, 39.210, -293.500, 0.000)
p4@checkpoint: (-0.280, 0.268, -0.853, 0.937)
raw GEN: n/a
raw SIM: SimTrack #1144 key=291237

Bremsstrahlung
Vertex 181
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (52.946, -22.091, -120.652, 4.836)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #383 key=304

e+ (-11)
Particle 220
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (3.604, -2.541, -9.009, 10.030)
nProdVtx: 1 nDecayVtx: 14
nParents: 1 nChildren: 14
nCheckpoints: 1
direct simHits: 12 simE=0.003097
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 114 simE=8.479095
subgraph recHits: 0 missing=114 recoE=0.000000
direct tracker simHits: 16 dE=16.499369
subgraph tracker simHits: 14 dE=16.499367
checkpointId: 0
x4@checkpoint: (88.111, -86.252, -255.709, 0.000)
p4@checkpoint: (0.025, -0.180, -0.379, 0.421)
raw GEN: n/a
raw SIM: SimTrack #1145 key=291238

Bremsstrahlung
Vertex 144
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (29.138, -20.889, -73.009, 3.067)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #326 key=247

Bremsstrahlung
Vertex 145
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (29.232, -20.960, -73.250, 3.076)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #327 key=248

Bremsstrahlung
Vertex 146
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (48.021, -36.416, -123.018, 4.924)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #328 key=249

Bremsstrahlung
Vertex 147
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (52.028, -40.078, -134.121, 5.337)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #329 key=250

Bremsstrahlung
Vertex 148
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (58.555, -46.375, -152.696, 6.026)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #330 key=251

Bremsstrahlung
Vertex 149
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (68.658, -57.129, -182.930, 7.148)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #331 key=252

Bremsstrahlung
Vertex 150
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.753, -70.928, -219.277, 8.497)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #332 key=253

Bremsstrahlung
Vertex 151
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.763, -70.941, -219.312, 8.498)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #333 key=254

Bremsstrahlung
Vertex 152
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.593, -76.348, -232.926, 9.003)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #334 key=255

Bremsstrahlung
Vertex 153
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.663, -76.450, -233.181, 9.013)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #336 key=257

Bremsstrahlung
Vertex 154
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.889, -78.259, -237.682, 9.180)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #337 key=258

Bremsstrahlung
Vertex 155
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (85.641, -79.449, -240.563, 9.287)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #338 key=259

Bremsstrahlung
Vertex 156
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (85.985, -80.050, -241.998, 9.340)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #339 key=260

Bremsstrahlung
Vertex 157
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (86.149, -80.350, -242.716, 9.366)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #340 key=261

gamma (22)
Particle 221
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.770, -0.581, -1.970, 2.193)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 31 simE=2.109008
direct recHits: 0 missing=31 recoE=0.000000
subgraph simHits: 34 simE=2.112308
subgraph recHits: 0 missing=34 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (98.956, -73.557, -251.602, 0.000)
p4@checkpoint: (0.770, -0.581, -1.970, 2.193)
raw GEN: n/a
raw SIM: SimTrack #1146 key=291241

Annihilation

Vertex 180
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-97.795, -61.958, -199.108, 17.673)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #381 key=302

gamma (22)
Particle 222
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.134, -0.101, -0.344, 0.383)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.392313
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=0.392313
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (98.961, -73.550, -251.645, 0.000)
p4@checkpoint: (0.134, -0.101, -0.344, 0.383)
raw GEN: n/a
raw SIM: SimTrack #1147 key=291242

gamma (22)
Particle 223
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.207, -0.185, -0.568, 0.633)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 19 simE=0.393649
subgraph recHits: 0 missing=19 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1148 key=291248

PairConversion

Vertex 176
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (90.887, -74.816, -240.735, 9.295)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #375 key=296

gamma (22)
Particle 224
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000174
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000174
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.283, -79.458, -252.444, 0.000)
p4@checkpoint: (0.001, -0.001, -0.004, 0.005)
raw GEN: n/a
raw SIM: SimTrack #1149 key=291249

gamma (22)
Particle 225
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.360, -0.359, -1.041, 1.159)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 21 simE=0.837359
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1150 key=291250

PairConversion

Vertex 170
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.152, -76.865, -241.173, 9.310)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #366 key=287

gamma (22)
Particle 226
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.152, -0.173, -0.471, 0.524)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.484940
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.484940
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.299, -82.869, -253.185, 0.000)
p4@checkpoint: (0.152, -0.173, -0.471, 0.524)
raw GEN: n/a
raw SIM: SimTrack #1151 key=291252

gamma (22)
Particle 227
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.047, -0.064, -0.163, 0.182)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 13 simE=0.024494
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1152 key=291254

PairConversion

Vertex 158
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.716, -76.310, -232.987, 9.006)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #341 key=262

gamma (22)
Particle 228
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001420
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001420
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.556, -82.586, -253.025, 0.000)
p4@checkpoint: (0.001, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1153 key=291255

gamma (22)
Particle 229
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.163, -0.239, -0.596, 0.662)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.614209
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=0.614209
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.428, -84.885, -254.211, 0.000)
p4@checkpoint: (0.163, -0.239, -0.596, 0.662)
raw GEN: n/a
raw SIM: SimTrack #1154 key=291256

gamma (22)
Particle 230
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.005854
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.005854
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.451, -84.861, -254.189, 0.000)
p4@checkpoint: (0.001, -0.001, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1156 key=291258

gamma (22)
Particle 231
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.916, -1.366, -3.389, 3.768)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 30 simE=3.556968
direct recHits: 0 missing=30 recoE=0.000000
subgraph simHits: 30 simE=3.556968
subgraph recHits: 0 missing=30 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.372, -84.945, -254.267, 0.000)
p4@checkpoint: (0.916, -1.366, -3.389, 3.768)
raw GEN: n/a
raw SIM: SimTrack #1157 key=291260

gamma (22)
Particle 232
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.007, -0.012, -0.029, 0.033)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.034982
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.034982
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.059, -85.273, -254.395, 0.000)
p4@checkpoint: (0.007, -0.012, -0.029, 0.033)
raw GEN: n/a
raw SIM: SimTrack #1158 key=291261

gamma (22)
Particle 233
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.004, -0.010, 0.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.009811
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.009811
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.991, -85.344, -254.607, 0.000)
p4@checkpoint: (0.002, -0.004, -0.010, 0.011)
raw GEN: n/a
raw SIM: SimTrack #1159 key=291262

gamma (22)
Particle 234
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.004, -0.009, 0.010)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.007518
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.007518
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.914, -85.423, -254.456, 0.000)
p4@checkpoint: (0.002, -0.004, -0.009, 0.010)
raw GEN: n/a
raw SIM: SimTrack #1160 key=291264

e- (11)
Particle 235
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.020, -0.028, -0.069, 0.078)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 7 simE=0.022018
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1161 key=293462

Bremsstrahlung
Vertex 165
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (85.805, -77.675, -237.283, 9.172)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #357 key=278

Bremsstrahlung
Vertex 166
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (85.811, -77.676, -237.293, 9.173)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #358 key=279

Bremsstrahlung
Vertex 167
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (85.887, -77.692, -237.431, 9.178)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #359 key=280

Bremsstrahlung
Vertex 168
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (86.042, -77.682, -237.655, 9.187)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #362 key=283

Bremsstrahlung
Vertex 169
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (86.094, -77.680, -237.631, 9.186)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #365 key=286

e+ (-11)
Particle 236
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.027, -0.035, -0.094, 0.104)
nProdVtx: 1 nDecayVtx: 6
nParents: 1 nChildren: 7
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 7 simE=0.002476
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1162 key=293463

Bremsstrahlung
Vertex 159
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.771, -76.378, -233.180, 9.013)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #342 key=263

Bremsstrahlung
Vertex 160
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.209, -79.563, -240.366, 9.277)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #345 key=266

Bremsstrahlung
Vertex 161
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.193, -79.852, -240.714, 9.292)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #346 key=267

Annihilation

Vertex 162
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.234, -80.281, -241.346, 9.473)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #351 key=272

Annihilation

Vertex 163
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.182, -79.684, -240.768, 9.297)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #352 key=273

Bremsstrahlung
Vertex 164
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.207, -79.863, -240.763, 9.294)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #353 key=274

gamma (22)
Particle 237
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000300
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000300
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (90.103, -84.168, -255.330, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1163 key=293464

gamma (22)
Particle 238
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.002, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000745
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000745
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.176, -90.096, -255.191, 0.000)
p4@checkpoint: (-0.000, -0.002, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1166 key=293472

gamma (22)
Particle 239
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000385
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000385
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (82.823, -91.341, -249.009, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1167 key=293482

gamma (22)
Particle 240
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000436
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000436
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.346, -73.030, -138.680, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1172 key=293493

gamma (22)
Particle 241
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000255
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000255
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.798, -76.374, -268.972, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1173 key=293494

gamma (22)
Particle 242
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000217
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000217
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.727, -84.570, -238.138, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1174 key=293503

gamma (22)
Particle 243
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000138
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000138
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (81.243, -92.749, -254.267, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1175 key=293505

gamma (22)
Particle 244
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.000, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001104
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001104
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.851, -79.968, -251.694, 0.000)
p4@checkpoint: (0.002, -0.000, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1179 key=293569

gamma (22)
Particle 245
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000177
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000177
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (101.318, -70.267, -210.265, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1180 key=293570

gamma (22)
Particle 246
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.005, -0.001, -0.008, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.011531
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.011531
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.914, -78.704, -253.012, 0.000)
p4@checkpoint: (0.005, -0.001, -0.008, 0.009)
raw GEN: n/a
raw SIM: SimTrack #1181 key=293572

gamma (22)
Particle 247
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.006, 0.001, -0.008, 0.010)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.008827
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.008827
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.375, -75.637, -251.563, 0.000)
p4@checkpoint: (0.006, 0.001, -0.008, 0.010)
raw GEN: n/a
raw SIM: SimTrack #1184 key=293575

gamma (22)
Particle 248
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000379
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000379
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.662, -85.685, -239.198, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1187 key=293634

e- (11)
Particle 249
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.259, -0.259, -0.751, 0.835)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.811272
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=0.811272
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.291, -80.621, -252.297, 0.000)
p4@checkpoint: (0.294, -0.227, -0.737, 0.825)
raw GEN: n/a
raw SIM: SimTrack #1188 key=293818

e+ (-11)
Particle 250
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.101, -0.100, -0.290, 0.323)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 7 simE=0.026087
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (90.538, -79.239, -293.500, 0.000)
p4@checkpoint: (-0.001, -0.001, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1189 key=293819

Bremsstrahlung
Vertex 171
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.177, -76.889, -241.243, 9.313)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #367 key=288

Bremsstrahlung
Vertex 172
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.184, -76.895, -241.262, 9.314)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #368 key=289

Bremsstrahlung
Vertex 173
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.215, -76.926, -241.350, 9.317)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #369 key=290

Bremsstrahlung
Vertex 174
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.552, -77.286, -242.334, 9.354)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #370 key=291

gamma (22)
Particle 251
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000973
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000973
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.169, -80.761, -252.492, 0.000)
p4@checkpoint: (0.000, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1190 key=293820

gamma (22)
Particle 252
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.006, -0.006, -0.017, 0.019)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.017125
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.017125
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.089, -80.854, -252.623, 0.000)
p4@checkpoint: (0.006, -0.006, -0.017, 0.019)
raw GEN: n/a
raw SIM: SimTrack #1191 key=293821

gamma (22)
Particle 253
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.003, -0.007, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.006669
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.006669
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.059, -80.888, -252.239, 0.000)
p4@checkpoint: (0.002, -0.003, -0.007, 0.008)
raw GEN: n/a
raw SIM: SimTrack #1192 key=293823

gamma (22)
Particle 254
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.079, -0.088, -0.235, 0.263)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.001320
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1193 key=293825

PairConversion

Vertex 175
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.833, -77.598, -243.166, 9.385)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #371 key=292

e- (11)
Particle 255
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.029, -0.031, -0.084, 0.094)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000789
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000789
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (92.251, -79.221, -293.500, 0.000)
p4@checkpoint: (0.041, -0.013, -0.081, 0.092)
raw GEN: n/a
raw SIM: SimTrack #1194 key=293839

e+ (-11)
Particle 256
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.050, -0.056, -0.150, 0.168)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000531
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000531
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.551, -82.591, -254.368, 0.000)
p4@checkpoint: (-0.007, -0.067, -0.141, 0.157)
raw GEN: n/a
raw SIM: SimTrack #1195 key=293840

e- (11)
Particle 257
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.128, -0.118, -0.351, 0.392)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 14 simE=0.354916
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 16 simE=0.355388
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.587, -77.885, -251.958, 0.000)
p4@checkpoint: (0.159, -0.069, -0.343, 0.384)
raw GEN: n/a
raw SIM: SimTrack #1199 key=294389

Bremsstrahlung
Vertex 178
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (90.892, -74.820, -240.748, 9.296)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #377 key=298

Bremsstrahlung
Vertex 179
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (91.633, -75.452, -242.710, 9.369)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #380 key=301

e+ (-11)
Particle 258
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.079, -0.070, -0.216, 0.241)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 2 simE=0.000356
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 4 simE=0.038260
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.560, -80.308, -254.062, 0.000)
p4@checkpoint: (0.002, -0.080, -0.170, 0.188)
raw GEN: n/a
raw SIM: SimTrack #1200 key=294390

Bremsstrahlung
Vertex 177
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (90.888, -74.817, -240.738, 9.295)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #376 key=297

gamma (22)
Particle 259
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.015, -0.013, -0.041, 0.045)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.037904
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.037904
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.087, -78.495, -252.195, 0.000)
p4@checkpoint: (0.015, -0.013, -0.041, 0.045)
raw GEN: n/a
raw SIM: SimTrack #1201 key=294391

gamma (22)
Particle 260
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000130
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000130
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.249, -82.925, -260.143, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1202 key=294589

gamma (22)
Particle 261
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000342
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000342
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.208, -78.347, -252.032, 0.000)
p4@checkpoint: (0.000, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1205 key=294593

gamma (22)
Particle 262
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001703
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001703
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-105.039, -64.574, -198.139, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1206 key=295034

gamma (22)
Particle 263
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001597
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001597
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.336, 21.921, -258.795, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1207 key=295035

gamma (22)
Particle 264
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.142, 0.016, -0.294, 0.327)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.002252
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.002252
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.466, -14.318, -264.558, 0.000)
p4@checkpoint: (0.142, 0.016, -0.294, 0.327)
raw GEN: n/a
raw SIM: SimTrack #1209 key=295610

gamma (22)
Particle 265
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.041, 0.058, -0.067, 0.097)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.080679
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=0.080679
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-70.291, 101.302, -124.152, 0.000)
p4@checkpoint: (-0.041, 0.058, -0.067, 0.097)
raw GEN: n/a
raw SIM: SimTrack #1211 key=297496

gamma (22)
Particle 266
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.022, -0.058, -0.107, 0.123)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.123936
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.123936
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.188, -114.721, -214.495, 0.000)
p4@checkpoint: (0.022, -0.058, -0.107, 0.123)
raw GEN: n/a
raw SIM: SimTrack #1212 key=297497

n (2112)
Particle 267
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.246, 0.268, 0.200, 1.027)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.025514
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.025514
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.630, -12.834, -127.463, 0.000)
p4@checkpoint: (-0.246, 0.268, 0.200, 1.027)
raw GEN: n/a
raw SIM: SimTrack #1213 key=297636

n (2112)
Particle 268
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.038, -0.082, -0.012, 0.944)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003219
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003219
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.480, -30.455, -137.103, 0.000)
p4@checkpoint: (-0.052, -0.065, -0.035, 0.944)
raw GEN: n/a
raw SIM: SimTrack #1214 key=297640

n (2112)
Particle 269
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.029, 0.042, 0.023, 0.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000268
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000268
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.944, -9.369, -127.264, 0.000)
p4@checkpoint: (-0.029, 0.042, 0.023, 0.941)
raw GEN: n/a
raw SIM: SimTrack #1215 key=297641

gamma (22)
Particle 270
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001433
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001433
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (62.952, 106.019, -197.190, 0.000)
p4@checkpoint: (0.001, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1216 key=297642

n (2112)
Particle 271
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.017, -0.403, -0.838, 1.322)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000492
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000492
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (0.483, -123.299, -262.365, 0.000)
p4@checkpoint: (-0.017, -0.403, -0.838, 1.322)
raw GEN: n/a
raw SIM: SimTrack #1217 key=297679

p (2212)
Particle 272
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.536, -0.424, -0.579, 1.297)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 30 dE=30.648197
subgraph tracker simHits: 30 dE=30.648197
raw GEN: n/a
raw SIM: SimTrack #1218 key=297680

pi- (-211)
Particle 273
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.107, 0.347, -0.267, 0.472)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 12 simE=0.000699
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 12 simE=0.000699
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 85 dE=85.559662
subgraph tracker simHits: 85 dE=85.559662
checkpointId: 0
x4@checkpoint: (-2.843, -10.554, -293.500, 0.000)
p4@checkpoint: (0.172, 0.247, -0.250, 0.416)
raw GEN: n/a
raw SIM: SimTrack #1219 key=297681

pi0 (111)
Particle 274
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.136, -0.283, 0.116, 0.361)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 19 simE=0.341259
subgraph recHits: 0 missing=19 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1220 key=297682

Decay
Vertex 200
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.516, -4.631, -15.573, 0.927)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #404 key=325

pi0 (111)
Particle 275
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.090, -0.283, -0.153, 0.361)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 25 simE=0.194793
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 35 dE=31.767265
subgraph tracker simHits: 11 dE=43.858242
raw GEN: n/a
raw SIM: SimTrack #1221 key=297683

Decay
Vertex 188
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.516, -4.631, -15.573, 0.927)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #391 key=312

pi- (-211)
Particle 276
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.095, -0.130, -0.139, 0.255)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000315
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 20 dE=17.002932
subgraph tracker simHits: 8 dE=19.412350
raw GEN: n/a
raw SIM: SimTrack #1222 key=297684

HadronInelastic

Vertex 187
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (14.276, -20.443, -175.159, 10.569)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #390 key=311

pi- (-211)
Particle 277
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.104, -0.135, -0.084, 0.236)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 3 simE=0.000760
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 23 dE=21.790575
subgraph tracker simHits: 8 dE=21.790575
raw GEN: n/a
raw SIM: SimTrack #1223 key=297686

HadronInelastic

Vertex 185
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.402, -25.190, -27.431, 2.009)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #388 key=309

Annihilation

Vertex 186
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-34.664, 17.578, 57.372, 5.488)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #389 key=310

n (2112)
Particle 278
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.056, -0.037, 0.033, 0.943)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000028
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000028
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.306, -22.084, 36.607, 0.000)
p4@checkpoint: (0.014, -0.001, 0.006, 0.940)
raw GEN: n/a
raw SIM: SimTrack #1224 key=297688

gamma (22)
Particle 279
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.000, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=1.000000
subgraph tracker simHits: 1 dE=1.000000
checkpointId: 0
x4@checkpoint: (26.458, -21.886, 293.500, 0.000)
p4@checkpoint: (0.000, -0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1225 key=297689

gamma (22)
Particle 280
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000150
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000150
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-20.360, 121.607, 37.039, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1226 key=297813

gamma (22)
Particle 281
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000281
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000281
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.460, 14.368, 83.521, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1227 key=297817

gamma (22)
Particle 282
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000330
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000330
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (98.192, -74.574, 96.418, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1228 key=297818

p (2212)
Particle 283
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.157, -0.287, -0.395, 1.069)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 8 dE=2.409417
subgraph tracker simHits: 8 dE=2.409417
checkpointId: 0
x4@checkpoint: (-34.660, -32.481, -293.500, 0.000)
p4@checkpoint: (0.032, 0.259, -0.332, 1.029)
raw GEN: n/a
raw SIM: SimTrack #1229 key=298192

gamma (22)
Particle 284
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000315
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000315
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.035, 82.052, -223.511, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1230 key=298195

e- (11)
Particle 285
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.094, -0.128, -0.053, 0.167)
nProdVtx: 1 nDecayVtx: 11
nParents: 1 nChildren: 11
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 16 simE=0.049852
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 8 dE=5.973486
subgraph tracker simHits: 6 dE=12.090977
raw GEN: n/a
raw SIM: SimTrack #1231 key=298298

Bremsstrahlung
Vertex 189
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (22.686, 3.105, -40.389, 3.912)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #392 key=313

Bremsstrahlung
Vertex 190
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (25.245, 11.400, -42.832, 4.229)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #393 key=314

Bremsstrahlung
Vertex 191
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (25.312, 7.638, -53.367, 5.711)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #394 key=315

Bremsstrahlung
Vertex 192
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.438, 12.143, -63.969, 7.126)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #395 key=316

Bremsstrahlung
Vertex 193
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (18.952, 8.784, -112.438, 12.239)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #396 key=317

Bremsstrahlung
Vertex 194
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.820, 18.993, -38.711, 4.168)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #397 key=318

Annihilation

Vertex 195
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.735, 19.025, -38.828, 4.173)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #398 key=319

Bremsstrahlung
Vertex 196
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (6.068, 15.851, -36.433, 3.294)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #399 key=320

Bremsstrahlung
Vertex 197
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (10.545, 14.437, -37.839, 3.506)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #400 key=321

Bremsstrahlung
Vertex 198
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.741, 13.528, -42.441, 4.448)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #401 key=322

Bremsstrahlung
Vertex 199
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (9.877, 14.745, -42.807, 4.510)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #402 key=323

gamma (22)
Particle 286
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.013, -0.154, -0.101, 0.184)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.144941
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.144941
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-4.293, -123.225, -93.708, 0.000)
p4@checkpoint: (-0.013, -0.154, -0.101, 0.184)
raw GEN: n/a
raw SIM: SimTrack #1232 key=298299

gamma (22)
Particle 287
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001156
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001156
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.843, 70.948, -67.333, 0.000)
p4@checkpoint: (0.001, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1233 key=298361

gamma (22)
Particle 288
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000037
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000037
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.220, 4.442, -147.241, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1234 key=298363

gamma (22)
Particle 289
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.007, 0.021, -0.006, 0.023)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.032389
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.032389
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (58.515, 108.531, -80.030, 0.000)
p4@checkpoint: (0.007, 0.021, -0.006, 0.023)
raw GEN: n/a
raw SIM: SimTrack #1235 key=298365

gamma (22)
Particle 290
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.001, -0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001588
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001588
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-108.406, -58.745, -74.171, 0.000)
p4@checkpoint: (-0.001, -0.001, -0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1236 key=298373

gamma (22)
Particle 291
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.002, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000177
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000177
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-5.032, -123.197, -178.802, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1237 key=298381

gamma (22)
Particle 292
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000475
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000475
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-104.087, 66.097, -124.666, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1238 key=298402

gamma (22)
Particle 293
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.002, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002405
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002405
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-24.847, 120.770, -99.758, 0.000)
p4@checkpoint: (-0.001, 0.002, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1239 key=298404

gamma (22)
Particle 294
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001397
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001397
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (25.865, -120.556, -69.929, 0.000)
p4@checkpoint: (0.000, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1240 key=298410

gamma (22)
Particle 295
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000300
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000300
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (60.053, 107.687, -53.114, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1241 key=298411

gamma (22)
Particle 296
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.000, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002272
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002272
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (119.423, 30.675, -68.587, 0.000)
p4@checkpoint: (0.002, 0.000, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1242 key=298415

gamma (22)
Particle 297
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.011, -0.003, 0.012)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.007656
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.007656
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 6 dE=6.117491
subgraph tracker simHits: 6 dE=6.117491
checkpointId: 0
x4@checkpoint: (17.197, 122.095, -70.450, 0.000)
p4@checkpoint: (-0.001, 0.008, -0.002, 0.008)
raw GEN: n/a
raw SIM: SimTrack #1243 key=298418

gamma (22)
Particle 298
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.018, -0.089, 0.009, 0.091)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.098494
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.098494
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-17.841, -122.002, -3.846, 0.000)
p4@checkpoint: (-0.018, -0.089, 0.009, 0.091)
raw GEN: n/a
raw SIM: SimTrack #1245 key=298433

gamma (22)
Particle 299
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.154, -0.194, 0.107, 0.269)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.242764
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.242764
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (77.724, -95.718, 34.627, 0.000)
p4@checkpoint: (0.154, -0.194, 0.107, 0.269)
raw GEN: n/a
raw SIM: SimTrack #1246 key=298434

gamma (22)
Particle 300
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, 0.001, -0.008, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.004393
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.004393
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (120.964, 23.887, -260.195, 0.000)
p4@checkpoint: (0.003, 0.000, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1249 key=315505

pi- (-211)
Particle 301
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.361, -0.185, -0.653, 0.781)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1251 key=390190

pi+ (211)
Particle 302
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.281, -0.637, -1.704, 1.847)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1252 key=390191

Ionisation

Vertex 203
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (179.066, -180.043, -572.823, 21.345)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #410 key=331

HadronInelastic

Vertex 204
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (169.700, -217.552, -688.829, 25.450)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #411 key=332

e- (11)
Particle 303
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.002, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1253 key=390677

pi+ (211)
Particle 304
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.215, -0.365, -0.475, 0.652)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1254 key=391747

HadronInelastic

Vertex 205
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (157.204, -236.561, -718.685, 26.735)
nIn: 1 nOut: 5
raw GEN: n/a
raw SIM: SimVertex #412 key=333

p (2212)
Particle 305
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.093, -0.255, -0.226, 1.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1255 key=392946

p (2212)
Particle 306
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.254, 0.103, 0.194, 0.996)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1256 key=392947

p (2212)
Particle 307
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.136, -0.107, -0.207, 0.976)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1257 key=392948

p (2212)
Particle 308
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.190, -0.145, 0.103, 0.974)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1258 key=392949

p (2212)
Particle 309
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.110, 0.087, 0.167, 0.963)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1259 key=392950

e- (11)
Particle 310
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.006, -0.012, 0.014)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1260 key=408265

pi0 (111)
Particle 311
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (20.965, -18.073, 37.664, 46.742)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 136 simE=45.764568
subgraph recHits: 0 missing=136 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1261 key=432993

Decay
Vertex 223
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.592, -74.824, 162.135, 6.987)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #436 key=357

K0_S (310)
Particle 312
pid: K0_S (310)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.055, -1.355, 1.972, 2.662)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 50 simE=0.624428
subgraph recHits: 0 missing=50 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1262 key=432994

Decay
Vertex 222
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (90.926, -76.537, 164.628, 7.099)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #435 key=356

pi0 (111)
Particle 313
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.441, 0.001, 0.383, 0.600)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 27 simE=0.602757
subgraph recHits: 0 missing=27 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1263 key=432995

Decay
Vertex 221
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.592, -74.823, 162.134, 6.987)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #434 key=355

pi+ (211)
Particle 314
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.336, -0.476, 0.572, 0.828)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 17 simE=0.413449
direct recHits: 0 missing=17 recoE=0.000000
subgraph simHits: 17 simE=0.413449
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.233, -80.687, 168.888, 0.000)
p4@checkpoint: (0.273, -0.512, 0.567, 0.823)
raw GEN: n/a
raw SIM: SimTrack #1264 key=432996

pi+ (211)
Particle 315
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.136, -0.049, 0.451, 0.494)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 6 simE=0.148044
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 7 simE=0.148375
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.775, -78.871, 183.919, 0.000)
p4@checkpoint: (0.078, -0.109, 0.442, 0.482)
raw GEN: n/a
raw SIM: SimTrack #1265 key=432997

Annihilation

Vertex 220
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (93.684, 67.941, 156.468, 33.150)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #433 key=354

pdg(3112)
Particle 316
caloBoundary
pid: pdg(3112)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.309, -0.734, 1.999, 2.772)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 67 simE=0.823840
direct recHits: 0 missing=67 recoE=0.000000
subgraph simHits: 69 simE=0.824680
subgraph recHits: 0 missing=69 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.483, -78.012, 171.047, 0.000)
p4@checkpoint: (1.333, -0.683, 1.992, 2.765)
raw GEN: n/a
raw SIM: SimTrack #1266 key=432998

HadronInelastic

Vertex 218
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.901, -21.196, 180.254, 25.288)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #431 key=352

HadronInelastic

Vertex 219
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (117.186, -9.639, 170.958, 30.113)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #432 key=353

pi0 (111)
Particle 317
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.006, -0.358, 1.535, 1.875)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 29 simE=1.859464
subgraph recHits: 0 missing=29 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1267 key=432999

Decay
Vertex 217
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.592, -74.823, 162.134, 6.987)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #429 key=350

pi+ (211)
Particle 318
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (2.115, -1.954, 4.348, 5.217)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 126 simE=2.932389
direct recHits: 0 missing=126 recoE=0.000000
subgraph simHits: 131 simE=2.940528
subgraph recHits: 0 missing=131 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.377, -79.346, 172.074, 0.000)
p4@checkpoint: (2.066, -2.002, 4.343, 5.211)
raw GEN: n/a
raw SIM: SimTrack #1268 key=433000

Bremsstrahlung
Vertex 213
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (78.412, -88.509, 207.089, 9.382)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #423 key=344

Bremsstrahlung
Vertex 214
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (96.015, -69.646, 206.263, 8.833)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #424 key=345

Bremsstrahlung
Vertex 215
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (96.544, -70.008, 207.390, 8.899)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #425 key=346

HadronInelastic

Vertex 216
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (90.217, -69.051, 206.477, 14.138)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #428 key=349

pi0 (111)
Particle 319
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.831, -1.259, 3.199, 3.897)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 46 simE=3.556572
subgraph recHits: 0 missing=46 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1269 key=433001

Decay
Vertex 211
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.592, -74.824, 162.134, 6.987)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #418 key=339

pi- (-211)
Particle 320
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.222, -0.456, 1.290, 1.394)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 25 simE=0.547040
direct recHits: 0 missing=25 recoE=0.000000
subgraph simHits: 29 simE=0.558573
subgraph recHits: 0 missing=29 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.078, -80.866, 179.778, 0.000)
p4@checkpoint: (0.285, -0.419, 1.280, 1.384)
raw GEN: n/a
raw SIM: SimTrack #1270 key=433002

NuclearCapture
Vertex 210
domain: SIM
reason: NuclearCapture
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (79.503, -83.824, 221.323, 385.129)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #417 key=338

pi- (-211)
Particle 321
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (2.580, -2.106, 3.897, 5.128)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 76 simE=1.630995
direct recHits: 0 missing=76 recoE=0.000000
subgraph simHits: 76 simE=1.630995
subgraph recHits: 0 missing=76 recoE=0.000000
direct tracker simHits: 3 dE=3.290227
subgraph tracker simHits: 3 dE=3.290227
checkpointId: 0
x4@checkpoint: (94.721, -78.936, 169.821, 0.000)
p4@checkpoint: (2.618, -2.054, 3.891, 5.122)
raw GEN: n/a
raw SIM: SimTrack #1271 key=433003

pi0 (111)
Particle 322
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.968, -1.824, 3.621, 4.509)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 47 simE=4.316197
subgraph recHits: 0 missing=47 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1272 key=433004

Decay
Vertex 209
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.592, -74.823, 162.134, 6.987)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #416 key=337

pi+ (211)
Particle 323
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (3.733, -3.030, 6.479, 8.069)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 26 simE=0.361236
direct recHits: 0 missing=26 recoE=0.000000
subgraph simHits: 26 simE=0.361236
subgraph recHits: 0 missing=26 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.663, -79.005, 171.003, 0.000)
p4@checkpoint: (3.679, -3.081, 6.477, 8.062)
raw GEN: n/a
raw SIM: SimTrack #1273 key=433005

p (2212)
Particle 324
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.066, -1.160, 1.900, 2.416)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 24 simE=0.505990
direct recHits: 0 missing=24 recoE=0.000000
subgraph simHits: 24 simE=0.505990
subgraph recHits: 0 missing=24 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.531, -84.777, 178.393, 0.000)
p4@checkpoint: (-0.066, -1.152, 1.886, 2.402)
raw GEN: n/a
raw SIM: SimTrack #1274 key=433007

pi+ (211)
Particle 325
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.215, -0.016, -0.069, 0.266)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.229245
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.229245
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.480, -76.776, 159.976, 0.000)
p4@checkpoint: (0.189, -0.097, -0.062, 0.261)
raw GEN: n/a
raw SIM: SimTrack #1275 key=433008

n (2112)
Particle 326
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.290, 0.157, -0.112, 1.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.004924
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.004924
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-115.441, 43.316, 81.397, 0.000)
p4@checkpoint: (-0.290, 0.157, -0.112, 1.002)
raw GEN: n/a
raw SIM: SimTrack #1277 key=433214

n (2112)
Particle 327
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.014, -0.076, 0.044, 0.944)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000126
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000126
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (93.323, -80.584, 179.575, 0.000)
p4@checkpoint: (0.005, -0.044, 0.075, 0.944)
raw GEN: n/a
raw SIM: SimTrack #1278 key=433216

n (2112)
Particle 328
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.010, -0.074, -0.044, 0.944)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000735
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000735
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (90.905, -83.302, 160.487, 0.000)
p4@checkpoint: (-0.007, -0.071, 0.004, 0.942)
raw GEN: n/a
raw SIM: SimTrack #1279 key=433217

gamma (22)
Particle 329
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000657
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000657
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-74.926, 97.924, -14.373, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1280 key=433219

gamma (22)
Particle 330
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.320, -0.348, 0.697, 0.842)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.801752
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.801752
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.076, -79.703, 171.907, 0.000)
p4@checkpoint: (0.320, -0.348, 0.697, 0.842)
raw GEN: n/a
raw SIM: SimTrack #1281 key=434540

gamma (22)
Particle 331
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.649, -1.476, 2.924, 3.667)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 39 simE=3.514446
direct recHits: 0 missing=39 recoE=0.000000
subgraph simHits: 39 simE=3.514446
subgraph recHits: 0 missing=39 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.491, -79.211, 170.823, 0.000)
p4@checkpoint: (1.649, -1.476, 2.924, 3.667)
raw GEN: n/a
raw SIM: SimTrack #1282 key=434541

gamma (22)
Particle 332
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.005780
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.005780
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.512, -79.186, 223.390, 0.000)
p4@checkpoint: (0.001, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1284 key=436829

gamma (22)
Particle 333
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003948
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003948
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.171, -90.100, 218.453, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1285 key=436830

gamma (22)
Particle 334
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001806
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001806
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-115.258, 43.801, -21.031, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1287 key=436832

gamma (22)
Particle 335
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.588, -0.335, 1.016, 1.221)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 25 simE=1.002714
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1288 key=437176

PairConversion

Vertex 212
domain: SIM
reason: PairConversion

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (90.495, -75.338, 163.694, 7.049)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #421 key=342

gamma (22)
Particle 336
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.243, -0.924, 2.183, 2.676)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 32 simE=2.553858
direct recHits: 0 missing=32 recoE=0.000000
subgraph simHits: 32 simE=2.553858
subgraph recHits: 0 missing=32 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.876, -78.749, 171.412, 0.000)
p4@checkpoint: (1.243, -0.924, 2.183, 2.676)
raw GEN: n/a
raw SIM: SimTrack #1289 key=437177

e- (11)
Particle 337
pid: e- (11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.180, -0.102, 0.311, 0.373)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.258189
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=0.258189
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.935, -77.456, 172.741, 0.000)
p4@checkpoint: (0.195, -0.042, 0.309, 0.368)
raw GEN: n/a
raw SIM: SimTrack #1292 key=438047

e+ (-11)
Particle 338
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.408, -0.233, 0.705, 0.848)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 15 simE=0.744524
direct recHits: 0 missing=15 recoE=0.000000
subgraph simHits: 15 simE=0.744524
subgraph recHits: 0 missing=15 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.104, -78.474, 172.055, 0.000)
p4@checkpoint: (0.368, -0.289, 0.702, 0.843)
raw GEN: n/a
raw SIM: SimTrack #1293 key=438048

gamma (22)
Particle 339
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000175
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000175
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (75.142, -97.758, 207.494, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1295 key=438630

gamma (22)
Particle 340
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000452
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000452
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.250, -77.064, 214.373, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1296 key=438724

gamma (22)
Particle 341
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000156
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000156
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (104.695, -65.130, 184.385, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1297 key=438725

gamma (22)
Particle 342
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.006, 0.001, -0.001, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.006952
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.006952
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.027, -67.737, 204.554, 0.000)
p4@checkpoint: (0.006, 0.001, -0.001, 0.007)
raw GEN: n/a
raw SIM: SimTrack #1301 key=438882

gamma (22)
Particle 343
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000403
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000403
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (102.675, -68.270, 207.677, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1302 key=438884

gamma (22)
Particle 344
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.311, -0.056, 0.394, 0.505)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 17 simE=0.459209
direct recHits: 0 missing=17 recoE=0.000000
subgraph simHits: 17 simE=0.459209
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.967, -76.160, 171.477, 0.000)
p4@checkpoint: (0.311, -0.056, 0.394, 0.505)
raw GEN: n/a
raw SIM: SimTrack #1303 key=439204

gamma (22)
Particle 345
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.695, -0.302, 1.142, 1.370)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 20 simE=1.400255
direct recHits: 0 missing=20 recoE=0.000000
subgraph simHits: 20 simE=1.400255
subgraph recHits: 0 missing=20 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.863, -77.544, 172.432, 0.000)
p4@checkpoint: (0.695, -0.302, 1.142, 1.370)
raw GEN: n/a
raw SIM: SimTrack #1304 key=439205

gamma (22)
Particle 346
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000427
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000427
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (65.012, 104.768, 147.999, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1306 key=439976

gamma (22)
Particle 347
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000414
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000414
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.079, -7.374, 167.741, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1308 key=439988

gamma (22)
Particle 348
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000330
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000330
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.546, 66.941, 147.975, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1309 key=440213

gamma (22)
Particle 349
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.155, 0.044, 0.078, 0.179)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.201289
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.201289
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.263, -71.766, 167.515, 0.000)
p4@checkpoint: (0.155, 0.044, 0.078, 0.179)
raw GEN: n/a
raw SIM: SimTrack #1310 key=440292

gamma (22)
Particle 350
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.286, -0.044, 0.305, 0.420)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.401469
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.401469
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.115, -75.971, 170.160, 0.000)
p4@checkpoint: (0.286, -0.044, 0.305, 0.420)
raw GEN: n/a
raw SIM: SimTrack #1311 key=440293

pi- (-211)
Particle 351
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.140, -0.458, 0.575, 0.761)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 32 simE=0.352417
direct recHits: 0 missing=32 recoE=0.000000
subgraph simHits: 32 simE=0.352417
subgraph recHits: 0 missing=32 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (92.711, -81.287, 170.762, 0.000)
p4@checkpoint: (0.194, -0.434, 0.574, 0.759)
raw GEN: n/a
raw SIM: SimTrack #1312 key=440489

pi+ (211)
Particle 352
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.914, -0.897, 1.397, 1.900)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.272011
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 18 simE=0.272011
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.057, -79.726, 169.504, 0.000)
p4@checkpoint: (0.879, -0.930, 1.396, 1.899)
raw GEN: n/a
raw SIM: SimTrack #1313 key=440490

gamma (22)
Particle 353
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (7.093, -6.169, 12.873, 15.940)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 81 simE=15.553347
direct recHits: 0 missing=81 recoE=0.000000
subgraph simHits: 83 simE=15.554026
subgraph recHits: 0 missing=83 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.552, -79.138, 171.137, 0.000)
p4@checkpoint: (7.093, -6.169, 12.873, 15.940)
raw GEN: n/a
raw SIM: SimTrack #1314 key=440810

Annihilation

Vertex 225
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (83.380, -83.146, 186.193, 8.528)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #438 key=359

gamma (22)
Particle 354
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (13.872, -11.905, 24.791, 30.802)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 103 simE=30.207237
direct recHits: 0 missing=103 recoE=0.000000
subgraph simHits: 106 simE=30.210537
subgraph recHits: 0 missing=106 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.580, -79.104, 171.048, 0.000)
p4@checkpoint: (13.872, -11.905, 24.791, 30.802)
raw GEN: n/a
raw SIM: SimTrack #1315 key=440811

Annihilation

Vertex 224
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (77.092, -86.025, 179.325, 8.814)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #437 key=358

gamma (22)
Particle 355
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000298
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000298
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (82.092, -91.999, 180.992, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1316 key=444963

gamma (22)
Particle 356
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.000, -0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003002
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003002
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-116.165, -41.336, 80.916, 0.000)
p4@checkpoint: (-0.003, 0.001, -0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1317 key=444964

gamma (22)
Particle 357
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000227
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000227
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (79.884, -93.923, 187.879, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1318 key=452635

gamma (22)
Particle 358
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000452
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000452
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (99.187, -73.246, 205.447, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1319 key=452636

gamma (22)
Particle 359
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000388
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000388
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (73.799, -98.776, 201.112, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1321 key=456351

n (2112)
Particle 360
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.014, 0.084, -0.037, 0.944)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001641
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001641
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.476, 75.507, 120.895, 0.000)
p4@checkpoint: (0.014, 0.084, -0.037, 0.944)
raw GEN: n/a
raw SIM: SimTrack #1322 key=461389

gamma (22)
Particle 361
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.002, 0.000, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001106
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001106
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.727, -88.621, 192.629, 0.000)
p4@checkpoint: (0.001, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1323 key=461390

gamma (22)
Particle 362
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000072
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000072
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.316, 15.550, -3.152, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1324 key=461391

n (2112)
Particle 363
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.026, 0.035, 0.056, 0.942)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000002
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000002
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.249, 81.811, 164.790, 0.000)
p4@checkpoint: (-0.017, 0.001, 0.004, 0.940)
raw GEN: n/a
raw SIM: SimTrack #1325 key=461691

gamma (22)
Particle 364
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.000, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000278
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000278
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (79.810, -93.985, 214.781, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1327 key=461693

gamma (22)
Particle 365
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.002, 0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001569
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001569
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (80.467, -93.424, 222.213, 0.000)
p4@checkpoint: (-0.000, -0.002, 0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1328 key=461694

n (2112)
Particle 366
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.101, 0.090, -0.048, 0.950)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.002895
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.002895
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-85.170, 89.157, 101.280, 0.000)
p4@checkpoint: (-0.101, 0.090, -0.048, 0.950)
raw GEN: n/a
raw SIM: SimTrack #1329 key=461712

n (2112)
Particle 367
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.016, 0.024, 0.031, 0.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (68.791, 102.326, 242.131, 0.000)
p4@checkpoint: (0.020, -0.006, 0.026, 0.940)
raw GEN: n/a
raw SIM: SimTrack #1330 key=461713

n (2112)
Particle 368
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.037, -0.034, -0.031, 0.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000009
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000009
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-83.200, -90.998, -6.910, 0.000)
p4@checkpoint: (-0.035, -0.041, -0.015, 0.941)
raw GEN: n/a
raw SIM: SimTrack #1331 key=463682

n (2112)
Particle 369
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.035, 0.117, 0.216, 0.972)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000026
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000026
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (81.671, 28.226, 293.500, 0.000)
p4@checkpoint: (0.035, 0.117, 0.216, 0.972)
raw GEN: n/a
raw SIM: SimTrack #1332 key=464734

gamma (22)
Particle 370
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.001, 0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003036
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003036
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (118.865, -32.772, 251.917, 0.000)
p4@checkpoint: (0.002, -0.001, 0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1333 key=464735

gamma (22)
Particle 371
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.001, -0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.017146
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.017146
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-123.298, 0.742, 189.477, 0.000)
p4@checkpoint: (-0.002, 0.001, -0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1336 key=465566

gamma (22)
Particle 372
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.003, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.005110
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.005110
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.845, -84.444, 196.960, 0.000)
p4@checkpoint: (0.001, -0.003, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1337 key=465567

gamma (22)
Particle 373
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002114
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002114
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.922, -70.836, 187.350, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1338 key=465569

pi- (-211)
Particle 374
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (3.544, -1.831, 5.137, 6.506)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 92 simE=1.952306
direct recHits: 0 missing=92 recoE=0.000000
subgraph simHits: 99 simE=1.982909
subgraph recHits: 0 missing=99 recoE=0.000000
direct tracker simHits: 18 dE=18.167643
subgraph tracker simHits: 16 dE=18.167643
checkpointId: 0
x4@checkpoint: (114.031, -46.903, 162.757, 0.000)
p4@checkpoint: (3.884, -0.742, 5.117, 6.468)
raw GEN: n/a
raw SIM: SimTrack #1341 key=468629

HadronInelastic

Vertex 234
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (11.963, 83.316, 56.668, 85.966)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #450 key=371

HadronInelastic

Vertex 235
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-8.396, 117.966, 112.998, 52.700)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #451 key=372

n (2112)
Particle 375
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.163, -0.430, 0.192, 1.064)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.043574
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.043574
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-21.543, -121.403, 80.563, 0.000)
p4@checkpoint: (-0.163, -0.430, 0.192, 1.064)
raw GEN: n/a
raw SIM: SimTrack #1342 key=468630

gamma (22)
Particle 376
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, 0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000813
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000813
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (77.879, 95.592, -62.733, 0.000)
p4@checkpoint: (0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1343 key=468632

gamma (22)
Particle 377
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000716
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000716
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (25.591, -120.615, 84.940, 0.000)
p4@checkpoint: (0.000, -0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1344 key=468633

gamma (22)
Particle 378
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.002, -0.000, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.025294
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.025294
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-5.145, 123.193, 52.524, 0.000)
p4@checkpoint: (-0.001, 0.002, -0.000, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1345 key=468753

gamma (22)
Particle 379
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.006, 0.000, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004814
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004814
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-9.458, 122.937, 113.222, 0.000)
p4@checkpoint: (-0.001, 0.006, 0.000, 0.006)
raw GEN: n/a
raw SIM: SimTrack #1347 key=468809

n (2112)
Particle 380
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.049, 0.062, -0.071, 0.946)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000104
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.000104
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-4.279, 123.226, 107.041, 0.000)
p4@checkpoint: (0.049, 0.062, -0.071, 0.946)
raw GEN: n/a
raw SIM: SimTrack #1348 key=468810

gamma (22)
Particle 381
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000391
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000391
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-39.072, 116.946, 117.117, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1349 key=468811

e- (11)
Particle 382
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.000, -0.007, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1350 key=510319

e- (11)
Particle 383
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1351 key=511762

e- (11)
Particle 384
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.003, -0.005, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1352 key=513749

e- (11)
Particle 385
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.000, 0.005, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1353 key=517673

e- (11)
Particle 386
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.020, -0.016, 0.073, 0.078)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1354 key=517863

Ionisation

Vertex 244
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-120.418, -207.286, 832.689, 29.291)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #460 key=381

e- (11)
Particle 387
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, 0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1355 key=518736

e- (11)
Particle 388
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.004, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1356 key=519459

e- (11)
Particle 389
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1357 key=520889

e- (11)
Particle 390
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, 0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1358 key=520956

gamma (22)
Particle 391
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.024, -0.002, 0.159, 0.160)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000178
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 8 dE=8.000000
subgraph tracker simHits: 4 dE=8.000000
raw GEN: n/a
raw SIM: SimTrack #1360 key=585741

PairConversion

Vertex 255
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-16.711, -1.651, 111.932, 4.095)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #486 key=407

gamma (22)
Particle 392
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.154, -0.013, 1.027, 1.039)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 35 simE=0.003615
direct recHits: 0 missing=35 recoE=0.000000
subgraph simHits: 35 simE=0.003615
subgraph recHits: 0 missing=35 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-43.996, -3.906, 293.500, 0.000)
p4@checkpoint: (-0.154, -0.013, 1.027, 1.039)
raw GEN: n/a
raw SIM: SimTrack #1361 key=585742

gamma (22)
Particle 393
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.016, 0.002, 0.108, 0.109)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000785
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 14 dE=13.443722
subgraph tracker simHits: 8 dE=14.000000
raw GEN: n/a
raw SIM: SimTrack #1362 key=585750

PairConversion

Vertex 253
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-17.683, -0.588, 118.660, 4.322)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #478 key=399

Annihilation

Vertex 254
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-16.743, -0.245, 133.847, 4.885)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #481 key=402

gamma (22)
Particle 394
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000107
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000107
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.130, -6.482, 141.660, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1363 key=585752

gamma (22)
Particle 395
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000272
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000272
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (82.146, 91.951, 136.595, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1364 key=585753

gamma (22)
Particle 396
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000407
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000407
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-112.653, -50.123, 187.594, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1367 key=585756

gamma (22)
Particle 397
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.026, 0.007, 0.177, 0.179)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.001372
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.001372
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-43.353, 6.027, 293.500, 0.000)
p4@checkpoint: (-0.026, 0.007, 0.177, 0.179)
raw GEN: n/a
raw SIM: SimTrack #1371 key=585765

gamma (22)
Particle 398
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.039, 0.014, 0.276, 0.279)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.000472
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.000472
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-43.139, 6.809, 293.500, 0.000)
p4@checkpoint: (-0.039, 0.014, 0.276, 0.279)
raw GEN: n/a
raw SIM: SimTrack #1373 key=585767

e- (11)
Particle 399
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.013, 0.001, 0.086, 0.087)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=0.556278
subgraph tracker simHits: 1 dE=0.556278
checkpointId: 0
x4@checkpoint: (-19.266, 0.028, 293.500, 0.000)
p4@checkpoint: (-0.012, -0.001, 0.048, 0.050)
raw GEN: n/a
raw SIM: SimTrack #1376 key=592876

gamma (22)
Particle 400
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000274
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000274
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (64.999, -104.776, 51.171, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1379 key=592883

gamma (22)
Particle 401
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-86.015, 88.342, 203.912, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1380 key=592884

e- (11)
Particle 402
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.022, -0.002, 0.152, 0.153)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000178
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-16.218, -1.964, 293.500, 0.000)
p4@checkpoint: (-0.019, 0.001, 0.067, 0.069)
raw GEN: n/a
raw SIM: SimTrack #1385 key=594602

Bremsstrahlung
Vertex 256
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-17.132, -1.778, 115.295, 4.208)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #490 key=411

gamma (22)
Particle 403
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, -0.002, 0.037, 0.037)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000178
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000178
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-39.912, -12.293, 293.500, 0.000)
p4@checkpoint: (-0.005, -0.003, 0.029, 0.030)
raw GEN: n/a
raw SIM: SimTrack #1389 key=594609

gamma (22)
Particle 404
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000151
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000151
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.514, -82.632, 177.805, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1394 key=594692

gamma (22)
Particle 405
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (116.456, 40.507, -27.591, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1395 key=594699

e- (11)
Particle 406
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.015, -0.014, 0.080, 0.082)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.001136
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 1 dE=0.662213
subgraph tracker simHits: 1 dE=0.662213
raw GEN: n/a
raw SIM: SimTrack #1396 key=594703

Bremsstrahlung
Vertex 260
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.152, -1.814, 18.711, 0.951)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #498 key=419

Bremsstrahlung
Vertex 261
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.717, -1.650, 140.576, 5.172)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #500 key=421

gamma (22)
Particle 407
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, -0.002, 0.013, 0.014)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000625
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000625
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-116.395, 40.682, 265.684, 0.000)
p4@checkpoint: (-0.001, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1397 key=654326

gamma (22)
Particle 408
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.000, 0.007, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (38.041, 7.233, 293.500, 0.000)
p4@checkpoint: (0.002, 0.000, 0.007, 0.007)
raw GEN: n/a
raw SIM: SimTrack #1399 key=654344

gamma (22)
Particle 409
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.651, 1.421, -15.955, 16.103)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 258 simE=0.064089
direct recHits: 0 missing=258 recoE=0.000000
subgraph simHits: 258 simE=0.064089
subgraph recHits: 0 missing=258 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-30.477, 26.144, -293.500, 0.000)
p4@checkpoint: (-1.651, 1.421, -15.955, 16.103)
raw GEN: n/a
raw SIM: SimTrack #1402 key=654400

gamma (22)
Particle 410
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.036, 0.032, -0.352, 0.355)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.001242
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=0.001242
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-30.099, 26.571, -293.500, 0.000)
p4@checkpoint: (-0.036, 0.032, -0.352, 0.355)
raw GEN: n/a
raw SIM: SimTrack #1403 key=654401

gamma (22)
Particle 411
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.057, 0.057, -0.584, 0.589)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 26 simE=0.003654
direct recHits: 0 missing=26 recoE=0.000000
subgraph simHits: 26 simE=0.003654
subgraph recHits: 0 missing=26 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-29.387, 27.535, -293.500, 0.000)
p4@checkpoint: (-0.057, 0.057, -0.584, 0.589)
raw GEN: n/a
raw SIM: SimTrack #1405 key=654405

gamma (22)
Particle 412
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.965, 0.958, -9.890, 9.983)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 205 simE=0.034905
direct recHits: 0 missing=205 recoE=0.000000
subgraph simHits: 205 simE=0.034905
subgraph recHits: 0 missing=205 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-29.398, 27.541, -293.500, 0.000)
p4@checkpoint: (-0.965, 0.958, -9.890, 9.983)
raw GEN: n/a
raw SIM: SimTrack #1406 key=654406

e- (11)
Particle 413
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.173, 0.142, -0.317, 0.388)
nProdVtx: 1 nDecayVtx: 15
nParents: 1 nChildren: 15
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 17 simE=0.174183
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 2 dE=5.277207
raw GEN: n/a
raw SIM: SimTrack #1417 key=713231

Bremsstrahlung
Vertex 280
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.092, 73.608, -162.971, 7.017)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #528 key=449

Bremsstrahlung
Vertex 281
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.130, 73.639, -163.041, 7.020)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #530 key=451

Bremsstrahlung
Vertex 282
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.137, 73.645, -163.053, 7.020)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #531 key=452

Bremsstrahlung
Vertex 283
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.165, 73.668, -163.108, 7.023)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #532 key=453

Bremsstrahlung
Vertex 284
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.225, 73.715, -163.228, 7.027)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #533 key=454

Bremsstrahlung
Vertex 285
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.248, 73.733, -163.271, 7.029)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #534 key=455

Bremsstrahlung
Vertex 286
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.298, 73.772, -163.368, 7.033)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #535 key=456

Bremsstrahlung
Vertex 287
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.718, 74.067, -164.160, 7.064)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #536 key=457

Bremsstrahlung
Vertex 288
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.723, 74.070, -164.168, 7.065)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #537 key=458

Bremsstrahlung
Vertex 289
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.725, 74.072, -164.172, 7.065)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #538 key=459

Bremsstrahlung
Vertex 290
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.806, 74.126, -164.311, 7.071)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #539 key=460

Bremsstrahlung
Vertex 291
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.819, 74.135, -164.333, 7.072)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #540 key=461

Bremsstrahlung
Vertex 292
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.357, 74.480, -168.004, 7.222)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #541 key=462

Bremsstrahlung
Vertex 293
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.297, 72.051, -172.921, 7.420)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #542 key=463

Bremsstrahlung
Vertex 294
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.061, 69.799, -176.889, 7.580)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #545 key=466

e+ (-11)
Particle 414
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-2.866, 2.370, -5.268, 6.448)
nProdVtx: 1 nDecayVtx: 10
nParents: 1 nChildren: 10
nCheckpoints: 1
direct simHits: 30 simE=1.743142
direct recHits: 0 missing=30 recoE=0.000000
subgraph simHits: 69 simE=6.196846
subgraph recHits: 0 missing=69 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.816, 78.821, -174.003, 0.000)
p4@checkpoint: (-0.748, 0.742, -1.503, 1.835)
raw GEN: n/a
raw SIM: SimTrack #1418 key=713232

Bremsstrahlung
Vertex 267
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.170, 73.673, -163.114, 7.023)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #515 key=436

Bremsstrahlung
Vertex 268
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.182, 73.682, -163.136, 7.024)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #516 key=437

Bremsstrahlung
Vertex 269
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.186, 73.685, -163.142, 7.024)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #517 key=438

Bremsstrahlung
Vertex 270
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.217, 73.712, -163.201, 7.026)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #518 key=439

Bremsstrahlung
Vertex 271
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.285, 73.768, -163.326, 7.032)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #519 key=440

Bremsstrahlung
Vertex 272
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.688, 74.107, -164.075, 7.062)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #520 key=441

Bremsstrahlung
Vertex 273
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.756, 74.165, -164.201, 7.067)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #521 key=442

Bremsstrahlung
Vertex 274
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.734, 75.010, -166.044, 7.142)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #522 key=443

Bremsstrahlung
Vertex 275
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.742, 75.017, -166.060, 7.143)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #523 key=444

Bremsstrahlung
Vertex 278
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.244, 73.726, -163.233, 7.028)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #526 key=447

gamma (22)
Particle 415
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.051, 0.869, -1.932, 2.365)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 43 simE=2.291061
direct recHits: 0 missing=43 recoE=0.000000
subgraph simHits: 43 simE=2.291061
subgraph recHits: 0 missing=43 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.051, 78.538, -173.928, 0.000)
p4@checkpoint: (-1.051, 0.869, -1.932, 2.365)
raw GEN: n/a
raw SIM: SimTrack #1419 key=713234

gamma (22)
Particle 416
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.216, 0.179, -0.398, 0.487)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 17 simE=0.498417
direct recHits: 0 missing=17 recoE=0.000000
subgraph simHits: 18 simE=0.498952
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.048, 78.541, -173.923, 0.000)
p4@checkpoint: (-0.216, 0.179, -0.398, 0.487)
raw GEN: n/a
raw SIM: SimTrack #1420 key=713235

Annihilation

Vertex 279
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-70.115, 92.378, -139.234, 9.887)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #527 key=448

gamma (22)
Particle 417
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.199, 0.164, -0.365, 0.447)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.466660
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=0.466660
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.051, 78.538, -173.932, 0.000)
p4@checkpoint: (-0.199, 0.164, -0.365, 0.447)
raw GEN: n/a
raw SIM: SimTrack #1421 key=713236

gamma (22)
Particle 418
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.011, 0.009, -0.020, 0.024)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.033621
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.033621
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.042, 78.549, -173.925, 0.000)
p4@checkpoint: (-0.011, 0.009, -0.020, 0.024)
raw GEN: n/a
raw SIM: SimTrack #1422 key=713237

gamma (22)
Particle 419
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.055, 0.046, -0.101, 0.123)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.125308
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.125308
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.026, 78.568, -173.959, 0.000)
p4@checkpoint: (-0.055, 0.046, -0.101, 0.123)
raw GEN: n/a
raw SIM: SimTrack #1423 key=713240

gamma (22)
Particle 420
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.502, 0.423, -0.934, 1.142)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 39 simE=1.030322
subgraph recHits: 0 missing=39 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1424 key=713241

PairConversion

Vertex 276
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.503, 75.638, -167.455, 7.200)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #524 key=445

gamma (22)
Particle 421
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001137
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001137
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.991, 78.611, -173.997, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1425 key=713242

gamma (22)
Particle 422
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001727
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001727
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.939, 78.673, -173.977, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1426 key=713244

gamma (22)
Particle 423
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.002, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004815
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004815
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.936, 78.676, -173.970, 0.000)
p4@checkpoint: (-0.002, 0.002, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1427 key=713245

e- (11)
Particle 424
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.383, 0.324, -0.715, 0.873)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 19 simE=0.779609
direct recHits: 0 missing=19 recoE=0.000000
subgraph simHits: 19 simE=0.780219
subgraph recHits: 0 missing=19 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.150, 78.417, -173.992, 0.000)
p4@checkpoint: (-0.413, 0.282, -0.713, 0.871)
raw GEN: n/a
raw SIM: SimTrack #1428 key=713814

Annihilation

Vertex 277
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-95.261, 66.008, -192.808, 8.637)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #525 key=446

e+ (-11)
Particle 425
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.119, 0.100, -0.220, 0.269)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.250103
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.250103
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.510, 79.188, -174.020, 0.000)
p4@checkpoint: (-0.078, 0.133, -0.216, 0.266)
raw GEN: n/a
raw SIM: SimTrack #1429 key=713815

gamma (22)
Particle 426
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000610
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000610
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.556, 76.680, -194.992, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1430 key=714107

gamma (22)
Particle 427
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000099
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000099
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.716, 72.524, -170.944, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1432 key=714198

gamma (22)
Particle 428
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000535
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000535
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-68.360, 102.615, -140.488, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1433 key=714437

gamma (22)
Particle 429
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000142
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000142
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.103, 78.475, -173.886, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1434 key=715255

gamma (22)
Particle 430
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.021, 0.018, -0.040, 0.049)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.045029
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.045029
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.063, 78.523, -174.205, 0.000)
p4@checkpoint: (-0.021, 0.018, -0.040, 0.049)
raw GEN: n/a
raw SIM: SimTrack #1436 key=715257

gamma (22)
Particle 431
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.051, 0.042, -0.097, 0.117)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 7 simE=0.012271
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 2 dE=5.277207
raw GEN: n/a
raw SIM: SimTrack #1437 key=715258

PairConversion

Vertex 295
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.257, 73.744, -163.282, 7.030)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #553 key=474

Bremsstrahlung
Vertex 301
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.969, 72.799, -166.709, 7.158)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #561 key=482

Annihilation

Vertex 302
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.165, 73.483, -164.919, 7.114)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #562 key=483

gamma (22)
Particle 432
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.008, 0.006, -0.015, 0.018)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.020067
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.020067
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.122, 78.452, -174.549, 0.000)
p4@checkpoint: (-0.008, 0.006, -0.015, 0.018)
raw GEN: n/a
raw SIM: SimTrack #1438 key=715260

gamma (22)
Particle 433
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001656
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001656
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.140, 78.430, -174.932, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1439 key=715261

gamma (22)
Particle 434
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.017, 0.013, -0.033, 0.039)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.030622
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.030622
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.224, 78.328, -174.880, 0.000)
p4@checkpoint: (-0.017, 0.013, -0.033, 0.039)
raw GEN: n/a
raw SIM: SimTrack #1440 key=715262

gamma (22)
Particle 435
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000091
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000091
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.264, 78.279, -174.641, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1441 key=715264

gamma (22)
Particle 436
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.012, 0.008, -0.021, 0.025)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.019857
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.019857
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.483, 78.012, -174.404, 0.000)
p4@checkpoint: (-0.012, 0.008, -0.021, 0.025)
raw GEN: n/a
raw SIM: SimTrack #1442 key=715266

gamma (22)
Particle 437
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000400
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000400
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.563, 77.914, -174.360, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1443 key=715267

gamma (22)
Particle 438
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.001, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004146
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004146
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.559, 77.919, -174.295, 0.000)
p4@checkpoint: (-0.002, 0.001, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1444 key=715268

gamma (22)
Particle 439
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.002, -0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004951
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004951
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.515, 77.973, -174.090, 0.000)
p4@checkpoint: (-0.002, 0.002, -0.004, 0.005)
raw GEN: n/a
raw SIM: SimTrack #1445 key=715269

gamma (22)
Particle 440
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.022, 0.015, -0.037, 0.046)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.034387
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.034387
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.503, 77.988, -173.831, 0.000)
p4@checkpoint: (-0.022, 0.015, -0.037, 0.046)
raw GEN: n/a
raw SIM: SimTrack #1446 key=715270

gamma (22)
Particle 441
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000166
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000166
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.301, 79.436, -180.777, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1447 key=715276

gamma (22)
Particle 442
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000065
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000065
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.376, 75.637, -175.361, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1448 key=715278

gamma (22)
Particle 443
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000333
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000333
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.228, 16.225, -227.233, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1451 key=715288

e+ (-11)
Particle 444
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.038, 0.031, -0.073, 0.088)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 6 simE=0.012010
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1459 key=715361

Bremsstrahlung
Vertex 296
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.266, 73.751, -163.299, 7.030)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #554 key=475

Bremsstrahlung
Vertex 297
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.419, 75.417, -166.372, 7.154)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #555 key=476

Bremsstrahlung
Vertex 298
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-86.671, 78.302, -174.787, 7.495)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #556 key=477

Annihilation

Vertex 299
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-86.751, 77.898, -175.593, 7.549)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #558 key=479

Annihilation

Vertex 300
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.746, 74.932, -139.187, 9.264)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #559 key=480

gamma (22)
Particle 445
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.003, -0.007, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.005649
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.005649
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.084, 78.498, -174.768, 0.000)
p4@checkpoint: (-0.004, 0.003, -0.007, 0.009)
raw GEN: n/a
raw SIM: SimTrack #1460 key=715362

gamma (22)
Particle 446
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.003, -0.005, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.005142
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.005142
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.496, 81.531, -177.796, 0.000)
p4@checkpoint: (-0.001, 0.003, -0.005, 0.006)
raw GEN: n/a
raw SIM: SimTrack #1461 key=715366

gamma (22)
Particle 447
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000602
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000602
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (119.224, 31.440, -182.168, 0.000)
p4@checkpoint: (0.001, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1462 key=715379

gamma (22)
Particle 448
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.295, 79.444, -177.865, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1464 key=715383

gamma (22)
Particle 449
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000106
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000106
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-93.965, 79.834, -155.110, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1465 key=715442

gamma (22)
Particle 450
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 5 dE=5.277207
subgraph tracker simHits: 5 dE=5.277207
checkpointId: 0
x4@checkpoint: (-52.912, 17.364, -293.500, 0.000)
p4@checkpoint: (0.000, -0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1468 key=715460

gamma (22)
Particle 451
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000262
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000262
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.247, 79.500, -167.967, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1469 key=715475

e- (11)
Particle 452
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.588, 0.452, -0.964, 1.216)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 13 simE=0.317825
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 29 simE=1.156450
subgraph recHits: 0 missing=29 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-98.373, 74.335, -159.884, 0.000)
p4@checkpoint: (-0.204, 0.060, -0.270, 0.343)
raw GEN: n/a
raw SIM: SimTrack #1470 key=715501

Bremsstrahlung
Vertex 305
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.181, 71.161, -150.936, 6.712)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #566 key=487

Bremsstrahlung
Vertex 306
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.493, 72.091, -153.016, 6.799)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #567 key=488

Bremsstrahlung
Vertex 307
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.831, 72.310, -153.534, 6.821)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #568 key=489

gamma (22)
Particle 453
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000413
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000413
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-93.258, 80.659, -161.238, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1472 key=715516

gamma (22)
Particle 454
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000093
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000093
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-110.877, -53.938, -100.797, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1473 key=715517

gamma (22)
Particle 455
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.370, 0.279, -0.606, 0.763)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 17 simE=0.725769
direct recHits: 0 missing=17 recoE=0.000000
subgraph simHits: 17 simE=0.725769
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.646, 75.287, -159.885, 0.000)
p4@checkpoint: (-0.370, 0.279, -0.606, 0.763)
raw GEN: n/a
raw SIM: SimTrack #1474 key=715528

gamma (22)
Particle 456
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.014, 0.009, -0.021, 0.026)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.023006
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.023006
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.907, 74.948, -159.807, 0.000)
p4@checkpoint: (-0.014, 0.009, -0.021, 0.026)
raw GEN: n/a
raw SIM: SimTrack #1475 key=715529

gamma (22)
Particle 457
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.040, 0.025, -0.060, 0.076)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.089849
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.089849
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.934, 74.913, -159.720, 0.000)
p4@checkpoint: (-0.040, 0.025, -0.060, 0.076)
raw GEN: n/a
raw SIM: SimTrack #1476 key=715531

e- (11)
Particle 458
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-13.674, 11.324, -24.347, 30.133)
nProdVtx: 1 nDecayVtx: 18
nParents: 1 nChildren: 18
nCheckpoints: 1
direct simHits: 44 simE=4.065349
direct recHits: 0 missing=44 recoE=0.000000
subgraph simHits: 273 simE=28.218266
subgraph recHits: 0 missing=273 recoE=0.000000
direct tracker simHits: 55 dE=55.829781
subgraph tracker simHits: 79 dE=259.602448
checkpointId: 0
x4@checkpoint: (-107.439, 60.496, -170.238, 0.000)
p4@checkpoint: (-2.340, 0.660, -3.344, 4.134)
raw GEN: n/a
raw SIM: SimTrack #1479 key=715871

Bremsstrahlung
Vertex 333
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-11.191, 9.210, -19.322, 1.158)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #599 key=520

Ionisation

Vertex 334
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-13.146, 10.811, -22.785, 1.301)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #600 key=521

Bremsstrahlung
Vertex 335
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-13.198, 10.853, -22.876, 1.305)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #601 key=522

Bremsstrahlung
Vertex 336
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-16.459, 13.473, -28.633, 1.542)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #602 key=523

Bremsstrahlung
Vertex 337
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.042, 17.019, -36.605, 1.871)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #603 key=524

Bremsstrahlung
Vertex 338
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.558, 17.410, -37.496, 1.908)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #604 key=525

Bremsstrahlung
Vertex 339
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-75.107, 48.803, -123.064, 5.437)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #606 key=527

Bremsstrahlung
Vertex 340
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-82.067, 51.727, -133.429, 5.865)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #607 key=528

Bremsstrahlung
Vertex 341
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-99.813, 58.192, -159.288, 6.933)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #608 key=529

Bremsstrahlung
Vertex 342
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-99.933, 58.231, -159.461, 6.940)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #609 key=530

Bremsstrahlung
Vertex 343
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-99.938, 58.233, -159.468, 6.940)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #610 key=531

Bremsstrahlung
Vertex 344
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-99.994, 58.251, -159.548, 6.944)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #611 key=532

Bremsstrahlung
Vertex 345
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-100.028, 58.262, -159.597, 6.946)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #612 key=533

Bremsstrahlung
Vertex 346
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-100.033, 58.264, -159.605, 6.946)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #613 key=534

Bremsstrahlung
Vertex 347
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-100.596, 58.442, -160.415, 6.979)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #614 key=535

Bremsstrahlung
Vertex 348
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-101.911, 58.858, -162.311, 7.058)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #615 key=536

Bremsstrahlung
Vertex 349
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-101.935, 58.866, -162.346, 7.059)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #616 key=537

Bremsstrahlung
Vertex 350
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-101.946, 58.869, -162.361, 7.060)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #617 key=538

e+ (-11)
Particle 459
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-3.419, 2.829, -6.083, 7.530)
nProdVtx: 1 nDecayVtx: 22
nParents: 1 nChildren: 24
nCheckpoints: 1
direct simHits: 28 simE=2.285794
direct recHits: 0 missing=28 recoE=0.000000
subgraph simHits: 94 simE=7.115697
subgraph recHits: 0 missing=94 recoE=0.000000
direct tracker simHits: 31 dE=30.149736
subgraph tracker simHits: 37 dE=67.455978
checkpointId: 0
x4@checkpoint: (-76.903, 96.379, -169.822, 0.000)
p4@checkpoint: (-0.541, 1.324, -1.919, 2.393)
raw GEN: n/a
raw SIM: SimTrack #1480 key=715872

Bremsstrahlung
Vertex 309
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-11.135, 9.190, -19.243, 1.155)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #571 key=492

Bremsstrahlung
Vertex 310
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-11.169, 9.219, -19.304, 1.158)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #572 key=493

Bremsstrahlung
Vertex 311
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-16.044, 13.422, -28.122, 1.522)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #573 key=494

Bremsstrahlung
Vertex 312
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-28.251, 24.949, -51.097, 2.471)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #574 key=495

Bremsstrahlung
Vertex 313
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-58.068, 61.619, -115.876, 5.147)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #576 key=497

Bremsstrahlung
Vertex 314
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-72.631, 86.978, -155.903, 6.801)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #577 key=498

Bremsstrahlung
Vertex 315
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-73.660, 89.093, -159.112, 6.934)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #578 key=499

Bremsstrahlung
Vertex 316
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-73.667, 89.106, -159.133, 6.935)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #579 key=500

Bremsstrahlung
Vertex 317
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-73.685, 89.145, -159.191, 6.937)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #580 key=501

Bremsstrahlung
Vertex 318
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-73.765, 89.312, -159.442, 6.948)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #581 key=502

Bremsstrahlung
Vertex 319
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-73.817, 89.421, -159.606, 6.954)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #582 key=503

Bremsstrahlung
Vertex 320
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-74.041, 89.892, -160.308, 6.984)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #583 key=504

Bremsstrahlung
Vertex 321
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-74.130, 90.082, -160.591, 6.995)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #584 key=505

Bremsstrahlung
Vertex 322
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-74.131, 90.084, -160.594, 6.995)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #585 key=506

Bremsstrahlung
Vertex 323
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-74.619, 91.127, -162.140, 7.060)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #586 key=507

Bremsstrahlung
Vertex 324
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-74.638, 91.169, -162.202, 7.062)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #587 key=508

Bremsstrahlung
Vertex 325
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-75.223, 92.455, -164.094, 7.141)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #588 key=509

Bremsstrahlung
Vertex 326
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-75.273, 92.567, -164.258, 7.148)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #589 key=510

Bremsstrahlung
Vertex 327
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-74.079, 90.184, -160.408, 6.994)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #590 key=511

Annihilation

Vertex 328
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-73.829, 90.412, -160.572, 7.006)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #591 key=512

Annihilation

Vertex 329
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-74.104, 90.384, -161.429, 7.035)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #592 key=513

Bremsstrahlung
Vertex 330
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-73.976, 89.960, -160.193, 6.982)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #594 key=515

gamma (22)
Particle 460
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.462, 0.390, -0.829, 1.026)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 17 simE=1.012605
direct recHits: 0 missing=17 recoE=0.000000
subgraph simHits: 17 simE=1.012605
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.357, 79.371, -168.539, 0.000)
p4@checkpoint: (-0.462, 0.390, -0.829, 1.026)
raw GEN: n/a
raw SIM: SimTrack #1481 key=715873

gamma (22)
Particle 461
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.789, 0.665, -1.415, 1.751)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 31 simE=1.701162
direct recHits: 0 missing=31 recoE=0.000000
subgraph simHits: 31 simE=1.701162
subgraph recHits: 0 missing=31 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.361, 79.366, -168.499, 0.000)
p4@checkpoint: (-0.789, 0.665, -1.415, 1.751)
raw GEN: n/a
raw SIM: SimTrack #1482 key=715875

gamma (22)
Particle 462
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.003, -0.005, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.004395
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.004395
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.789, 81.198, -168.247, 0.000)
p4@checkpoint: (-0.003, 0.003, -0.005, 0.006)
raw GEN: n/a
raw SIM: SimTrack #1483 key=715876

gamma (22)
Particle 463
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.060, 0.060, -0.116, 0.144)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.153103
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.153103
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.604, 85.745, -168.477, 0.000)
p4@checkpoint: (-0.060, 0.060, -0.116, 0.144)
raw GEN: n/a
raw SIM: SimTrack #1484 key=715881

gamma (22)
Particle 464
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.033, 0.049, -0.081, 0.100)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.005376
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 38 dE=33.777889
subgraph tracker simHits: 18 dE=37.306248
raw GEN: n/a
raw SIM: SimTrack #1486 key=715885

PairConversion

Vertex 331
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-59.905, 64.400, -120.429, 5.336)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #595 key=516

gamma (22)
Particle 465
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.009, -0.014, 0.017)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.014881
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.014881
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.166, 96.168, -169.899, 0.000)
p4@checkpoint: (-0.004, 0.009, -0.014, 0.017)
raw GEN: n/a
raw SIM: SimTrack #1487 key=715886

gamma (22)
Particle 466
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.008, 0.018, -0.027, 0.033)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.029064
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.029064
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.086, 96.232, -169.905, 0.000)
p4@checkpoint: (-0.008, 0.018, -0.027, 0.033)
raw GEN: n/a
raw SIM: SimTrack #1488 key=715888

gamma (22)
Particle 467
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.047, 0.098, -0.148, 0.184)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.185206
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.185206
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.084, 96.234, -169.901, 0.000)
p4@checkpoint: (-0.047, 0.098, -0.148, 0.184)
raw GEN: n/a
raw SIM: SimTrack #1489 key=715889

gamma (22)
Particle 468
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000365
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000365
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.076, 96.240, -169.913, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1490 key=715890

gamma (22)
Particle 469
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000350
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000350
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-80.265, 93.597, -157.943, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1491 key=715892

gamma (22)
Particle 470
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.028, 0.059, -0.089, 0.110)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 12 simE=0.108978
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 12 simE=0.108978
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.071, 96.244, -169.788, 0.000)
p4@checkpoint: (-0.028, 0.059, -0.089, 0.110)
raw GEN: n/a
raw SIM: SimTrack #1492 key=715893

gamma (22)
Particle 471
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.107, 0.226, -0.336, 0.419)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.420156
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.420156
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.056, 96.257, -169.795, 0.000)
p4@checkpoint: (-0.107, 0.226, -0.336, 0.419)
raw GEN: n/a
raw SIM: SimTrack #1493 key=715894

gamma (22)
Particle 472
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.002, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.004466
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.004466
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.040, 96.269, -169.794, 0.000)
p4@checkpoint: (-0.001, 0.002, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1494 key=715895

gamma (22)
Particle 473
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000309
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000309
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.039, 96.270, -169.791, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1495 key=715896

gamma (22)
Particle 474
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.312, 0.673, -0.998, 1.244)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=1.177170
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=1.177170
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.014, 96.290, -169.795, 0.000)
p4@checkpoint: (-0.312, 0.673, -0.998, 1.244)
raw GEN: n/a
raw SIM: SimTrack #1496 key=715897

gamma (22)
Particle 475
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000292
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000292
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.003, 96.299, -169.766, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1497 key=715898

gamma (22)
Particle 476
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000920
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000920
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-76.960, 96.333, -169.789, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1498 key=715900

gamma (22)
Particle 477
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.002, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.003271
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.003271
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-76.958, 96.335, -169.799, 0.000)
p4@checkpoint: (-0.001, 0.002, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1499 key=715901

gamma (22)
Particle 478
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000132
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000132
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-70.520, 101.143, -170.687, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1500 key=717237

gamma (22)
Particle 479
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.003, -0.001, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003181
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003181
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-47.977, 113.583, -167.246, 0.000)
p4@checkpoint: (0.003, 0.003, -0.001, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1501 key=717240

gamma (22)
Particle 480
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.003548
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.003548
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (113.374, 48.469, -271.711, 0.000)
p4@checkpoint: (0.001, -0.000, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1502 key=717241

gamma (22)
Particle 481
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000166
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000166
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-65.151, 104.681, -143.663, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1503 key=717249

gamma (22)
Particle 482
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000168
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000168
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-65.539, 33.383, -293.500, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1504 key=717250

gamma (22)
Particle 483
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000641
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000641
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-76.415, 96.766, -164.419, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1506 key=717261

e+ (-11)
Particle 484
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.024, 0.037, -0.060, 0.074)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 1 simE=0.000173
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 2 simE=0.005376
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 7 dE=3.528353
subgraph tracker simHits: 5 dE=3.528353
checkpointId: 0
x4@checkpoint: (-60.933, 69.791, -293.500, 0.000)
p4@checkpoint: (0.010, 0.030, -0.043, 0.053)
raw GEN: n/a
raw SIM: SimTrack #1507 key=717351

Bremsstrahlung
Vertex 332
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-60.499, 66.115, -123.061, 5.443)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #596 key=517

gamma (22)
Particle 485
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.004, -0.006, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.005202
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.005202
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-61.435, 106.905, -180.322, 0.000)
p4@checkpoint: (0.001, 0.004, -0.005, 0.007)
raw GEN: n/a
raw SIM: SimTrack #1508 key=717352

gamma (22)
Particle 486
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-10.836, 8.927, -19.253, 23.828)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 199 simE=22.496471
subgraph recHits: 0 missing=199 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 28 dE=82.807076
raw GEN: n/a
raw SIM: SimTrack #1511 key=718355

PairConversion

Vertex 359
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-12.020, 9.893, -20.794, 1.219)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #627 key=548

e- (11)
Particle 487
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.133, 0.094, -0.202, 0.260)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 14 simE=0.241870
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 65 dE=66.702370
subgraph tracker simHits: 20 dE=66.702370
raw GEN: n/a
raw SIM: SimTrack #1512 key=718357

Bremsstrahlung
Vertex 356
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-23.317, 13.276, -36.356, 1.879)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #624 key=545

Bremsstrahlung
Vertex 357
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-22.019, 9.193, -45.300, 2.259)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #625 key=546

Bremsstrahlung
Vertex 358
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.553, 10.681, -72.682, 3.973)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #626 key=547

gamma (22)
Particle 488
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.214, 0.175, -0.381, 0.471)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 19 simE=0.038690
subgraph recHits: 0 missing=19 recoE=0.000000
direct tracker simHits: 28 dE=26.538025
subgraph tracker simHits: 21 dE=54.263226
raw GEN: n/a
raw SIM: SimTrack #1513 key=718358

PairConversion

Vertex 351
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-65.628, 53.671, -116.065, 5.147)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #619 key=540

gamma (22)
Particle 489
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000592
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000592
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.549, 71.363, -172.355, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1514 key=718359

gamma (22)
Particle 490
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.156, 0.118, -0.270, 0.333)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.318936
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.318936
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.685, 75.236, -169.162, 0.000)
p4@checkpoint: (-0.156, 0.118, -0.270, 0.333)
raw GEN: n/a
raw SIM: SimTrack #1515 key=718361

gamma (22)
Particle 491
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.055, 0.041, -0.095, 0.117)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.107959
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.107959
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.821, 75.060, -169.031, 0.000)
p4@checkpoint: (-0.055, 0.041, -0.095, 0.117)
raw GEN: n/a
raw SIM: SimTrack #1516 key=718362

gamma (22)
Particle 492
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, 0.003, -0.011, 0.014)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.008580
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.008580
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-106.330, 62.424, -169.862, 0.000)
p4@checkpoint: (-0.007, 0.003, -0.011, 0.014)
raw GEN: n/a
raw SIM: SimTrack #1518 key=718366

gamma (22)
Particle 493
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001607
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001607
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-106.755, 61.695, -169.960, 0.000)
p4@checkpoint: (-0.001, 0.000, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1519 key=718367

gamma (22)
Particle 494
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.376, 0.123, -0.543, 0.672)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 23 simE=0.726236
direct recHits: 0 missing=23 recoE=0.000000
subgraph simHits: 23 simE=0.726236
subgraph recHits: 0 missing=23 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.351, 60.651, -170.171, 0.000)
p4@checkpoint: (-0.376, 0.123, -0.543, 0.672)
raw GEN: n/a
raw SIM: SimTrack #1520 key=718371

gamma (22)
Particle 495
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.012, 0.004, -0.018, 0.022)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.021669
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.021669
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.365, 60.627, -170.176, 0.000)
p4@checkpoint: (-0.012, 0.004, -0.018, 0.022)
raw GEN: n/a
raw SIM: SimTrack #1521 key=718372

gamma (22)
Particle 496
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001080
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001080
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.365, 60.627, -170.174, 0.000)
p4@checkpoint: (-0.001, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1522 key=718373

gamma (22)
Particle 497
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.011, 0.004, -0.016, 0.020)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.014498
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.014498
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.362, 60.632, -170.167, 0.000)
p4@checkpoint: (-0.011, 0.004, -0.016, 0.020)
raw GEN: n/a
raw SIM: SimTrack #1523 key=718374

gamma (22)
Particle 498
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.001, -0.006, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.007967
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.007967
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.364, 60.628, -170.151, 0.000)
p4@checkpoint: (-0.004, 0.001, -0.006, 0.007)
raw GEN: n/a
raw SIM: SimTrack #1524 key=718375

gamma (22)
Particle 499
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, 0.002, -0.010, 0.013)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.008869
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.008869
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.365, 60.628, -170.155, 0.000)
p4@checkpoint: (-0.007, 0.002, -0.010, 0.013)
raw GEN: n/a
raw SIM: SimTrack #1525 key=718376

gamma (22)
Particle 500
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.001, -0.006, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.007461
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.007461
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.388, 60.586, -170.204, 0.000)
p4@checkpoint: (-0.004, 0.001, -0.006, 0.008)
raw GEN: n/a
raw SIM: SimTrack #1526 key=718377

gamma (22)
Particle 501
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.057, 0.018, -0.082, 0.101)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.089826
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.089826
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.397, 60.570, -170.207, 0.000)
p4@checkpoint: (-0.057, 0.018, -0.082, 0.101)
raw GEN: n/a
raw SIM: SimTrack #1527 key=718380

gamma (22)
Particle 502
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.009, 0.003, -0.013, 0.016)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.018255
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.018255
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.398, 60.569, -170.200, 0.000)
p4@checkpoint: (-0.009, 0.003, -0.013, 0.016)
raw GEN: n/a
raw SIM: SimTrack #1528 key=718381

gamma (22)
Particle 503
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.017, 0.005, -0.024, 0.030)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.042348
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.042348
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.396, 60.572, -170.212, 0.000)
p4@checkpoint: (-0.017, 0.005, -0.024, 0.030)
raw GEN: n/a
raw SIM: SimTrack #1529 key=718382

e- (11)
Particle 504
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.173, 0.141, -0.306, 0.379)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 10 simE=0.002126
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 12 simE=0.002521
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 25 dE=24.745125
subgraph tracker simHits: 11 dE=24.745121
checkpointId: 0
x4@checkpoint: (-70.160, 57.383, -293.500, 0.000)
p4@checkpoint: (-0.193, 0.088, -0.293, 0.362)
raw GEN: n/a
raw SIM: SimTrack #1531 key=720102

Bremsstrahlung
Vertex 355
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-58.734, 41.774, -265.503, 11.245)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #623 key=544

e+ (-11)
Particle 505
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.041, 0.034, -0.074, 0.092)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 7 simE=0.036169
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 5 dE=2.980082
subgraph tracker simHits: 3 dE=2.980082
raw GEN: n/a
raw SIM: SimTrack #1532 key=720103

Bremsstrahlung
Vertex 352
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-65.632, 53.674, -116.071, 5.147)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #620 key=541

Bremsstrahlung
Vertex 353
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-65.795, 57.854, -123.106, 5.431)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #621 key=542

Annihilation

Vertex 354
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-66.738, 58.316, -262.598, 17.160)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #622 key=543

gamma (22)
Particle 506
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.017, 0.014, -0.031, 0.038)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.034312
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.034312
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.956, 78.652, -169.680, 0.000)
p4@checkpoint: (-0.017, 0.014, -0.031, 0.038)
raw GEN: n/a
raw SIM: SimTrack #1533 key=720104

gamma (22)
Particle 507
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.014, 0.012, -0.029, 0.034)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001346
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001346
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (9.334, 122.946, -284.096, 0.000)
p4@checkpoint: (0.014, 0.012, -0.029, 0.034)
raw GEN: n/a
raw SIM: SimTrack #1534 key=720105

gamma (22)
Particle 508
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.119, -79.652, -195.873, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1535 key=720112

gamma (22)
Particle 509
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.002, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000395
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000395
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-61.061, 61.763, -293.500, 0.000)
p4@checkpoint: (-0.000, 0.002, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1537 key=720176

gamma (22)
Particle 510
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.131, -0.020, -0.169, 0.215)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 11 simE=0.229887
direct recHits: 0 missing=11 recoE=0.000000
subgraph simHits: 11 simE=0.229887
subgraph recHits: 0 missing=11 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-123.284, -1.989, -164.610, 0.000)
p4@checkpoint: (-0.131, -0.020, -0.169, 0.215)
raw GEN: n/a
raw SIM: SimTrack #1538 key=720658

gamma (22)
Particle 511
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, 0.002, -0.005, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.007042
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.007042
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (108.827, 57.961, -208.747, 0.000)
p4@checkpoint: (0.004, 0.002, -0.005, 0.007)
raw GEN: n/a
raw SIM: SimTrack #1539 key=720661

gamma (22)
Particle 512
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.005, -0.003, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004940
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004940
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-2.035, 123.283, -127.366, 0.000)
p4@checkpoint: (0.001, 0.005, -0.003, 0.006)
raw GEN: n/a
raw SIM: SimTrack #1540 key=720663

e- (11)
Particle 513
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-5.472, 4.507, -9.722, 12.032)
nProdVtx: 1 nDecayVtx: 15
nParents: 1 nChildren: 17
nCheckpoints: 1
direct simHits: 22 simE=1.419927
direct recHits: 0 missing=22 recoE=0.000000
subgraph simHits: 121 simE=11.187084
subgraph recHits: 0 missing=121 recoE=0.000000
direct tracker simHits: 27 dE=26.966602
subgraph tracker simHits: 20 dE=41.777466
checkpointId: 0
x4@checkpoint: (-101.159, 70.496, -169.217, 0.000)
p4@checkpoint: (-0.772, 0.335, -1.172, 1.442)
raw GEN: n/a
raw SIM: SimTrack #1541 key=720759

Bremsstrahlung
Vertex 403
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-31.960, 25.634, -55.654, 2.658)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #675 key=596

Bremsstrahlung
Vertex 404
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-67.779, 50.796, -115.778, 5.139)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #676 key=597

Bremsstrahlung
Vertex 405
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-70.820, 52.765, -120.754, 5.344)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #677 key=598

Bremsstrahlung
Vertex 406
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-72.278, 53.701, -123.135, 5.442)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #678 key=599

Bremsstrahlung
Vertex 407
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.166, 65.873, -155.189, 6.765)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #679 key=600

Bremsstrahlung
Vertex 408
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.219, 65.904, -155.273, 6.768)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #680 key=601

Bremsstrahlung
Vertex 409
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.118, 67.007, -158.291, 6.893)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #681 key=602

Bremsstrahlung
Vertex 410
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.175, 67.040, -158.381, 6.896)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #682 key=603

Bremsstrahlung
Vertex 411
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.201, 67.055, -158.424, 6.898)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #683 key=604

Bremsstrahlung
Vertex 412
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.213, 67.062, -158.442, 6.899)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #684 key=605

Bremsstrahlung
Vertex 413
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.399, 67.167, -158.734, 6.911)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #685 key=606

Bremsstrahlung
Vertex 414
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-96.066, 68.062, -161.357, 7.019)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #686 key=607

Annihilation

Vertex 415
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-111.045, 32.415, -204.697, 9.571)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #687 key=608

Bremsstrahlung
Vertex 416
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-109.466, 39.360, -200.266, 9.291)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #688 key=609

Annihilation

Vertex 473
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.269, 69.553, -158.096, 6.885)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #768 key=689

e+ (-11)
Particle 514
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-5.364, 4.420, -9.530, 11.796)
nProdVtx: 1 nDecayVtx: 21
nParents: 1 nChildren: 22
nCheckpoints: 1
direct simHits: 38 simE=4.901062
direct recHits: 0 missing=38 recoE=0.000000
subgraph simHits: 105 simE=11.309379
subgraph recHits: 0 missing=105 recoE=0.000000
direct tracker simHits: 30 dE=29.615206
subgraph tracker simHits: 18 dE=41.029621
checkpointId: 0
x4@checkpoint: (-87.725, 86.644, -168.936, 0.000)
p4@checkpoint: (-1.803, 2.381, -4.115, 5.085)
raw GEN: n/a
raw SIM: SimTrack #1542 key=720760

Bremsstrahlung
Vertex 360
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-16.228, 13.394, -28.297, 1.529)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #628 key=549

Bremsstrahlung
Vertex 361
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.451, 17.832, -37.686, 1.917)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #629 key=550

Bremsstrahlung
Vertex 362
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-40.256, 34.784, -72.389, 3.350)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #630 key=551

Bremsstrahlung
Vertex 363
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-40.278, 34.805, -72.431, 3.351)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #631 key=552

Bremsstrahlung
Vertex 364
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-43.219, 37.604, -77.996, 3.581)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #632 key=553

Bremsstrahlung
Vertex 365
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-64.880, 59.664, -120.380, 5.331)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #633 key=554

Bremsstrahlung
Vertex 366
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-64.885, 59.669, -120.390, 5.332)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #634 key=555

Bremsstrahlung
Vertex 367
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-82.996, 80.587, -158.347, 6.898)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #635 key=556

Bremsstrahlung
Vertex 368
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.051, 80.656, -158.468, 6.903)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #636 key=557

Bremsstrahlung
Vertex 369
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.112, 80.731, -158.600, 6.909)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #637 key=558

Bremsstrahlung
Vertex 370
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.120, 80.741, -158.618, 6.910)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #638 key=559

Bremsstrahlung
Vertex 371
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.145, 80.772, -158.672, 6.912)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #639 key=560

Bremsstrahlung
Vertex 372
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.644, 81.391, -159.769, 6.957)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #640 key=561

Bremsstrahlung
Vertex 373
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.649, 81.396, -159.779, 6.957)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #641 key=562

Bremsstrahlung
Vertex 374
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.729, 81.497, -159.957, 6.965)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #642 key=563

Bremsstrahlung
Vertex 375
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.317, 82.237, -161.261, 7.018)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #643 key=564

Bremsstrahlung
Vertex 376
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.724, 82.753, -162.166, 7.056)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #644 key=565

Annihilation

Vertex 377
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-76.995, 86.200, -198.534, 8.785)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #645 key=566

Bremsstrahlung
Vertex 378
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-76.968, 86.194, -198.567, 8.786)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #646 key=567

Bremsstrahlung
Vertex 394
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.600, 79.364, -158.394, 6.905)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #664 key=585

Bremsstrahlung
Vertex 395
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.236, 80.750, -159.757, 6.991)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #666 key=587

gamma (22)
Particle 515
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.459, 0.385, -0.821, 1.016)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 19 simE=1.008669
direct recHits: 0 missing=19 recoE=0.000000
subgraph simHits: 19 simE=1.008669
subgraph recHits: 0 missing=19 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.551, 79.139, -168.324, 0.000)
p4@checkpoint: (-0.459, 0.385, -0.821, 1.016)
raw GEN: n/a
raw SIM: SimTrack #1543 key=720761

gamma (22)
Particle 516
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.150, 0.129, -0.271, 0.335)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.320036
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=0.320036
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-93.805, 80.022, -168.448, 0.000)
p4@checkpoint: (-0.150, 0.129, -0.271, 0.335)
raw GEN: n/a
raw SIM: SimTrack #1544 key=720766

gamma (22)
Particle 517
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=1.060141
subgraph tracker simHits: 1 dE=1.060141
checkpointId: 0
x4@checkpoint: (-24.025, 33.135, -293.500, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1545 key=720768

gamma (22)
Particle 518
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.170, 0.160, -0.320, 0.396)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 18 simE=0.326568
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=2.041312
raw GEN: n/a
raw SIM: SimTrack #1546 key=720769

PairConversion

Vertex 396
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-85.854, 77.823, -158.348, 6.899)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #667 key=588

gamma (22)
Particle 519
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.008, 0.008, -0.016, 0.019)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.020229
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.020229
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-90.882, 83.327, -168.530, 0.000)
p4@checkpoint: (-0.008, 0.008, -0.016, 0.019)
raw GEN: n/a
raw SIM: SimTrack #1547 key=720770

gamma (22)
Particle 520
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.040, 0.043, -0.080, 0.099)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.098958
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.098958
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.808, 85.534, -168.659, 0.000)
p4@checkpoint: (-0.040, 0.043, -0.080, 0.099)
raw GEN: n/a
raw SIM: SimTrack #1548 key=720775

gamma (22)
Particle 521
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.783, 0.847, -1.579, 1.955)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 39 simE=1.898807
subgraph recHits: 0 missing=39 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1549 key=720776

PairConversion

Vertex 380
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-82.979, 79.235, -156.891, 6.839)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #650 key=571

gamma (22)
Particle 522
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.081, 0.100, -0.176, 0.218)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 11 simE=0.216561
direct recHits: 0 missing=11 recoE=0.000000
subgraph simHits: 11 simE=0.216561
subgraph recHits: 0 missing=11 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-87.817, 86.551, -168.870, 0.000)
p4@checkpoint: (-0.081, 0.100, -0.176, 0.218)
raw GEN: n/a
raw SIM: SimTrack #1550 key=720780

gamma (22)
Particle 523
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.007, 0.008, -0.014, 0.018)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000147
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (74.583, 81.658, -293.500, 0.000)
p4@checkpoint: (0.001, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1551 key=720781

Bremsstrahlung
Vertex 379
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.294, 82.216, -166.116, 7.282)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #648 key=569

gamma (22)
Particle 524
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.018, 0.022, -0.039, 0.049)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.041233
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.041233
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-87.826, 86.542, -168.885, 0.000)
p4@checkpoint: (-0.018, 0.022, -0.039, 0.049)
raw GEN: n/a
raw SIM: SimTrack #1552 key=720782

gamma (22)
Particle 525
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.839, 1.033, -1.829, 2.262)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 26 simE=2.227546
direct recHits: 0 missing=26 recoE=0.000000
subgraph simHits: 26 simE=2.227546
subgraph recHits: 0 missing=26 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-87.828, 86.539, -168.887, 0.000)
p4@checkpoint: (-0.839, 1.033, -1.829, 2.262)
raw GEN: n/a
raw SIM: SimTrack #1553 key=720783

gamma (22)
Particle 526
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.045, 0.056, -0.098, 0.122)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.105626
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.105626
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-87.820, 86.548, -168.892, 0.000)
p4@checkpoint: (-0.045, 0.056, -0.098, 0.122)
raw GEN: n/a
raw SIM: SimTrack #1554 key=720784

gamma (22)
Particle 527
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000661
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000661
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-87.797, 86.571, -168.925, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1555 key=720788

gamma (22)
Particle 528
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.006, 0.008, -0.013, 0.017)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.012209
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.012209
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-87.797, 86.571, -168.928, 0.000)
p4@checkpoint: (-0.006, 0.008, -0.013, 0.017)
raw GEN: n/a
raw SIM: SimTrack #1556 key=720789

gamma (22)
Particle 529
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001296
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001296
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-87.783, 86.586, -168.932, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1557 key=720791

gamma (22)
Particle 530
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.042, 0.054, -0.094, 0.117)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.127796
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.127796
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-87.763, 86.605, -168.932, 0.000)
p4@checkpoint: (-0.042, 0.054, -0.094, 0.117)
raw GEN: n/a
raw SIM: SimTrack #1558 key=720792

gamma (22)
Particle 531
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000136
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000136
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-87.756, 86.612, -168.921, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1559 key=720793

gamma (22)
Particle 532
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000143
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000143
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (64.396, -105.148, -236.680, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1560 key=721884

gamma (22)
Particle 533
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000364
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000364
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-75.209, 97.706, -203.016, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1561 key=721885

gamma (22)
Particle 534
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, -0.002, -0.004, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000241
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000241
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 8 dE=8.312955
subgraph tracker simHits: 8 dE=8.312955
checkpointId: 0
x4@checkpoint: (24.652, 67.929, -293.500, 0.000)
p4@checkpoint: (0.003, -0.000, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1562 key=721889

gamma (22)
Particle 535
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000147
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000147
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-84.533, 89.762, -165.961, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1564 key=723174

e- (11)
Particle 536
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.164, 0.178, -0.332, 0.410)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 12 simE=0.411366
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1566 key=723249

Bremsstrahlung
Vertex 390
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.630, 79.912, -158.168, 6.892)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #660 key=581

Bremsstrahlung
Vertex 391
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.806, 80.078, -158.482, 6.905)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #661 key=582

Bremsstrahlung
Vertex 392
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.816, 80.088, -158.500, 6.906)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #662 key=583

Bremsstrahlung
Vertex 393
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.273, 80.470, -159.305, 6.939)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #663 key=584

e+ (-11)
Particle 537
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.619, 0.669, -1.248, 1.545)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 9
nCheckpoints: 1
direct simHits: 25 simE=0.932427
direct recHits: 0 missing=25 recoE=0.000000
subgraph simHits: 37 simE=1.487441
subgraph recHits: 0 missing=37 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.311, 86.046, -168.679, 0.000)
p4@checkpoint: (-0.292, 0.448, -0.729, 0.904)
raw GEN: n/a
raw SIM: SimTrack #1567 key=723250

Bremsstrahlung
Vertex 381
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.613, 79.927, -158.177, 6.892)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #651 key=572

Bremsstrahlung
Vertex 382
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.615, 79.930, -158.182, 6.892)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #652 key=573

Bremsstrahlung
Vertex 383
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.667, 79.988, -158.289, 6.897)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #653 key=574

Bremsstrahlung
Vertex 384
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.677, 79.999, -158.309, 6.898)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #654 key=575

Bremsstrahlung
Vertex 385
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.769, 80.101, -158.498, 6.905)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #655 key=576

Bremsstrahlung
Vertex 386
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.810, 80.147, -158.583, 6.909)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #656 key=577

Bremsstrahlung
Vertex 387
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.813, 80.151, -158.590, 6.909)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #657 key=578

Bremsstrahlung
Vertex 388
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-83.820, 80.159, -158.604, 6.910)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #658 key=579

Bremsstrahlung
Vertex 389
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.186, 80.576, -159.355, 6.941)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #659 key=580

gamma (22)
Particle 538
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.167, 0.185, -0.341, 0.422)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.411389
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.411389
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.747, 85.597, -168.651, 0.000)
p4@checkpoint: (-0.167, 0.185, -0.341, 0.422)
raw GEN: n/a
raw SIM: SimTrack #1568 key=723251

gamma (22)
Particle 539
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.004, -0.007, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.007847
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.007847
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.759, 85.585, -168.634, 0.000)
p4@checkpoint: (-0.004, 0.004, -0.007, 0.009)
raw GEN: n/a
raw SIM: SimTrack #1569 key=723252

gamma (22)
Particle 540
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.026, 0.029, -0.053, 0.066)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.074826
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.074826
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.721, 85.624, -168.648, 0.000)
p4@checkpoint: (-0.026, 0.029, -0.053, 0.066)
raw GEN: n/a
raw SIM: SimTrack #1570 key=723253

gamma (22)
Particle 541
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.024, 0.026, -0.048, 0.060)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.009185
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.009185
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.702, 85.643, -168.617, 0.000)
p4@checkpoint: (-0.024, 0.026, -0.048, 0.060)
raw GEN: n/a
raw SIM: SimTrack #1571 key=723254

gamma (22)
Particle 542
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, 0.006, -0.010, 0.013)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.007031
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.007031
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.721, 85.624, -168.663, 0.000)
p4@checkpoint: (-0.005, 0.006, -0.010, 0.013)
raw GEN: n/a
raw SIM: SimTrack #1572 key=723255

gamma (22)
Particle 543
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.002, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003165
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003165
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.710, 85.635, -168.700, 0.000)
p4@checkpoint: (-0.001, 0.002, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1573 key=723257

gamma (22)
Particle 544
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000190
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000190
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-81.982, 92.097, -188.934, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1574 key=723259

gamma (22)
Particle 545
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.017, 0.020, -0.036, 0.045)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.040145
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.040145
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.662, 85.685, -168.623, 0.000)
p4@checkpoint: (-0.017, 0.020, -0.036, 0.045)
raw GEN: n/a
raw SIM: SimTrack #1575 key=723260

gamma (22)
Particle 546
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.002, -0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001237
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001237
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.645, 85.702, -168.537, 0.000)
p4@checkpoint: (-0.002, 0.002, -0.004, 0.005)
raw GEN: n/a
raw SIM: SimTrack #1576 key=723263

gamma (22)
Particle 547
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.081, 0.081, -0.151, 0.189)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.203670
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.203670
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.042, 85.290, -168.209, 0.000)
p4@checkpoint: (-0.081, 0.081, -0.151, 0.189)
raw GEN: n/a
raw SIM: SimTrack #1577 key=723766

gamma (22)
Particle 548
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001121
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001121
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.844, 85.496, -168.257, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1578 key=723768

gamma (22)
Particle 549
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.031, 0.033, -0.060, 0.075)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.068582
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.068582
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.877, 85.462, -168.278, 0.000)
p4@checkpoint: (-0.031, 0.033, -0.060, 0.075)
raw GEN: n/a
raw SIM: SimTrack #1579 key=723769

gamma (22)
Particle 550
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.056, 0.043, -0.095, 0.119)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.137993
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.137993
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.680, 84.619, -168.494, 0.000)
p4@checkpoint: (-0.056, 0.043, -0.095, 0.119)
raw GEN: n/a
raw SIM: SimTrack #1580 key=723771

gamma (22)
Particle 551
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000780
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000780
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-85.439, 88.899, -163.353, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1581 key=723936

gamma (22)
Particle 552
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000310
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000310
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-85.698, 88.650, -148.007, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1583 key=723943

e- (11)
Particle 553
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.138, 0.129, -0.257, 0.319)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 11 simE=0.254856
direct recHits: 0 missing=11 recoE=0.000000
subgraph simHits: 14 simE=0.281807
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.785, 81.203, -168.618, 0.000)
p4@checkpoint: (-0.169, 0.035, -0.225, 0.283)
raw GEN: n/a
raw SIM: SimTrack #1584 key=723970

Bremsstrahlung
Vertex 400
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-86.018, 77.968, -158.649, 6.911)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #671 key=592

Bremsstrahlung
Vertex 401
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-86.032, 77.979, -158.674, 6.912)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #672 key=593

Bremsstrahlung
Vertex 402
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-86.430, 78.280, -159.352, 6.940)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #673 key=594

e+ (-11)
Particle 554
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.033, 0.031, -0.063, 0.078)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.044761
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=2.041312
raw GEN: n/a
raw SIM: SimTrack #1585 key=723971

Bremsstrahlung
Vertex 397
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-85.972, 77.960, -158.591, 6.909)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #668 key=589

Bremsstrahlung
Vertex 398
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-85.977, 77.967, -158.603, 6.909)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #669 key=590

Annihilation

Vertex 399
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-85.364, 77.911, -159.909, 7.011)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #670 key=591

gamma (22)
Particle 555
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.018, 0.021, -0.041, 0.049)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.035742
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.035742
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-90.702, 83.522, -169.142, 0.000)
p4@checkpoint: (-0.018, 0.021, -0.041, 0.049)
raw GEN: n/a
raw SIM: SimTrack #1586 key=723976

gamma (22)
Particle 556
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.006, -0.010, 0.012)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.008326
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.008326
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.807, 84.485, -170.500, 0.000)
p4@checkpoint: (-0.003, 0.006, -0.010, 0.012)
raw GEN: n/a
raw SIM: SimTrack #1587 key=723977

gamma (22)
Particle 557
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000183
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000183
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 2 dE=2.041312
subgraph tracker simHits: 2 dE=2.041312
checkpointId: 0
x4@checkpoint: (-112.514, 50.433, -77.343, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1588 key=723982

gamma (22)
Particle 558
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.345, 78.180, -160.086, 0.000)
p4@checkpoint: (-0.001, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1589 key=723983

gamma (22)
Particle 559
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, 0.005, -0.012, 0.015)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.018146
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.018146
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-91.733, 82.389, -168.803, 0.000)
p4@checkpoint: (-0.007, 0.005, -0.012, 0.015)
raw GEN: n/a
raw SIM: SimTrack #1590 key=723994

gamma (22)
Particle 560
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002428
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002428
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-91.559, 82.582, -166.657, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1591 key=723995

gamma (22)
Particle 561
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.003, -0.006, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.006377
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.006377
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-91.811, 82.302, -168.528, 0.000)
p4@checkpoint: (-0.004, 0.003, -0.006, 0.008)
raw GEN: n/a
raw SIM: SimTrack #1592 key=723996

gamma (22)
Particle 562
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000369
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000369
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-102.740, 68.172, -170.000, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1594 key=724541

gamma (22)
Particle 563
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.388, 0.253, -0.635, 0.786)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 26 simE=0.678225
subgraph recHits: 0 missing=26 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1595 key=724545

PairConversion

Vertex 465
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.698, 67.702, -158.261, 6.892)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #758 key=679

gamma (22)
Particle 564
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.083, 0.054, -0.136, 0.168)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 18 simE=0.030354
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 3 dE=3.277124
raw GEN: n/a
raw SIM: SimTrack #1596 key=724546

PairConversion

Vertex 449
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.039, 67.071, -157.056, 6.842)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #738 key=659

Bremsstrahlung
Vertex 462
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.002, 67.385, -158.343, 6.897)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #754 key=675

Bremsstrahlung
Vertex 463
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.217, 67.310, -158.731, 6.912)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #755 key=676

Annihilation

Vertex 464
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.694, 66.850, -160.939, 6.989)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #757 key=678

gamma (22)
Particle 565
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.089, 0.057, -0.145, 0.180)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.170291
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.170291
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.345, 71.651, -168.910, 0.000)
p4@checkpoint: (-0.089, 0.057, -0.145, 0.180)
raw GEN: n/a
raw SIM: SimTrack #1597 key=724547

gamma (22)
Particle 566
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.033, 0.019, -0.052, 0.064)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.060165
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.060165
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.845, 70.945, -169.006, 0.000)
p4@checkpoint: (-0.033, 0.019, -0.052, 0.064)
raw GEN: n/a
raw SIM: SimTrack #1598 key=724552

gamma (22)
Particle 567
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.035, 0.021, -0.056, 0.069)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.075522
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.075522
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.848, 70.941, -169.003, 0.000)
p4@checkpoint: (-0.035, 0.021, -0.056, 0.069)
raw GEN: n/a
raw SIM: SimTrack #1599 key=724553

gamma (22)
Particle 568
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.167, 0.096, -0.265, 0.327)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 34 simE=0.159537
subgraph recHits: 0 missing=34 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=11.533735
raw GEN: n/a
raw SIM: SimTrack #1600 key=724554

PairConversion

Vertex 434
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.266, 67.093, -158.527, 6.902)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #711 key=632

gamma (22)
Particle 569
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.530, 0.306, -0.841, 1.040)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 19 simE=0.993839
direct recHits: 0 missing=19 recoE=0.000000
subgraph simHits: 20 simE=0.994297
subgraph recHits: 0 missing=20 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.871, 70.909, -168.997, 0.000)
p4@checkpoint: (-0.530, 0.306, -0.841, 1.040)
raw GEN: n/a
raw SIM: SimTrack #1601 key=724558

Annihilation

Vertex 433
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-107.644, 46.896, -206.767, 9.366)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #710 key=631

gamma (22)
Particle 570
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-4.019, 2.321, -6.370, 7.881)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 65 simE=7.586254
subgraph recHits: 0 missing=65 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1602 key=724559

PairConversion

Vertex 417
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.248, 67.082, -158.498, 6.901)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #689 key=610

gamma (22)
Particle 571
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000198
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000198
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-98.393, 74.308, -171.234, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1603 key=724560

gamma (22)
Particle 572
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, 0.003, -0.008, 0.010)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.006460
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.006460
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.330, 71.671, -173.709, 0.000)
p4@checkpoint: (-0.003, 0.002, -0.007, 0.008)
raw GEN: n/a
raw SIM: SimTrack #1604 key=724563

gamma (22)
Particle 573
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000230
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000230
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-101.038, 70.670, -169.157, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1605 key=724564

gamma (22)
Particle 574
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000286
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000286
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-116.867, 39.306, -202.791, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1606 key=724754

gamma (22)
Particle 575
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.000, -0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001814
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001814
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-118.690, 33.399, -206.275, 0.000)
p4@checkpoint: (-0.002, 0.000, -0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1607 key=724755

gamma (22)
Particle 576
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.001, 0.000, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.002745
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.002745
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-118.275, 34.840, -199.772, 0.000)
p4@checkpoint: (-0.003, -0.001, 0.000, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1608 key=724758

e- (11)
Particle 577
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-2.698, 1.561, -4.276, 5.291)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 47 simE=4.965097
direct recHits: 0 missing=47 recoE=0.000000
subgraph simHits: 52 simE=5.052016
subgraph recHits: 0 missing=52 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.946, 70.802, -169.026, 0.000)
p4@checkpoint: (-2.632, 1.411, -4.108, 5.079)
raw GEN: n/a
raw SIM: SimTrack #1609 key=725010

Bremsstrahlung
Vertex 426
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.317, 67.122, -158.607, 6.906)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #698 key=619

e+ (-11)
Particle 578
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.321, 0.763, -2.093, 2.590)
nProdVtx: 1 nDecayVtx: 8
nParents: 1 nChildren: 8
nCheckpoints: 1
direct simHits: 32 simE=1.959232
direct recHits: 0 missing=32 recoE=0.000000
subgraph simHits: 35 simE=2.534237
subgraph recHits: 0 missing=35 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.739, 71.096, -169.213, 0.000)
p4@checkpoint: (-0.928, 0.623, -1.574, 1.931)
raw GEN: n/a
raw SIM: SimTrack #1610 key=725011

Bremsstrahlung
Vertex 418
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.252, 67.084, -158.503, 6.901)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #690 key=611

Bremsstrahlung
Vertex 419
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.284, 67.103, -158.554, 6.903)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #691 key=612

Bremsstrahlung
Vertex 420
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.366, 67.150, -158.685, 6.909)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #692 key=613

Bremsstrahlung
Vertex 421
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.376, 67.156, -158.700, 6.909)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #693 key=614

Bremsstrahlung
Vertex 422
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.431, 67.188, -158.788, 6.913)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #694 key=615

Bremsstrahlung
Vertex 423
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-96.014, 68.110, -161.339, 7.018)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #695 key=616

Bremsstrahlung
Vertex 424
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-96.034, 68.122, -161.371, 7.019)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #696 key=617

Bremsstrahlung
Vertex 425
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-97.278, 68.876, -163.424, 7.103)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #697 key=618

gamma (22)
Particle 579
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, 0.003, -0.007, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.008981
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.008981
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.875, 70.903, -168.976, 0.000)
p4@checkpoint: (-0.005, 0.003, -0.007, 0.009)
raw GEN: n/a
raw SIM: SimTrack #1611 key=725012

gamma (22)
Particle 580
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.251, 0.145, -0.398, 0.492)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.448997
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.448997
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.877, 70.899, -168.997, 0.000)
p4@checkpoint: (-0.251, 0.145, -0.398, 0.492)
raw GEN: n/a
raw SIM: SimTrack #1612 key=725013

gamma (22)
Particle 581
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.002, -0.005, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.006626
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.006626
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.865, 70.917, -169.051, 0.000)
p4@checkpoint: (-0.003, 0.002, -0.005, 0.006)
raw GEN: n/a
raw SIM: SimTrack #1613 key=725015

gamma (22)
Particle 582
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.022, 0.013, -0.035, 0.043)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.034964
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.034964
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.866, 70.916, -169.062, 0.000)
p4@checkpoint: (-0.022, 0.013, -0.035, 0.043)
raw GEN: n/a
raw SIM: SimTrack #1614 key=725016

gamma (22)
Particle 583
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000521
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000521
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.455, 72.881, -158.765, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1615 key=725018

gamma (22)
Particle 584
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.014, 0.008, -0.022, 0.027)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.018136
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.018136
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.822, 70.977, -169.180, 0.000)
p4@checkpoint: (-0.014, 0.008, -0.022, 0.027)
raw GEN: n/a
raw SIM: SimTrack #1616 key=725021

gamma (22)
Particle 585
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.029, 0.017, -0.047, 0.058)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.048931
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.048931
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.826, 70.972, -169.220, 0.000)
p4@checkpoint: (-0.029, 0.017, -0.047, 0.058)
raw GEN: n/a
raw SIM: SimTrack #1617 key=725022

gamma (22)
Particle 586
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.002, -0.006, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.007850
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.007850
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.792, 71.021, -169.215, 0.000)
p4@checkpoint: (-0.004, 0.002, -0.006, 0.007)
raw GEN: n/a
raw SIM: SimTrack #1618 key=725025

gamma (22)
Particle 587
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.101, 0.058, -0.159, 0.197)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 8 simE=0.086919
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1619 key=725801

PairConversion

Vertex 427
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.898, 67.459, -159.524, 6.944)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #699 key=620

e+ (-11)
Particle 588
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.098, 0.057, -0.155, 0.192)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 7
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 8 simE=0.086919
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1620 key=727384

Annihilation

Vertex 428
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-96.850, 69.476, -163.320, 7.102)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #700 key=621

Bremsstrahlung
Vertex 429
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-96.801, 69.715, -163.633, 7.114)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #702 key=623

Bremsstrahlung
Vertex 430
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.314, 70.391, -178.640, 7.950)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #703 key=624

Annihilation

Vertex 431
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.974, 69.540, -177.931, 8.068)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #704 key=625

Bremsstrahlung
Vertex 432
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-99.065, 66.214, -178.068, 7.800)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #709 key=630

gamma (22)
Particle 589
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001548
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001548
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.690, 71.165, -164.448, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1621 key=727389

gamma (22)
Particle 590
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.029, 0.053, -0.081, 0.101)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.083215
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.083215
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.039, 73.444, -169.450, 0.000)
p4@checkpoint: (-0.029, 0.053, -0.081, 0.101)
raw GEN: n/a
raw SIM: SimTrack #1622 key=727390

gamma (22)
Particle 591
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000971
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000971
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.552, 75.409, -173.220, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1624 key=727416

gamma (22)
Particle 592
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000216
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000216
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-91.041, -83.153, -282.412, 0.000)
p4@checkpoint: (-0.000, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1625 key=727424

gamma (22)
Particle 593
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.365, 71.622, -178.820, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1626 key=727430

gamma (22)
Particle 594
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000261
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000261
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (79.828, 93.970, -200.412, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1627 key=727431

gamma (22)
Particle 595
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000197
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000197
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.045, 61.189, -190.751, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1632 key=727452

gamma (22)
Particle 596
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000458
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000458
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (113.269, 48.714, -126.841, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1633 key=727699

e- (11)
Particle 597
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.070, 0.040, -0.111, 0.137)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 13 simE=0.081243
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1635 key=727821

Bremsstrahlung
Vertex 444
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-95.269, 67.514, -160.338, 6.973)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #728 key=649

Bremsstrahlung
Vertex 445
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-96.640, 67.281, -162.800, 7.068)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #729 key=650

Bremsstrahlung
Vertex 446
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-96.659, 67.272, -162.842, 7.069)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #730 key=651

Bremsstrahlung
Vertex 447
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-96.727, 67.227, -162.977, 7.074)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #731 key=652

Bremsstrahlung
Vertex 448
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.943, 65.280, -171.897, 7.415)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #737 key=658

e+ (-11)
Particle 598
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.097, 0.056, -0.154, 0.191)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 10
nCheckpoints: 1
direct simHits: 11 simE=0.060313
direct recHits: 0 missing=11 recoE=0.000000
subgraph simHits: 21 simE=0.078294
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=11.533735
checkpointId: 0
x4@checkpoint: (-97.902, 74.954, -172.174, 0.000)
p4@checkpoint: (0.012, 0.088, -0.135, 0.161)
raw GEN: n/a
raw SIM: SimTrack #1636 key=727822

Bremsstrahlung
Vertex 435
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.931, 67.570, -159.716, 6.950)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #712 key=633

Bremsstrahlung
Vertex 436
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-96.422, 69.095, -162.823, 7.076)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #714 key=635

Bremsstrahlung
Vertex 437
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-87.326, 76.192, -194.367, 8.363)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #717 key=638

Bremsstrahlung
Vertex 438
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-87.327, 76.044, -194.573, 8.371)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #719 key=640

Bremsstrahlung
Vertex 439
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-87.326, 75.910, -194.709, 8.378)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #721 key=642

Annihilation

Vertex 440
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.289, 74.522, -195.155, 8.453)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #724 key=645

Bremsstrahlung
Vertex 441
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-97.482, -57.919, -255.376, 13.291)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #725 key=646

Annihilation

Vertex 442
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-97.723, -57.815, -255.781, 13.328)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #726 key=647

Annihilation

Vertex 443
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-87.224, 75.554, -195.188, 8.406)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #727 key=648

gamma (22)
Particle 599
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, 0.006, -0.013, 0.016)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.015823
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.015823
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.137, 71.941, -169.568, 0.000)
p4@checkpoint: (-0.007, 0.006, -0.013, 0.016)
raw GEN: n/a
raw SIM: SimTrack #1637 key=727825

gamma (22)
Particle 600
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000222
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000222
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.351, 73.023, -170.037, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1639 key=727829

gamma (22)
Particle 601
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000264
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000264
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (114.308, -46.222, -207.576, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1642 key=727844

gamma (22)
Particle 602
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.002, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 12 dE=11.533735
subgraph tracker simHits: 12 dE=11.533735
checkpointId: 0
x4@checkpoint: (-58.793, 10.870, -293.500, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1644 key=727848

gamma (22)
Particle 603
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000526
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000526
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-67.724, -51.040, -293.500, 0.000)
p4@checkpoint: (0.000, -0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1646 key=727854

gamma (22)
Particle 604
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.656, 79.014, -194.956, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1649 key=727865

gamma (22)
Particle 605
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000098
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000098
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.118, -114.748, -117.487, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1651 key=727873

gamma (22)
Particle 606
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000093
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000093
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.804, -77.617, -251.607, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1652 key=727874

gamma (22)
Particle 607
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000292
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000292
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-109.060, -57.523, -265.787, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1653 key=727875

gamma (22)
Particle 608
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000152
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000152
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.017, 82.072, -195.792, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1654 key=727885

gamma (22)
Particle 609
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.037, 0.006, -0.066, 0.076)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.073798
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.073798
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-102.436, 68.627, -173.267, 0.000)
p4@checkpoint: (-0.037, 0.006, -0.066, 0.076)
raw GEN: n/a
raw SIM: SimTrack #1655 key=727964

gamma (22)
Particle 610
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.001, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001894
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001894
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-101.623, 69.826, -172.970, 0.000)
p4@checkpoint: (-0.001, 0.000, -0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1656 key=727966

gamma (22)
Particle 611
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.001, -0.004, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.005214
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.005214
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-104.910, 64.782, -180.970, 0.000)
p4@checkpoint: (-0.002, -0.001, -0.004, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1657 key=727967

gamma (22)
Particle 612
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000059
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000059
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-106.467, 62.191, -180.771, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1658 key=727968

gamma (22)
Particle 613
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000277
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000277
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.171, 81.898, -186.477, 0.000)
p4@checkpoint: (0.000, 0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1664 key=727978

e+ (-11)
Particle 614
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.067, 0.042, -0.110, 0.136)
nProdVtx: 1 nDecayVtx: 10
nParents: 1 nChildren: 11
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 13 simE=0.021242
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 3 dE=3.277124
raw GEN: n/a
raw SIM: SimTrack #1665 key=728122

Bremsstrahlung
Vertex 450
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.756, 67.577, -158.293, 6.893)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #739 key=660

Bremsstrahlung
Vertex 451
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.834, 67.627, -158.429, 6.898)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #740 key=661

Bremsstrahlung
Vertex 452
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.897, 67.674, -158.541, 6.903)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #741 key=662

Bremsstrahlung
Vertex 453
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.963, 67.722, -158.647, 6.907)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #742 key=663

Bremsstrahlung
Vertex 454
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.970, 67.726, -158.658, 6.907)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #743 key=664

Bremsstrahlung
Vertex 455
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.415, 68.083, -159.455, 6.940)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #744 key=665

Bremsstrahlung
Vertex 456
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-95.254, 69.184, -161.254, 7.016)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #745 key=666

Annihilation

Vertex 457
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.959, 74.756, -170.498, 7.402)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #747 key=668

Annihilation

Vertex 460
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.398, 74.171, -175.369, 7.673)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #750 key=671

Bremsstrahlung
Vertex 461
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.356, 75.422, -173.381, 7.540)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #752 key=673

gamma (22)
Particle 615
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.001, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004094
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004094
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.326, 71.677, -169.523, 0.000)
p4@checkpoint: (-0.002, 0.001, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1666 key=728125

gamma (22)
Particle 616
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=1.126060
subgraph tracker simHits: 1 dE=1.126060
checkpointId: 0
x4@checkpoint: (-109.997, 55.709, -164.141, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1667 key=728127

gamma (22)
Particle 617
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000675
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000675
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.025, 72.097, -168.461, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1668 key=728128

gamma (22)
Particle 618
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.012, 0.009, -0.020, 0.025)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003264
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003264
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-102.839, 68.023, -177.530, 0.000)
p4@checkpoint: (-0.002, -0.000, -0.004, 0.005)
raw GEN: n/a
raw SIM: SimTrack #1669 key=728129

gamma (22)
Particle 619
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000299
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000299
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.086, 72.012, -169.060, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1670 key=728130

gamma (22)
Particle 620
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.002, -0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.005897
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.005897
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.580, 72.710, -168.502, 0.000)
p4@checkpoint: (-0.002, 0.002, -0.004, 0.005)
raw GEN: n/a
raw SIM: SimTrack #1671 key=728133

gamma (22)
Particle 621
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000238
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000238
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-98.400, 74.299, -169.665, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1672 key=728136

gamma (22)
Particle 622
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.023, 0.010, -0.033, 0.041)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000590
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (64.482, 12.296, -293.500, 0.000)
p4@checkpoint: (0.001, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1674 key=728140

Bremsstrahlung
Vertex 458
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.859, 75.687, -173.555, 7.529)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #748 key=669

Bremsstrahlung
Vertex 459
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.954, 77.380, -176.224, 7.641)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #749 key=670

gamma (22)
Particle 623
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.001, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000181
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000181
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-80.497, 93.397, -201.522, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1675 key=728145

gamma (22)
Particle 624
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000408
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000408
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-90.571, 83.664, -185.101, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1676 key=728153

gamma (22)
Particle 625
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.002, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002403
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002403
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.667, 81.337, -170.332, 0.000)
p4@checkpoint: (-0.000, 0.002, 0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1677 key=728175

gamma (22)
Particle 626
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.007, -0.001, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003634
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003634
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.845, 85.495, -176.285, 0.000)
p4@checkpoint: (0.001, 0.007, -0.001, 0.007)
raw GEN: n/a
raw SIM: SimTrack #1678 key=728176

gamma (22)
Particle 627
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000147
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000147
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 2 dE=2.151064
subgraph tracker simHits: 2 dE=2.151064
checkpointId: 0
x4@checkpoint: (-123.244, -3.731, -84.108, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1680 key=728181

gamma (22)
Particle 628
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, 0.000, -0.006, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.008174
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.008174
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-103.170, 67.519, -168.953, 0.000)
p4@checkpoint: (-0.005, 0.000, -0.006, 0.008)
raw GEN: n/a
raw SIM: SimTrack #1682 key=728223

gamma (22)
Particle 629
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.003, -0.005, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000281
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000281
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-104.868, 64.850, -163.218, 0.000)
p4@checkpoint: (-0.001, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1683 key=728227

gamma (22)
Particle 630
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000440
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000440
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.262, 75.782, -156.320, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1685 key=728232

gamma (22)
Particle 631
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000216
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000216
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-98.740, -73.847, -191.894, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1686 key=728233

e- (11)
Particle 632
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.030, 0.020, -0.049, 0.060)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 3 simE=0.004383
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1687 key=728244

Bremsstrahlung
Vertex 470
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.451, 67.966, -159.037, 6.929)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #764 key=685

Bremsstrahlung
Vertex 471
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-97.538, 64.447, -164.601, 7.211)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #765 key=686

Bremsstrahlung
Vertex 472
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-96.799, 63.803, -165.703, 7.261)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #767 key=688

e+ (-11)
Particle 633
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.357, 0.233, -0.587, 0.726)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 21 simE=0.664861
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 23 simE=0.673842
subgraph recHits: 0 missing=23 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.687, 72.563, -168.751, 0.000)
p4@checkpoint: (-0.295, 0.293, -0.562, 0.699)
raw GEN: n/a
raw SIM: SimTrack #1688 key=728245

Bremsstrahlung
Vertex 466
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.090, 67.961, -158.911, 6.919)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #759 key=680

Bremsstrahlung
Vertex 467
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-94.648, 68.345, -159.843, 6.957)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #761 key=682

Bremsstrahlung
Vertex 468
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-95.431, 68.916, -161.161, 7.012)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #762 key=683

Bremsstrahlung
Vertex 469
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-95.882, 69.256, -161.926, 7.044)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #763 key=684

gamma (22)
Particle 634
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.001, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001566
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001566
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-101.137, 70.528, -166.500, 0.000)
p4@checkpoint: (-0.002, 0.001, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1689 key=728247

gamma (22)
Particle 635
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.001, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.005054
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.005054
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.002, 72.129, -168.756, 0.000)
p4@checkpoint: (-0.002, 0.001, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1691 key=728250

gamma (22)
Particle 636
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.002205
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.002205
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.904, 72.264, -168.713, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1692 key=728251

gamma (22)
Particle 637
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000156
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000156
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.880, 72.297, -168.735, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1693 key=728252

gamma (22)
Particle 638
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.001, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003762
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003762
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-101.814, 69.547, -166.164, 0.000)
p4@checkpoint: (-0.003, 0.001, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1694 key=728482

gamma (22)
Particle 639
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000347
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000347
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (98.298, -74.434, -215.186, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1695 key=728490

gamma (22)
Particle 640
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000273
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000273
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (57.146, 109.258, -273.462, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1697 key=728505

gamma (22)
Particle 641
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000277
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000277
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-98.696, 73.906, -175.907, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1698 key=728526

gamma (22)
Particle 642
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000135
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000135
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-21.862, 121.346, -70.271, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1699 key=728527

e- (11)
Particle 643
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-12.164, 9.956, -21.481, 26.618)
nProdVtx: 1 nDecayVtx: 21
nParents: 1 nChildren: 21
nCheckpoints: 1
direct simHits: 29 simE=1.679835
direct recHits: 0 missing=29 recoE=0.000000
subgraph simHits: 126 simE=25.152225
subgraph recHits: 0 missing=126 recoE=0.000000
direct tracker simHits: 15 dE=13.206404
subgraph tracker simHits: 10 dE=13.206404
checkpointId: 0
x4@checkpoint: (-97.317, 75.711, -167.965, 0.000)
p4@checkpoint: (-0.864, 0.521, -1.366, 1.698)
raw GEN: n/a
raw SIM: SimTrack #1700 key=728530

Bremsstrahlung
Vertex 491
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-28.726, 23.486, -50.181, 2.434)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #789 key=710

Bremsstrahlung
Vertex 492
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-66.029, 52.956, -115.151, 5.120)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #790 key=711

Bremsstrahlung
Vertex 493
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-69.214, 55.377, -120.618, 5.346)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #791 key=712

Bremsstrahlung
Vertex 494
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-75.822, 60.352, -131.924, 5.813)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #792 key=713

Bremsstrahlung
Vertex 495
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-89.083, 70.139, -154.450, 6.744)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #793 key=714

Bremsstrahlung
Vertex 496
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.736, 71.341, -157.243, 6.860)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #794 key=715

Bremsstrahlung
Vertex 497
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.747, 71.349, -157.262, 6.860)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #795 key=716

Bremsstrahlung
Vertex 498
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.756, 71.355, -157.276, 6.861)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #796 key=717

Bremsstrahlung
Vertex 499
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.798, 71.386, -157.347, 6.864)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #797 key=718

Bremsstrahlung
Vertex 500
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.834, 71.412, -157.409, 6.866)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #798 key=719

Bremsstrahlung
Vertex 501
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.858, 71.430, -157.450, 6.868)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #799 key=720

Bremsstrahlung
Vertex 502
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.894, 71.455, -157.509, 6.871)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #800 key=721

Bremsstrahlung
Vertex 503
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.898, 71.459, -157.518, 6.871)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #801 key=722

Bremsstrahlung
Vertex 504
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.903, 71.462, -157.526, 6.871)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #802 key=723

Bremsstrahlung
Vertex 505
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.933, 71.484, -157.575, 6.873)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #803 key=724

Bremsstrahlung
Vertex 506
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.960, 71.503, -157.621, 6.875)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #804 key=725

Bremsstrahlung
Vertex 507
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.979, 71.518, -157.654, 6.877)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #805 key=726

Bremsstrahlung
Vertex 508
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.468, 71.868, -158.472, 6.910)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #806 key=727

Bremsstrahlung
Vertex 509
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.737, 72.059, -158.919, 6.929)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #807 key=728

Bremsstrahlung
Vertex 525
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.741, 71.779, -157.714, 6.878)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #832 key=753

Annihilation

Vertex 526
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.717, 71.780, -157.738, 6.879)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #833 key=754

e+ (-11)
Particle 644
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-3.516, 2.873, -6.212, 7.695)
nProdVtx: 1 nDecayVtx: 13
nParents: 1 nChildren: 13
nCheckpoints: 1
direct simHits: 24 simE=1.480491
direct recHits: 0 missing=24 recoE=0.000000
subgraph simHits: 77 simE=7.299109
subgraph recHits: 0 missing=77 recoE=0.000000
direct tracker simHits: 13 dE=10.924579
subgraph tracker simHits: 8 dE=10.924578
checkpointId: 0
x4@checkpoint: (-89.077, 85.254, -168.388, 0.000)
p4@checkpoint: (-0.532, 0.768, -1.269, 1.576)
raw GEN: n/a
raw SIM: SimTrack #1701 key=728531

Bremsstrahlung
Vertex 475
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-50.787, 42.969, -90.452, 4.098)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #770 key=691

Bremsstrahlung
Vertex 476
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-66.223, 58.468, -120.366, 5.335)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #771 key=692

Bremsstrahlung
Vertex 477
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-80.980, 75.006, -150.597, 6.585)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #772 key=693

Bremsstrahlung
Vertex 478
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.435, 79.159, -157.969, 6.890)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #773 key=694

Bremsstrahlung
Vertex 479
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.441, 79.167, -157.982, 6.890)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #774 key=695

Bremsstrahlung
Vertex 480
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.464, 79.195, -158.032, 6.892)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #775 key=696

Bremsstrahlung
Vertex 481
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.512, 79.254, -158.136, 6.897)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #776 key=697

Bremsstrahlung
Vertex 482
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.627, 79.395, -158.383, 6.907)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #777 key=698

Bremsstrahlung
Vertex 483
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.655, 79.429, -158.443, 6.909)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #778 key=699

Bremsstrahlung
Vertex 484
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.991, 79.842, -159.166, 6.939)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #779 key=700

Bremsstrahlung
Vertex 488
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.694, 79.151, -157.977, 6.894)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #786 key=707

Bremsstrahlung
Vertex 489
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.701, 79.153, -157.985, 6.894)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #787 key=708

Bremsstrahlung
Vertex 490
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-84.740, 79.159, -158.024, 6.896)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #788 key=709

gamma (22)
Particle 645
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000714
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000714
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-1.102, 123.295, -180.906, 0.000)
p4@checkpoint: (0.000, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1702 key=728533

gamma (22)
Particle 646
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.040, 0.042, -0.080, 0.099)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.096003
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.096003
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-90.297, 83.961, -168.324, 0.000)
p4@checkpoint: (-0.040, 0.042, -0.080, 0.099)
raw GEN: n/a
raw SIM: SimTrack #1703 key=728534

gamma (22)
Particle 647
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.254, 0.302, -0.538, 0.667)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.652046
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.652046
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.370, 84.947, -168.346, 0.000)
p4@checkpoint: (-0.254, 0.302, -0.538, 0.667)
raw GEN: n/a
raw SIM: SimTrack #1704 key=728539

gamma (22)
Particle 648
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.167, 0.204, -0.360, 0.446)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 12 simE=0.436864
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 12 simE=0.436864
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.267, 85.055, -168.381, 0.000)
p4@checkpoint: (-0.167, 0.204, -0.360, 0.446)
raw GEN: n/a
raw SIM: SimTrack #1705 key=728544

gamma (22)
Particle 649
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.002, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003793
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003793
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.264, 85.058, -168.378, 0.000)
p4@checkpoint: (-0.002, 0.002, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #1706 key=728545

gamma (22)
Particle 650
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.015, 0.019, -0.033, 0.041)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.038073
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.038073
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.264, 85.058, -168.379, 0.000)
p4@checkpoint: (-0.015, 0.019, -0.033, 0.041)
raw GEN: n/a
raw SIM: SimTrack #1707 key=728546

gamma (22)
Particle 651
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.404, 0.495, -0.871, 1.080)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 22 simE=1.031405
direct recHits: 0 missing=22 recoE=0.000000
subgraph simHits: 22 simE=1.031405
subgraph recHits: 0 missing=22 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.258, 85.064, -168.361, 0.000)
p4@checkpoint: (-0.404, 0.495, -0.871, 1.080)
raw GEN: n/a
raw SIM: SimTrack #1708 key=728547

gamma (22)
Particle 652
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.103, 0.127, -0.222, 0.275)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.261234
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 18 simE=0.261234
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.248, 85.074, -168.329, 0.000)
p4@checkpoint: (-0.103, 0.127, -0.222, 0.275)
raw GEN: n/a
raw SIM: SimTrack #1709 key=728549

gamma (22)
Particle 653
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000519
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000519
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.253, 85.070, -168.335, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1710 key=728550

gamma (22)
Particle 654
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.278, 1.573, -2.754, 3.419)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 38 simE=3.260118
direct recHits: 0 missing=38 recoE=0.000000
subgraph simHits: 41 simE=3.261950
subgraph recHits: 0 missing=41 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-89.244, 85.079, -168.335, 0.000)
p4@checkpoint: (-1.278, 1.573, -2.754, 3.419)
raw GEN: n/a
raw SIM: SimTrack #1711 key=728551

Bremsstrahlung
Vertex 485
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-46.853, 105.887, -160.885, 9.507)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #782 key=703

Bremsstrahlung
Vertex 486
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-47.078, 105.716, -160.903, 9.523)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #784 key=705

Bremsstrahlung
Vertex 487
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-47.081, 105.718, -160.925, 9.524)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #785 key=706

gamma (22)
Particle 655
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.002, -0.000, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001256
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001256
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-64.771, 104.917, -170.262, 0.000)
p4@checkpoint: (-0.001, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1714 key=729122

gamma (22)
Particle 656
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000188
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000188
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (20.555, 101.556, -293.500, 0.000)
p4@checkpoint: (0.000, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1716 key=729129

gamma (22)
Particle 657
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000388
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000388
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-54.088, 110.803, -156.055, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1717 key=729131

gamma (22)
Particle 658
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, 0.001, -0.005, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.006071
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.006071
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.520, 81.504, -166.595, 0.000)
p4@checkpoint: (-0.005, 0.001, -0.005, 0.007)
raw GEN: n/a
raw SIM: SimTrack #1718 key=730706

gamma (22)
Particle 659
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000091
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000091
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-93.634, 80.222, -124.242, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1719 key=730707

gamma (22)
Particle 660
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.017, 0.006, -0.017, 0.025)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.029856
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.029856
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.430, 81.607, -165.418, 0.000)
p4@checkpoint: (-0.017, 0.006, -0.017, 0.025)
raw GEN: n/a
raw SIM: SimTrack #1720 key=730708

gamma (22)
Particle 661
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, 0.002, -0.005, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001410
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001410
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.615, 71.271, -167.429, 0.000)
p4@checkpoint: (-0.001, -0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1721 key=730955

gamma (22)
Particle 662
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.260, 0.198, -0.446, 0.553)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 20 simE=0.325038
subgraph recHits: 0 missing=20 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1722 key=730957

PairConversion

Vertex 527
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.201, 72.148, -158.414, 6.908)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #834 key=755

gamma (22)
Particle 663
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.344, 0.261, -0.590, 0.731)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 22 simE=0.648253
subgraph recHits: 0 missing=22 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1723 key=730959

PairConversion

Vertex 519
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.271, 72.109, -158.471, 6.910)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #823 key=744

gamma (22)
Particle 664
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.003, -0.007, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000413
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000413
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.964, 76.163, -168.009, 0.000)
p4@checkpoint: (-0.004, 0.003, -0.007, 0.009)
raw GEN: n/a
raw SIM: SimTrack #1724 key=730960

gamma (22)
Particle 665
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.052, 0.038, -0.087, 0.108)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.091825
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.091825
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.106, 75.983, -168.013, 0.000)
p4@checkpoint: (-0.052, 0.038, -0.087, 0.108)
raw GEN: n/a
raw SIM: SimTrack #1725 key=730961

gamma (22)
Particle 666
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, 0.005, -0.012, 0.015)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.016817
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.016817
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.116, 75.970, -168.021, 0.000)
p4@checkpoint: (-0.007, 0.005, -0.012, 0.015)
raw GEN: n/a
raw SIM: SimTrack #1726 key=730963

gamma (22)
Particle 667
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001657
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001657
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.117, 75.968, -168.023, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1727 key=730965

gamma (22)
Particle 668
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-6.633, 4.810, -11.203, 13.879)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 59 simE=13.492324
direct recHits: 0 missing=59 recoE=0.000000
subgraph simHits: 62 simE=13.493008
subgraph recHits: 0 missing=62 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.117, 75.968, -168.021, 0.000)
p4@checkpoint: (-6.633, 4.810, -11.203, 13.879)
raw GEN: n/a
raw SIM: SimTrack #1728 key=730966

Annihilation

Vertex 517
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.292, 107.874, 132.217, 19.345)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #821 key=742

Bremsstrahlung
Vertex 518
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-95.921, 68.312, -164.854, 8.960)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #822 key=743

gamma (22)
Particle 669
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.911, 0.662, -1.540, 1.908)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 36 simE=1.668662
subgraph recHits: 0 missing=36 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1729 key=730967

PairConversion

Vertex 511
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.968, 71.509, -157.635, 6.876)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #812 key=733

gamma (22)
Particle 670
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-2.261, 1.640, -3.815, 4.728)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 41 simE=4.480053
direct recHits: 0 missing=41 recoE=0.000000
subgraph simHits: 42 simE=4.482757
subgraph recHits: 0 missing=42 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.117, 75.969, -168.008, 0.000)
p4@checkpoint: (-2.261, 1.640, -3.815, 4.728)
raw GEN: n/a
raw SIM: SimTrack #1730 key=730968

Bremsstrahlung
Vertex 510
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.016, 72.360, -193.412, 8.699)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #809 key=730

gamma (22)
Particle 671
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.048, 0.035, -0.082, 0.101)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.014965
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.014965
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.112, 75.974, -168.007, 0.000)
p4@checkpoint: (-0.048, 0.035, -0.082, 0.101)
raw GEN: n/a
raw SIM: SimTrack #1731 key=730969

gamma (22)
Particle 672
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001196
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001196
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.061, 77.299, -166.122, 0.000)
p4@checkpoint: (-0.000, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1732 key=730970

gamma (22)
Particle 673
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.745, 0.542, -1.260, 1.561)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 28 simE=1.447625
direct recHits: 0 missing=28 recoE=0.000000
subgraph simHits: 28 simE=1.447625
subgraph recHits: 0 missing=28 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.108, 75.980, -168.016, 0.000)
p4@checkpoint: (-0.745, 0.542, -1.260, 1.561)
raw GEN: n/a
raw SIM: SimTrack #1733 key=730971

gamma (22)
Particle 674
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.021, 0.015, -0.035, 0.044)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.044291
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.044291
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.109, 75.978, -168.042, 0.000)
p4@checkpoint: (-0.021, 0.015, -0.035, 0.044)
raw GEN: n/a
raw SIM: SimTrack #1734 key=730972

gamma (22)
Particle 675
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.290, 0.210, -0.489, 0.606)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.599482
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.599482
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.118, 75.967, -168.014, 0.000)
p4@checkpoint: (-0.290, 0.210, -0.489, 0.606)
raw GEN: n/a
raw SIM: SimTrack #1735 key=730973

gamma (22)
Particle 676
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001935
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001935
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.116, 75.970, -168.003, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1736 key=730974

gamma (22)
Particle 677
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.051, 0.037, -0.085, 0.106)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 12 simE=0.105316
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 12 simE=0.105316
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.123, 75.961, -167.989, 0.000)
p4@checkpoint: (-0.051, 0.037, -0.085, 0.106)
raw GEN: n/a
raw SIM: SimTrack #1737 key=730975

gamma (22)
Particle 678
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.170, 0.121, -0.285, 0.353)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.347870
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.347870
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.154, 75.921, -168.019, 0.000)
p4@checkpoint: (-0.170, 0.121, -0.285, 0.353)
raw GEN: n/a
raw SIM: SimTrack #1738 key=730977

gamma (22)
Particle 679
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.076, 0.054, -0.126, 0.156)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.179112
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.179112
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.168, 75.904, -167.955, 0.000)
p4@checkpoint: (-0.076, 0.054, -0.126, 0.156)
raw GEN: n/a
raw SIM: SimTrack #1739 key=730978

gamma (22)
Particle 680
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.000, -0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002704
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002704
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (109.762, 56.171, -276.264, 0.000)
p4@checkpoint: (0.002, -0.000, -0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1741 key=732716

e- (11)
Particle 681
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.862, 0.625, -1.455, 1.803)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 33 simE=1.591834
direct recHits: 0 missing=33 recoE=0.000000
subgraph simHits: 34 simE=1.664430
subgraph recHits: 0 missing=34 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.323, 75.705, -167.966, 0.000)
p4@checkpoint: (-0.852, 0.514, -1.349, 1.676)
raw GEN: n/a
raw SIM: SimTrack #1744 key=733834

Bremsstrahlung
Vertex 516
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.972, 71.512, -157.642, 6.876)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #819 key=740

e+ (-11)
Particle 682
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.047, 0.038, -0.086, 0.105)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.004232
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1745 key=733835

Bremsstrahlung
Vertex 512
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-90.986, 71.524, -157.669, 6.877)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #813 key=734

Bremsstrahlung
Vertex 513
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.002, 71.536, -157.697, 6.878)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #814 key=735

Bremsstrahlung
Vertex 514
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-92.157, 75.565, -166.557, 7.211)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #816 key=737

Annihilation

Vertex 515
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.067, 75.086, -175.125, 7.657)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #818 key=739

gamma (22)
Particle 683
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002449
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002449
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.945, 76.188, -168.512, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #1746 key=733837

gamma (22)
Particle 684
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001439
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001439
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.376, 75.636, -168.066, 0.000)
p4@checkpoint: (-0.001, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1747 key=733839

gamma (22)
Particle 685
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000174
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000174
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-93.305, 80.604, -162.462, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1749 key=733843

gamma (22)
Particle 686
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000170
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000170
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-108.925, 57.777, -175.886, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1751 key=733853

gamma (22)
Particle 687
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.055, 0.040, -0.093, 0.116)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.072596
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.072596
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.124, 75.960, -168.025, 0.000)
p4@checkpoint: (-0.055, 0.040, -0.093, 0.116)
raw GEN: n/a
raw SIM: SimTrack #1752 key=733891

gamma (22)
Particle 688
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000427
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000427
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (41.396, 116.143, 145.959, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1754 key=735288

gamma (22)
Particle 689
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000257
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000257
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-102.293, 68.841, -162.556, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1755 key=736220

e- (11)
Particle 690
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.030, 0.022, -0.051, 0.063)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.009442
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #1756 key=738717

Bremsstrahlung
Vertex 521
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.755, 72.359, -159.169, 6.940)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #827 key=748

Bremsstrahlung
Vertex 522
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.463, 72.607, -161.206, 7.029)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #828 key=749

Annihilation

Vertex 523
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.823, 69.990, -173.281, 7.550)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #830 key=751

Annihilation

Vertex 524
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.627, 67.798, -169.954, 7.408)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #831 key=752

e+ (-11)
Particle 691
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.314, 0.238, -0.539, 0.668)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 16 simE=0.637471
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 17 simE=0.638811
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.374, 76.909, -168.144, 0.000)
p4@checkpoint: (-0.255, 0.292, -0.534, 0.660)
raw GEN: n/a
raw SIM: SimTrack #1757 key=738718

Bremsstrahlung
Vertex 520
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.553, 74.033, -162.595, 7.080)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #824 key=745

gamma (22)
Particle 692
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001340
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001340
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.522, 76.723, -168.088, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1758 key=738720

gamma (22)
Particle 693
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, 0.002, -0.007, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.008088
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.008088
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-98.006, 74.817, -168.062, 0.000)
p4@checkpoint: (-0.005, 0.002, -0.007, 0.008)
raw GEN: n/a
raw SIM: SimTrack #1761 key=738959

gamma (22)
Particle 694
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000207
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000207
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-99.037, 73.448, -171.497, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1762 key=738960

gamma (22)
Particle 695
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000203
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000203
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.794, 11.156, -251.213, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1764 key=738985

gamma (22)
Particle 696
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-101.020, 70.695, -169.452, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1766 key=738996

gamma (22)
Particle 697
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000432
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000432
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.223, 4.370, -116.958, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1767 key=738997

gamma (22)
Particle 698
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000251
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000251
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-93.169, 80.761, -161.069, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1768 key=739005

gamma (22)
Particle 699
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.023, 86.341, -161.126, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #1769 key=739006

e- (11)
Particle 700
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.153, 0.117, -0.263, 0.326)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 14 simE=0.230961
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 18 simE=0.259364
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-97.950, 74.892, -168.312, 0.000)
p4@checkpoint: (-0.171, 0.024, -0.231, 0.289)
raw GEN: n/a
raw SIM: SimTrack #1770 key=739012

Bremsstrahlung
Vertex 531
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.368, 72.270, -158.695, 6.920)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #844 key=765

Bremsstrahlung
Vertex 532
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.435, 72.317, -158.803, 6.924)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #845 key=766

e+ (-11)
Particle 701
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.107, 0.081, -0.183, 0.227)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.065674
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.443, 79.268, -169.160, 0.000)
p4@checkpoint: (0.005, 0.097, -0.128, 0.161)
raw GEN: n/a
raw SIM: SimTrack #1771 key=739013

Bremsstrahlung
Vertex 528
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-91.290, 72.218, -158.565, 6.914)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #835 key=756

Bremsstrahlung
Vertex 529
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-80.107, 83.453, -198.565, 8.560)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #841 key=762

Bremsstrahlung
Vertex 530
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-80.010, 83.253, -198.216, 8.551)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #843 key=764

gamma (22)
Particle 702
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.022, 0.018, -0.038, 0.048)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.065278
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.065278
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.674, 76.531, -167.738, 0.000)
p4@checkpoint: (-0.022, 0.018, -0.038, 0.048)
raw GEN: n/a
raw SIM: SimTrack #1772 key=739016

gamma (22)
Particle 703
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000149
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000149
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-98.224, 74.532, -182.986, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1778 key=739052

gamma (22)
Particle 704
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000248
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000248
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-106.042, 6.445, -293.500, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1780 key=739123

gamma (22)
Particle 705
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.015, 0.010, -0.024, 0.029)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.028248
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.028248
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.969, 76.157, -167.734, 0.000)
p4@checkpoint: (-0.015, 0.010, -0.024, 0.029)
raw GEN: n/a
raw SIM: SimTrack #1781 key=739190

gamma (22)
Particle 706
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000155
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000155
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.926, 76.212, -167.814, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #1782 key=739191

gamma (22)
Particle 707
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001512
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001512
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-0.063, 123.300, -208.179, 0.000)
p4@checkpoint: (0.001, 0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #1784 key=745541

connector

Particle 708
genEvent: 0
hasGen: no hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 1
nParents: 0 nChildren: 10
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 6979 simE=184.465042
subgraph recHits: 0 missing=6979 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 599 dE=1817.824463
raw GEN: n/a
raw SIM: n/a

initial state

Vertex 535
initial state (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: no isSink: no
x4: (0.001, -0.000, 0.551, 0.338)
nIn: 1 nOut: 10
raw GEN: n/a
raw SIM: n/a

interaction

Vertex 534
interaction (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: yes isSink: no
x4: (0.001, -0.000, 0.551, 0.338)
nIn: 0 nOut: 1
raw GEN: n/a
raw SIM: n/a